

O-RAN WG7**Hardware Reference Design Specification for Indoor Picocell
(FR1) with Split Architecture Option 7-2**

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Chapter 1 Introductory Material

1.1 Scope

This Technical Specification has been produced by the O-RAN.org.

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- y the second digit is incremented when editorial only changes have been incorporated in the document.
- z the third digit included only in working versions of the document indicating incremental changes during the editing process. This variable is for internal WG7 use only.

The present document specifies system requirements and high-level architecture for the FR1 Picocell Indoor deployment scenario as specified in the Deployment Scenarios and Base Station Classes document [1].

In the main body of this specification (in any “chapter”) the information contained therein is informative, unless explicitly described as normative. Information contained in an “Annex” to this specification is always informative unless otherwise marked as normative.

1.2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- [1] ORAN-WG7.DSC.0-V01.00 Technical Specification, ‘Deployment Scenarios and Base Station Classes for White Box Hardware’. <https://www.o-ran.org/specifications>
- [2] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".
- [3] 3GPP TR 38.104: "NR; Base Station (BS) radio transmission and reception".
http://www.3gpp.org/ftp/Specs/archive/38_series/38.104/38104-g10.zip
- [4] ORAN-WG4.CUS.0-v01.00 Technical Specification, ‘O-RAN Fronthaul Working Group Control, User and Synchronization Plane Specification’. <https://www.o-ran.org/specifications>
- [5] 3GPP TS 38.113: "NR: Base Station (BS) Electromagnetic Compatibility (EMC)".
http://www.3gpp.org/ftp/Specs/archive/38_series/38.113/38113-f80.zip
- [6] ORAN-WG7. IPC. HAR-v01.00 Technical Specification, ‘Indoor Pico Cell Hardware Architecture and Requirement Specification’. <https://www.o-ran.org/specifications>.

[7] ORAN-WG7. IPC. HRD.O8-v03.00 Technical Specification, ‘Indoor Pico Cell BS Hardware Reference Design Specifications with Fronthaul Split Option 8 and FR1’. <https://www.o-ran.org/specifications>

1.3 Definitions and Abbreviations

1.3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [2]. For the base station classes of Pico, Micro and Macro, the definitions are given in 3GPP TR 38.104 [3].

Carrier Frequency: Center frequency of the cell.

F1 interface: The open interface between O-CU and O-DU₇₋₂ as defined by 3GPP TS 38.473 between CU and DU.

Integrated architecture: In the integrated architecture, the O-RU and O-DU_x are implemented on one platform. Each O-RU and RF front end is associated with one O-DU₇₋₂. They are then aggregated to O-CU and connected by F1 interface.

Split architecture: The O-RU_x and O-DU_x are physically separated from one another in this architecture. A switch may aggregate multiple O-RU_xs to one O-DU_x. O-DU_x switch and O-RU_xs are connected by the fronthaul interface as defined in WG4.

Transmission Reception Point (TRxP): Antenna array with one or more antenna elements available to the network located at a specific geographical location for a specific area.

1.3.2 Abbreviations

For the purposes of the present document, the abbreviations given in [2] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, as in [2].

7-2	Fronthaul interface split option as defined by O-RAN WG4, also referred to as 7-2x
3GPP	Third Generation Partnership Project
5G	Fifth-Generation Mobile Communications
5GC	5G Core
ACS	Adjacent Channel Selectivity
ADC	Analog to Digital Converter
ASIC	Application Specific Integrated Circuit
ATA	Advanced Technology Attachment
BBDEV	Baseband Device
BH	Backhaul
BMC	Baseboard Management Controller
BPSK	Binary Phase Shift Keying
BS	Base Station
CISPR	International Special Committee on Radio Interference
CFR	Crest Factor Reduction

1	CU	Centralized Unit as defined by 3GPP
2	COM	Cluster Communication
3	CPRI	Common Public Radio Interface
4	CPU	Central Processing Unit
5	CRC	Cyclic Redundancy Check
6	DAC	Digital to Analog Converter
7	DDC	Digital Down Conversion
8	DDR	Double Data Rate
9	DIMM	Dual-Inline-Memory-Modules
10	DL	Downlink
11	DPD	Digital Pre-Distortion
12	DPDK	Data Plane Development Kit
13	DSP	Digital Signal Processor
14	DU	Distributed Unit as defined by 3GPP
15	DUC	Digital Up Conversion
16	ECC	Error Correcting Code
17	eCPRI	evolved Common Public Radio Interface
18	EMC	Electro Magnetic Compatibility
19	EVM	Error Vector Magnitude
20	FCC	Federal Communications Commission
21	FEC	Forward Error Correction
22	FFT	Fast Fourier Transform
23	FH	Fronthaul
24	FHGW	Fronthaul Gateway
25	FHM _x	Fronthaul Multiplexer with no FH protocol translation, supporting an O-DU _x with split option x and an O-RU _x with split option x, with currently available options 6→6, 7-2→7-2 and 8→8
26		
27	FHGW _{x→y}	Fronthaul Gateway that can translate fronthaul protocol from an O-DU _x with split option x to an O-RU _y with split option y, with currently available option 7-2→8.
28		
29	FHHL	Full Height Half Length
30	FPGA	Field Programmable Gate Array
31	GbE	Gigabit Ethernet
32	GNSS	Global Navigation Satellite System
33	GPP	General Purpose Processor
34	GPS	Global Positioning System
35	HARQ	Hybrid Automatic Repeat request
36	HHHL	Half Height Half Length
37	IEEE	Institute of Electrical and Electronics Engineers
38	IFFT	Inverse Fast Fourier Transform
39	IMD	Inter Modulation Distortion
40	I/O	Input/Output
41	JTAG	Joint Test Action Group
42	L1	Layer 1
43	LDPC	Low-Density Parity Codes

1	LRDIMM	Load-Reduced Dual In-line Memory Module
2	LTE	Long Term Evolution
3	LVDS	Low-Voltage Differential Signaling
4	MAC	Media Access Control
5	MCP	Multi-Chip Package
6	MH	Midhaul
7	MIG	Memory Interface Generator
8	MII	Media-Independent interface
9	MIMO	Multiple Input Multiple Output
10	MU-MIMO	Multiple User MIMO
11	NEBS	Network Equipment-Building System
12	NetConf	Network Configuration Protocol
13	NFV	Network Functions Virtualization
14	NIC	Network Interface Controller
15	NR	New Radio
16	O-CU	O-RAN Centralized Unit as defined by O-RAN
17	O-DU _x	A specific O-RAN Distributed Unit having fronthaul split option x where x may be 6, 7-2 (as
18		defined by WG4) or 8
19	O-RU _x	A specific O-RAN Radio Unit having fronthaul split option x, where x is 6, 7-2 (as defined by
20		WG4) or 8, and which is used in a configuration where the fronthaul interface is the same at the O-
21		DU _x
22	OCXO	Oven Controlled Crystal Oscillator
23	PCIe	Peripheral Component Interface Express
24	PDCP	Packet Data Convergence Protocol
25	PHY	Physical Layer
26	PMBus	Power Management Bus
27	POE	Power over Ethernet
28	PPS	Pulse Per Second
29	PRACH	Physical Random Access Channel
30	QAM	Quadrature Amplitude Modulation
31	QPSK	Quadrature Phase Shift Keying
32	QSFP	Quad Small Form-factor Pluggable
33	RAN	Radio Access Network
34	RDIMM	Registered Dual In-line Memory Module
35	RF	Radio Frequency
36	RoE	Radio over Ethernet
37	RU	Radio Unit as defined by 3GPP
38	RX	Receiver
39	SATA	Serial ATA
40	SDU	Service Data Unit
41	SFP	Small Form-factor Pluggable
42	SFP+	Small Form-factor Pluggable plus
43	SOC	System On Chip

1	SPI	Serial Peripheral Interface
2	SSD	Solid State Drive
3	TCXO	Temperature Compensate Crystal Oscillator
4	TDP	Thermal Design Power
5	TR	Technical Report
6	TS	Technical Specification
7	TX	Transmitter
8	UL	Uplink
9	USB	Universal Serial Bus
10	WG	Working Group
11		

Chapter 2 Hardware Reference Design 1

This chapter describes a white box hardware reference design example for indoor Picocell deployment scenario. It includes O-CU, O-DU₇₋₂ and FHM₇₋₂/FHGW_{7-2→8} for IPC deployment scenario.

2.1 O-CU Hardware Reference Design

The O-CU white box hardware is the platform that perform the O-CU function of upper L2 and L3. The hardware systems specified in this document meet the computing, power and environmental requirements of use cases configurations and feature sets of RAN physical node. These requirements are described in the early hardware requirement specification as well as in the use cases document. The O-CU hardware includes the chassis platform, mother board, peripheral devices and cooling devices. The mother board contains processing unit, memory, the internal I/O interfaces, and external connection ports. The midhaul (MH) and backhaul (BH) interface are used to carry the traffic between O-CU and O-DU₇₋₂ as well as O-CU and core network. The other hardware functional components include: the storage for software, hardware and system debugging interfaces, board management controller, just to name a few; the O-CU designer will make decision based on the specific needs of the implementation.

The HW reference design of O-CU is the same as O-DU₇₋₂ except for the need of HW accelerator, thus detail design will be described in O-DU₇₋₂ section 2.2.

2.2 O-DU₇₋₂ Hardware Reference Design

The O-DU₇₋₂ white box hardware is the platform that performs the O- DU₇₋₂ functions such as upper L1 and lower L2 functions. The hardware systems specified in this document meet the computing, power and environmental requirements of use case's configurations and feature sets of RAN physical node. These requirements are described in the early hardware requirement specification as well as in the use cases document. The O-DU₇₋₂ hardware includes the chassis platform, mother board, peripheral devices and cooling devices. The mother board contains processing unit, memory, the internal I/O interfaces, and external connection ports. The fronthaul and backhaul interface are used to carry the traffic between O-RU₇₋₂/FHM₇₋₂/FHGW_{7-2→8} and O-DU₇₋₂ as well as O-CU and O-DU₇₋₂. The O-DU₇₋₂ design may also provide an interface for hardware accelerator if that option is preferred. The other hardware functional components include: the storage for software, hardware and system debugging interfaces, board management controller, just to name a few; the O-DU₇₋₂ designer will make decision based on the specific needs of the implementation.

Note that the O-DU₇₋₂ HW reference design is also feasible for O-CU and integrated O-CU/ O-DU₇₋₂.

2.2.1 O-DU₇₋₂ High-Level Functional Block Diagram

Figure 2-1 shows the major functional blocks of O-DU₇₋₂. The digital processing unit handles the baseband processing workload. To make the processing more efficient, an accelerator can be used to assist with the baseband workload processing. The memory devices include the random-access memory (RAM) for temporary storage of data while flash memory is used for codes and logs. The storage device is for persistent storage. The external network cards can be used for fronthaul or backhaul connection. The baseboard management controller (BMC) is a microcontroller which

monitors hardware operation on motherboard. The clock circuits provide digital processing unit with required clock signals.

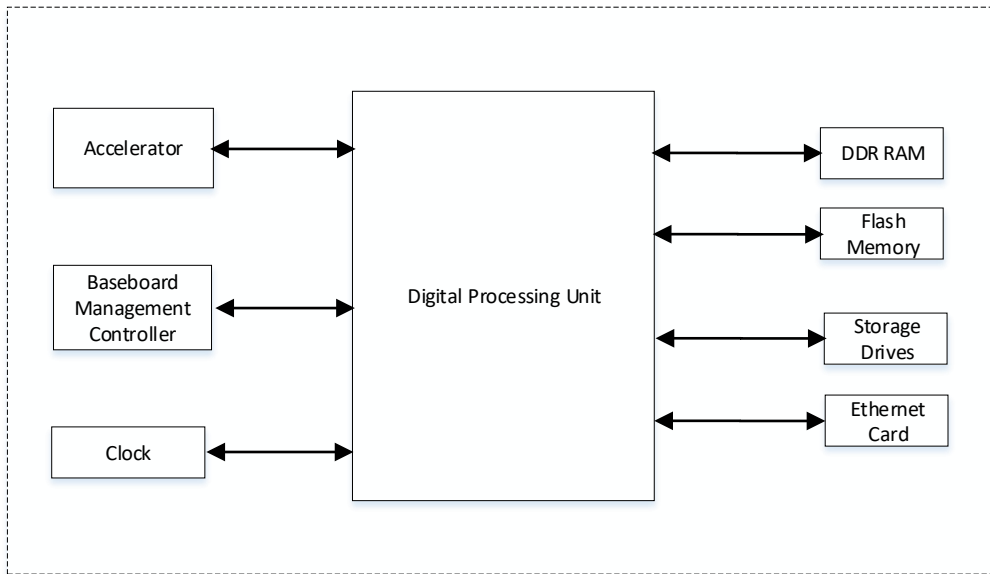


Figure 2-1: O-DU₇₋₂ Functional Block Diagram.

2.2.2 O-DU₇₋₂ Hardware Design Description

Figure 2-2 describes the various components and connections inside the O-DU₇₋₂ white box.

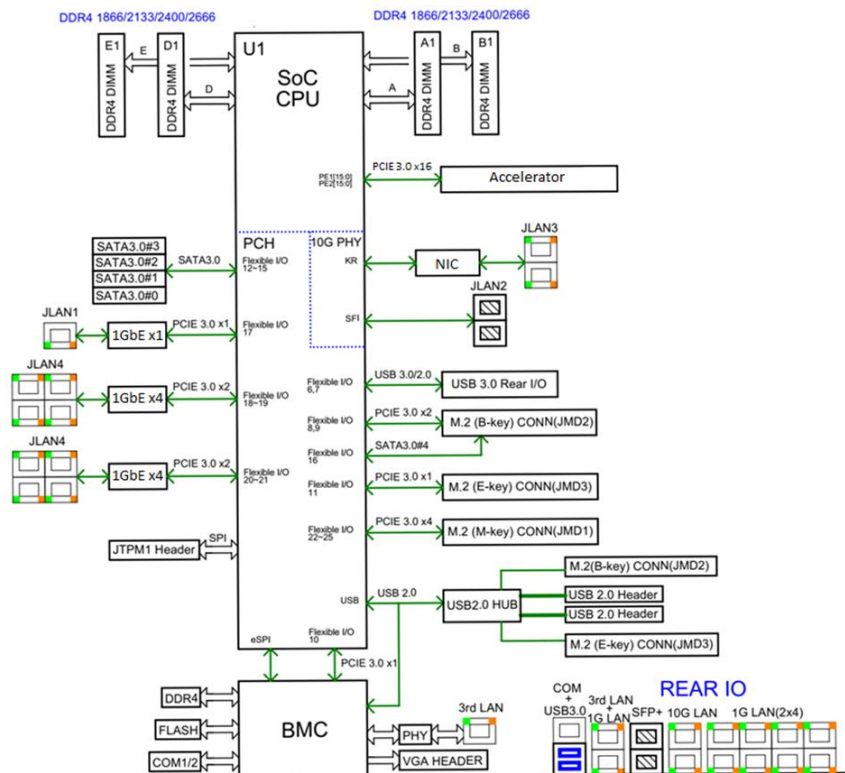


Figure 2-2: O-DU₇₋₂ Hardware Block Diagram

As described in the previous section, the O-DU₇₋₂ hardware can be implemented with different design choices. Here, we use a SoC design-based as an example which performs most of the workload for O-DU₇₋₂. The accelerator can be used to perform other functions based on the overall performance requirement. Several Ethernet controllers are also used for front haul link, back haul link and remote console control connections. The other parts include RAM, flash memory, and hard drive storage. The JTAG and USB ports are provided for hardware debug and local connection if needed. Finally, BMC block is mainly responsible for monitoring the hardware status of the motherboard.

2.2.3 O-DU₇₋₂ Hardware Components

In this section, we describe the details of the O-DU₇₋₂ hardware component's requirements, their features and parameters. The component selection is based on the use case requirements which are listed in the hardware architecture and requirements document [6].

2.2.3.1 Digital Processing Unit

This example of the digital processing unit in O-DU₇₋₂ is based on the General-Purpose Processor (GPP).

a. *Digital Processing Unit Requirement*

The GPP requirements are listed in the following table.

Table 2-1: The Processor Feature List

Item Name	Description
# of Cores	16
# of Threads	32
Base Frequency	2.20 GHz
Max Turbo Frequency	3.00 GHz
Cache	22 MB
Thermal Design Power (TDP)	100W
Max Memory Size (dependent on memory type)	512 GB
Memory Types	DDR4
Max # of Memory Channels	4

Interfaces: The interface specifications on the main board are as follows:

Memory Channel Interfaces: The system memory capacity, type and related information are described in the following table.

Table 2-2: The Memory Channel Feature List

Item Name	Description
Memory Types	DDR4
# of Memory Channels	4
ECC LRDIMM	Up to 512GB
ECC RDIMM	Up to 256GB
Memory Speed	2666/2400/2133MHz
DIMM Sizes	128GB, 64GB, 32GB, 16GB
Memory Voltage	1.2 V

PCIe: PCIe Gen 3 should be supported by the processor. There are total of 32 PCIe lanes with 128 Gb/s bandwidth. The 32 PCIe lanes can be divided into two x16 slots by using a riser card.

Ethernet: The system should be capable to offer aggregated 48 Gb/s Ethernet bandwidth. The breakout the ports are discussed in later section. When higher Ethernet bandwidth required, an Ethernet card can be installed in one of the PCIe slot.

b. Digital Processing Unit Design

The digital processing unit is a system-on-a-chip (SoC) device which is a 64-bit multi-core server class processor. This SoC includes an integrated Platform Controller Hub (PCH), integrated high-speed I/O, Integrated Memory Controllers (IMC), and four integrated 10 Gigabit Ethernet ports.

The SOC supports 512-bit wide vector processing instruction set. It also supports hardware virtualization to enable dynamic provisioning of services as communication service providers extend network functions virtualization (NFV). Figure 2-2 shows the major functional blocks of the digital processing unit.

2.2.3.2 Hardware Accelerator

Hardware accelerators can be used in O-DU₇₋₂ to offload computationally intensive functions and to optimize the performance under varying traffic and loading conditions. The acceleration functional requirements and implementation are system designer's choice; however, the O-DU₇₋₂ shall meet the minimum system performance requirements under various loading and deployment conditions. In most cases, a Field Programmable Gate Array (FPGA) or Application Specific Integrated Circuit (ASIC) based PCIe card can be used to optimize the system performance. The FPGA(s) are part of a Network Interface Controller (NIC) that further provides connectivity services.

2.2.3.2.1 Accelerator Design Solution 1

Channel coding for Low-Density Parity Codes (LDPC) and fronthaul compression requires a significant amount of bit level processing and is well suited to a fine-grained FPGA architecture and/or low cost/power structured ASIC. Options include:

- Virtualization: Multi-channel DMA, Virtual machine integration, load balancing functions
- Look-aside FEC: LDPC (de)coding, Polar (de)coding, Rate (De)Matcher, (De) Interleaver, Cyclic Redundancy Check (CRC), Hybrid Automatic Repeat request (HARQ) retransmission
- Bump-in-wire Fronthaul: compression / decompression for latency and bandwidth reduction

a. Accelerator Requirements

Hardware and firmware requirements for this accelerator design are given in Table 2-3 and Table 2-4, respectively.

Table 2-3: Accelerator Hardware Feature List

Item Name	Description
PCIe (Interface with digital processing unit)	Gen4 x16 (and lower)
Form factor	FHHL
Connectivity	2x QSFP28/56
FPGA	Logic Elements: 1437K

	M20K Memory: 139Mb Quad A53 Hard Processor Sub-system
NIC Device	100Gb xHAUL for FH, BH & MH traffic shaping. Optional FPGA co-processing.
DDR Main	8+8+1GB DDR4
Flash (FPGA images)	>=1 Gbit
BMC	Telemetry, Security, remote upgrade
Clocking	For O-RAN C1, C2, C3 & C4
Fronthaul	eCPRI, RoE IEEE1914.3, O-RAN WG4
GPS	SMA for 1 PPS & 10MHz (in/out)
Operating Temperature (ambient)	NEBS Compliant
Power	<75W (without optional NIC device)
Clock Accuracy	Low-Jitter, configurable clock ranging from 10MHz to 750MHz. Option for OCXO (TCXO as standard)

Table 2-4: Accelerator Firmware Feature List

Item Name	Description
Remote system upgrade	Securely upgradable FPGA flash image
Queuing	64 Queues supported equally split between UL & DL.
LDPC Acceleration	NR LDPC Encoding with, interleaving and rate-matching. NR LDPC Decoding with sub-block de-interleaving function of reverse rate matching. Early Termination, CRC attachment and HARQ buffering. 5G Throughput: DL 14.8Gbps, UL 3.2Gbps
Load Balancing (channel coding)	Load balancer distributes the pending encoder/decoder requests to encoder/decoders
Descriptor Format (channel coding)	Code block based interface. Software enablement by BBDEV API (DPDK) https://www.dpdk.org/ https://doc.dpdk.org/guides/prog_guide/bbdev.html
Fronthaul Compression	In-line compression/decompression for Mu-Law, block-floating point and quantization according to the O-RAN WG4 specification.
Open programmable acceleration environment	Support for: - FPGA Flashing upgrade - Firmware version reporting - PCIe diagnostics - Ethernet diagnostics - Temperature and voltage telemetry information

b. Accelerator Design

Figure 2-3 illustrates Accelerator Design 1 without optional NIC Device. Figure 2-4 illustrates the Accelerator Design 1 with optional NIC Device.

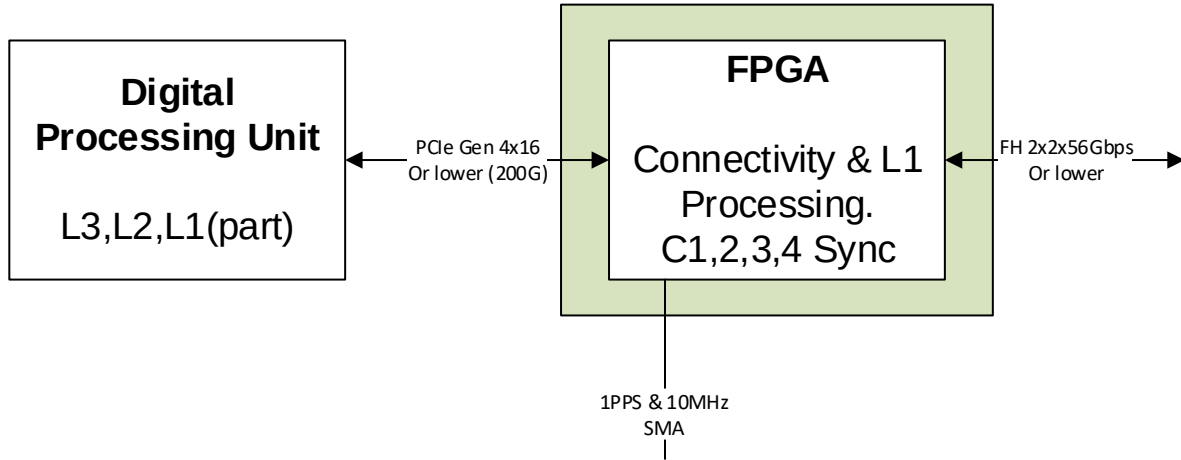


Figure 2-3: Accelerator Design 1 without optional NIC Device

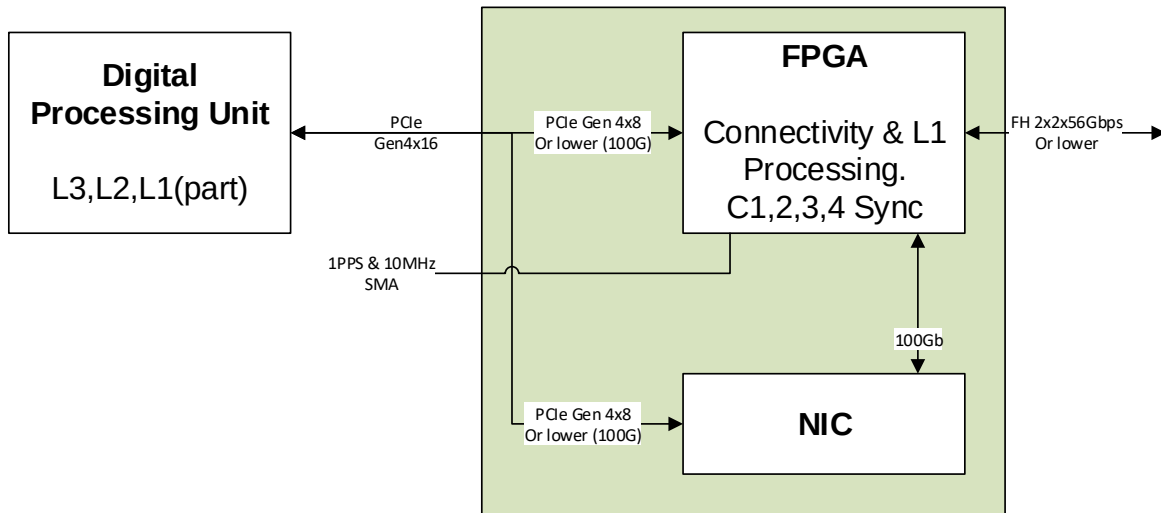


Figure 2-4: Accelerator Design 1 with optional NIC Device

2.2.3.2.2 Accelerator Design Solution 2

The O-DU₇₋₂ system is typically implemented using a multi-core processor and one or more hardware accelerators. Parts of O-DU₇₋₂ protocol stack can be implemented in software running on the multi-core processors, some of the computationally intensive L1 and L2 functions are offloaded to FPGA-based or similar hardware accelerators. This is a programmable hardware, which provides both flexibility and high computing capabilities.

a. Accelerator Requirement

The accelerator unit comprises one or more FPGAs (e.g., two FPGAs), sufficient amount of DDR4 memory, and synchronization circuitry where one of FPGAs is used for L1 functional offload and the other one is used to perform fronthaul connectivity functions/protocols. The FPGA for L1 offload uses dedicated cores for channel encoding/decoding as well as FPGA and processing resources for running L1 functions such as but not

limited to rate matching and de-matching, interleaving and scrambling, demodulation and HARQ buffer management as well as OFDM modulation/demodulation and channel estimation.

Key features of the FPGA-based accelerator include:

- 2X10/25G eCPRI or Ethernet open fronthaul interface
- Built-in SyncE/IEEE 1588v2 synchronization + external reference timing
- L1 offloading options
 - LDPC encoding and decoding
 - Polar encoding and decoding
 - HARQ management with on board DDR memory (including DDR controller and interfaces)
 - Other L1 offloading candidates include PRACH detection, MIMO encoding and decoding, channel estimation
 - Partial or full L1 functions can be offloaded. It is recommended to offload the user-plane channel coding chain and part of or the entire control-plane channel coding chain to the hardware accelerator.
- PCIe Gen3 x16, two Gen4 x8, or PCIe Gen3 x16 bifurcated to two Gen3 x8
- GPP supported
- Standard PCIe FHHL card (It is assumed that the hardware accelerators further perform NIC functions).
- 2x SFP28 for 25G Ethernet
- 8GB DDR4 memory for buffering
- Power consumption not to exceed 75W

The accelerator requirements are summarized in Table 2-5.

Table 2-5: Accelerator Hardware Component List

Item Name	Description for FPGA 1	Description for FPGA 2
SoC Resources	System Logic cells - 930K CLB LUT - 425K SDFEC -8 DSP Slices - 4,272 BRAM - 38.0Mb URAM - 22.5Mb	System Logic cells - 1,143K CLB LUT - 523K CLB Flip-Flops -1,045K DSP Slices - 1,968 BRAM - 34.6Mb URAM - 36.0Mb
Form Factor	FHHL PCIe Form Factor	
PCIe Interface	x8 Gen1, Gen2, Gen3 interface to FPGA 2 and x8 Gen1, Gen2, Gen3 interface to FPGA 1 have x16 to two x8 bifurcation	
On Board Memory	Total Capacity 4 GB in PL, upgradeable to 8GB Total Capacity 2 GB in PS, upgradeable to 4GB	Total Capacity 4 GB in PL, upgradeable to 8GB Total Capacity 2 GB in PS, upgradeable to 4GB
Network Interface(s)		2xSFP28 optical interfaces to FPGA2 (User Configurable, includes 10/25 Ethernet)

Other External Interface(s)	Micro USB for JTAG support (FPGA programming and debug) and access to BMC
Graphical User interface	GUI for monitoring the basic board parameters, monitoring temperature alerts, firmware upgrades for BMC
Board Management Controller	• Power sequencing and reset • Field upgrades • FPGA configuration and control • Clock configuration • I2C bus access • USB 2.0
Operating Temperature	-5°C - 55°C
Power	< 75 W
Clocking Options	Low-Jitter, configurable clock ranging from 10MHz to 750MHz 1 PPS input and output with assembly option for OCXO and TCXO

The following are the accelerator's functional and interface requirements.

Functional offload requirements:

One of the candidate functions for offloading is the LDPC encoder and decoder, which typically consists of computationally intensive and relatively highly power consuming functions. It must be noted that neither software implementation in CPU nor soft FPGA logic implementation would provide a highly power efficient solution while meeting/exceeding 3GPP NR user-plane encoder/decoder throughput and latency requirements, rather a hardened implementation of the FEC functions would be very power efficient. Downlink and uplink throughputs of up to 40Gbps and 18Gbps, respectively, are shown feasible with this architecture. Other candidate L1 functions for acceleration include CRC generation, segmentation, bit-level/sub-block interleaving and scrambling as well as FFT/IFFT processing, for which an FPGA can be used.

For other symbol processing L1 functions which require heavy multiply and accumulation operations, FPGAs 1 and 2 have DSP blocks that can efficiently perform these operations. Polar encoding and/or decoding on the control-plane can also be offloaded to FPGA 1 resulting in high throughputs and low latencies.

Interface requirements:

PCIe: PCIe interface is widely used to provide interface between the GPP and hardware accelerators. FPGA devices have dedicated PCIe hard IP which facilitate implementation and quick setup of PCIe interface. They support both PCIe Gen3 x16 or PCIe Gen4 x8, which allow the FPGA device to interface with any GPP supporting either PCIe Gen3 x16 or PCIe Gen4 x8 interface.

Fronthaul: FPGA devices can support various speed grades and any fronthaul protocols, customers can use off-the-shelf eCPRI or Ethernet IPs to quickly implement and configure any fronthaul interface protocol.

Serial transceivers: FPGA devices have SerDes resources to implement various connectivity speeds (e.g., 33 Gb/s) per SerDes, 10G/25G/50G/100G Ethernet connections can use these SerDes resources.

Ethernet MAC speed: FPGA devices have hardened implementations of Ethernet MAC that support speeds of 100 Gbps and above. The Ethernet MAC IP allows power-efficient implementation of high-speed Ethernet connectivity. In the example shown in Figure 2-3, FPGA 1 and FPGA 2 can use the hard 100Gbps MAC IPs to connect each other, allowing the L1 and fronthaul functions to be distributed across these two FPGAs with less

connectivity overhead. For other Ethernet MACs such as 10G/25G Ethernet, they provide soft Ethernet MAC IPs, so when implementing eCPRI or Ethernet fronthaul functions, 10G/25G Ethernet MAC can be used.

b. Accelerator Design

The hardware accelerator supports GPP. Figure 2-5 illustrates a two-chip acceleration architecture comprising two FPGAs with multi-lane PCIe interfaces toward the CPU and external connectivity toward O-RU₇₋₂(s) via eCPRI and O-CU(s) through GbE connectivity. The example architecture further depicts multi-lane Gen3 or Gen4 PCIe interfaces between each FPGA and the CPU. The FPGAs communicate through high-bandwidth Ethernet (GbE) transport.

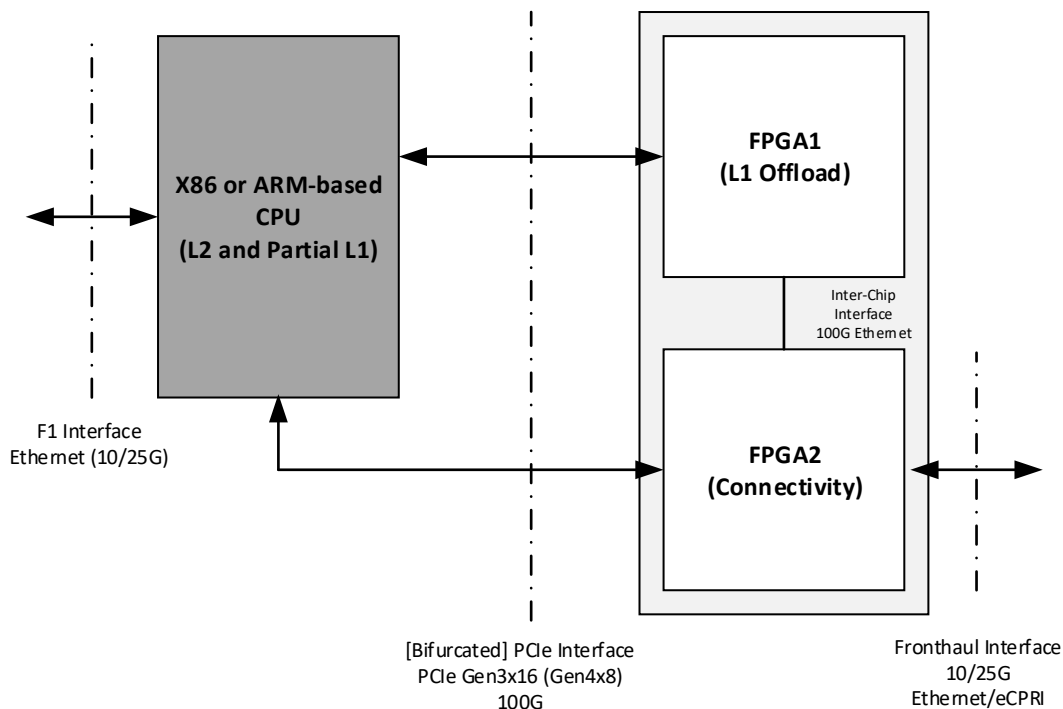


Figure 2-5: Example 2-chip FPGA-based Hardware Acceleration in O-DU₇₋₂

2.2.3.2.3 Accelerator Design Solution 3

Channel coding for LDPC and fronthaul compression requires a significant amount of bit level processing and is well suited to a fine-grained structured ASIC. Features include:

- Virtualization: Multi-channel DMA, Virtual machine integration, load balancing functions
- Look-aside FEC: Turbo (de)coding, LDPC (de)coding, Rate (De)Matcher, (De) Interleaver, CRC, HARQ retransmission & (de) interleaver.

a. Accelerator Requirement

Accelerator Hardware and Firmware features are listed in Table 2-6 and Table 2-7, respectively.

Table 2-6: Accelerator Hardware Feature List

Item Name	Description
PCIe	Gen3 x16
Form factor	HHHL
DDR	DDR4 (64-bit +ECC), 2667Mbps Interface for HARQ buffering

Board Management Controller	Telemetry, Security.
Power	<35W

Table 2-7: Accelerator Firmware Feature List

Item Name	Description
Queuing	64 Queues supported equally split between UL & DL.
LDPC Acceleration	NR LDPC Encoding with, interleaving and rate-matching. NR LDPC Decoding with sub-block de-interleaving function of reverse rate matching. Early Termination, CRC attachment and HARQ buffering. 5G Throughput: DL 23Gbps, UL 8Gbps
Load Balancing (channel coding)	Load balancer distributes the pending encoder/decoder requests to encoder/decoders
Descriptor Format (channel coding)	Code block and transport block based interface. Software enablement by BBDEV API (DPDK) https://www.dpdk.org/ https://doc.dpdk.org/guides/prog_guide/bbdev.html

b. Accelerator Design

The following diagram shows the structured ASIC based accelerator design.

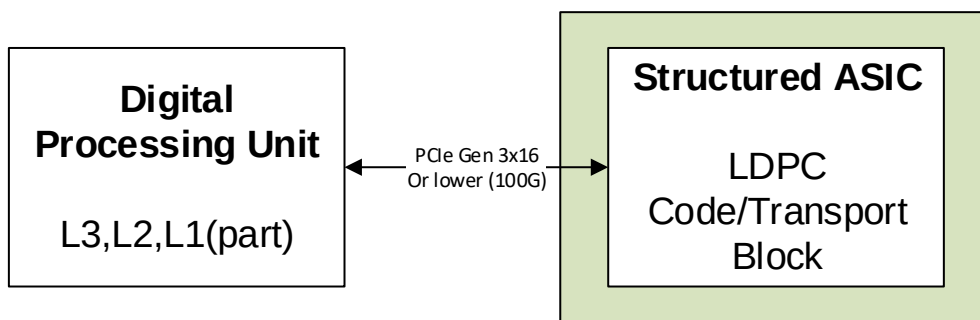


Figure 2-6: Structured ASIC Accelerator Design

2.2.3.3 Baseboard Management Controller

BMC is used to perform hardware power control (power on, power off and power cycle), monitor hardware status (temperatures, voltages, etc), monitor Basic I/O System (BIOS)/ Unified Extensible Firmware Interface (UEFI) firmware status, and log system events. It provides remote access via shared or dedicated NIC. System user can do

console access via serial or physical/Kernel-based Virtual Machine (KVM). The BMC has dedicated RAM and flash memory. It provides access via serial port or Ethernet port. Figure 2-2 describes the BMC connections with related components.

2.2.4 Synchronization and Timing

2.2.4.1 Synchronization and Timing Design 1

This section describes the synchronization and timing mechanism that is used in the O-DU₇₋₂.

a. Hardware Requirements

The O-DU₇₋₂ shall support following timing synchronization methods:

1. GPS Synchronization
2. Ethernet based IEEE1588V2 Synchronization
3. BeiDou Synchronization
4. BeiDou and GPS switching

b. Hardware Design

Depending on the timing distribution topologies used, the O-DU₇₋₂ system clock is able to synchronize with the Grand Master Clock (GMC) using IEEE1588 via either the front haul NIC or backhaul NIC, or synchronizing timing using Global Navigation Satellite System (GNSS). In the case of IEEE1588, the Physical Hardware Clock (PHC) within the NIC is synchronized with the GMC first, then the O-DU₇₋₂ system clock is synchronized with the PHC. The O-DU₇₋₂ is also capable to provide clock to the O-RU₇₋₂ via front haul if needed. When GNSS becomes available to O-DU₇₋₂, it will be able to synchronize the system clock to Coordinated Universal Time (UTC). Figure 2-7 outlines the O-DU₇₋₂ timing synchronization mechanisms.

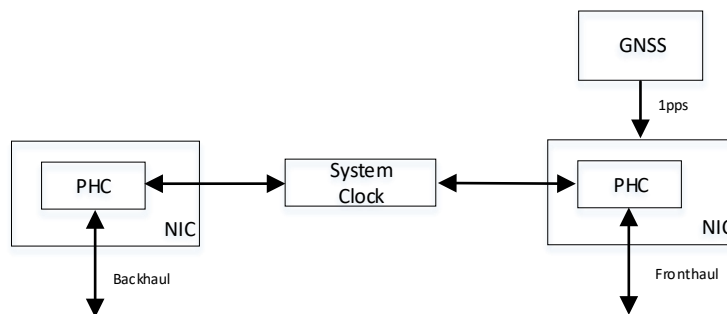


Figure 2-7: O-DU₇₋₂ Timing Synchronization

2.2.4.2 Synchronization and Timing Design 2

This section describes the synchronization and timing mechanism that is used in the hardware accelerator board.

a. Hardware Requirements

Each hardware accelerator board that supports connectivity functions (e.g., O-RAN fronthaul) in O-DU₇₋₂ must be able to support external synchronization I/O and to receive or transmit 1 PPS reference clock source in order to ensure synchronization across network.

b. Hardware Design

The accelerator board can operate in the master or slave mode when supporting IEEE 1588v2 timing/synchronization. It can generate 1 PPS reference clock for synchronization in the master mode and can receive the 1 PPS reference clock in the slave mode for internal synchronization. The timing circuitry of the accelerator board is shown in Figure 2-8. Two FPGA SerDes transceivers are used to receive and transmit SyncE TX and RX clocks.

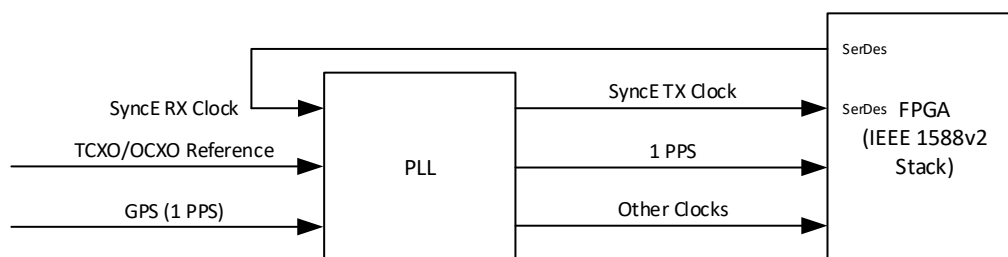


Figure 2-8: HW Acceleration Board Timing and Synchronization

2.2.5 External Interface Ports

The external interfaces of O-DU₇₋₂ are described below.

a. Hardware Requirements

The following table shows the external ports or slots that the system provided.

Table 2-8: External Port List

Item Name	Description
Ethernet	Octave Gigabit Ethernet LAN connectors
	Dual 10GbE Base-T Ethernet connectors
	Dual 10GbE SFP+ Faber Ethernet connectors
	2x100G QSFP28 or 2x25G SFP28
USB	2 USB 3.0 ports
Serial Port	1 COM port via RJ45
Antenna port	1 SMA connector for GNSS

b. Hardware Design

The digital processing unit is a SOC device which provides the external ports described in the hardware requirement section. The system includes 2 USB 3.0 ports, and the serial RS232 port that can be used for Console Redirection, e.g. Out-of-Band Management. The system provides eight 1Gbps and four 10Gbps Ethernet ports. There are two or four 25G eCPRI ports in system depends on the accelerator card used. The system also provides a RF interface to connect GNSS antenna. The following diagram outlines the external interfaces that supported.

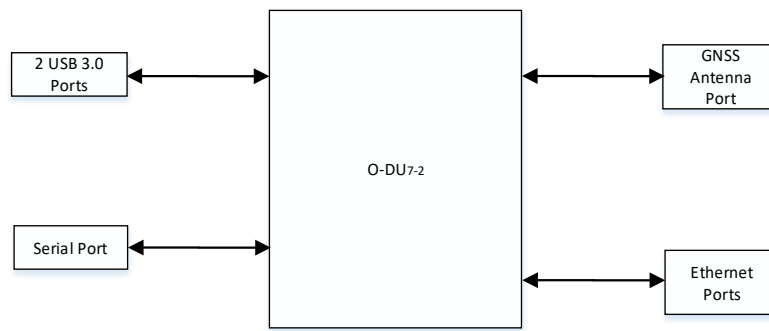


Figure 2-9: External interfaces reference design

2.2.6 O-DU₇₋₂ Firmware

BIOS and BMC firmware are needed in the system and shall be installed.

2.2.7 Mechanical

The mechanical design for mother board, chassis, and cooling are listed in this section.

- **Mother Board**

The mechanical layout of the mother board shows the location of major components and interface ports. The following diagram also provides the dimension of the board.

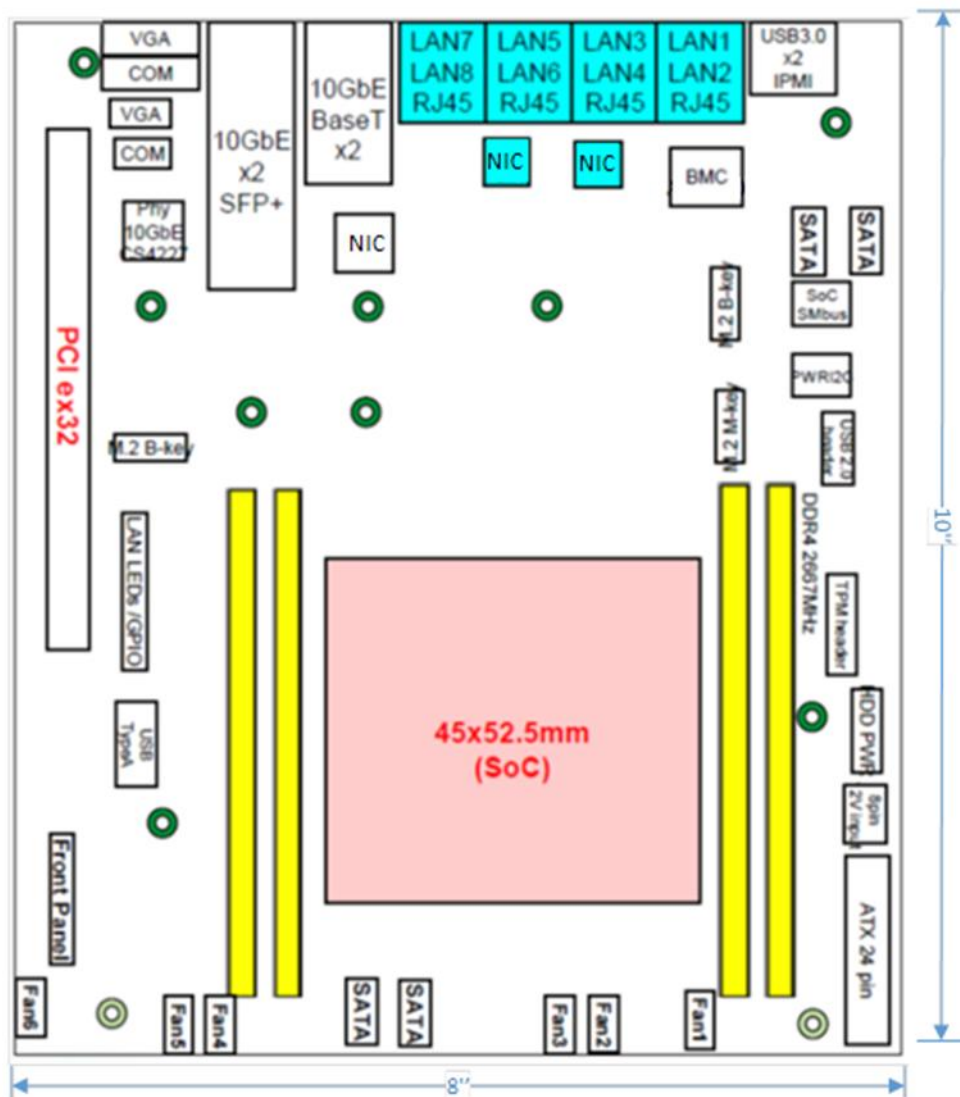


Figure 2-10: Mother Board Layout Diagram

- Chassis
- The 1U rack mount chassis contains the layout of the power supply, Solid State Drive (SSD) and fans. The chassis dimension is showed in following figure.

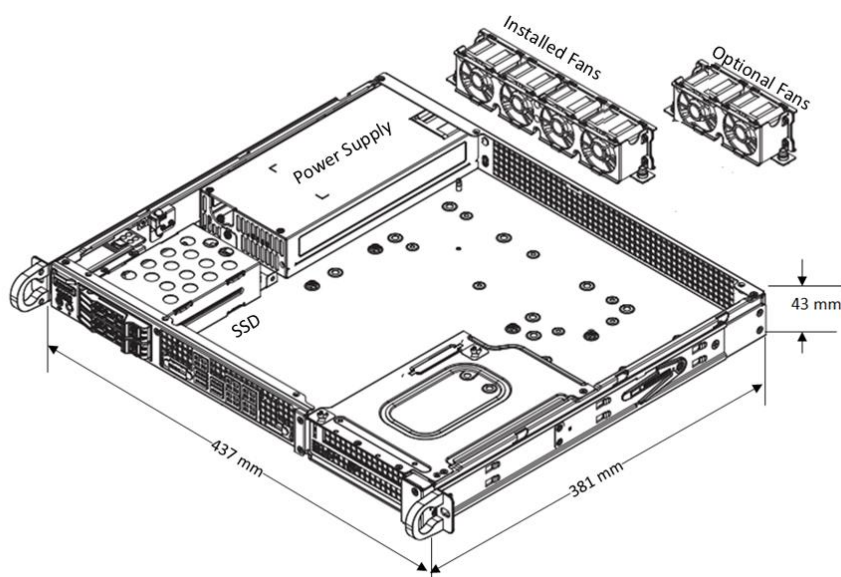


Figure 2-11: Chassis Mechanical Diagram

c. Cooling

The system installs 4x 40x28mm PWM fans for the cooling. Up to 6 fans can be installed if needed.

2.2.8 Power Unit

In a fully loaded system with two PCIe slots populated with 75W each, the system power consumption should be less than 400W. The total system power requirement shall be kept less than 80% of the power supply capacity.

a. Hardware Requirements

The power is provided by 500W High-Efficiency power supply with Power Management Bus (PMBus) 1.2. The power support input and output power rails are listed below.

AC Input: 100-240V, 50-60Hz, 6.6A max

DC Output: +3.3V: 12A

+5V: 15A

+5V standby: 3A

+12V: 41A

-12V: 0.2A

b. Hardware Requirements

The O-DU_{7.2} chassis includes one 500W power supply unit. The power supply unit is auto-switching capable, which enables it to automatically sense and operate at a 100v to 240v input voltage. The power supply unit features are listed in the following table.

Table 2-9: Power supply unit feature list

Item Name	Description
Output connectors	24pin/8pin/4+4pin/HDD/I2C
Dimension (W x D x H)	3.9 x 7.1 x 1.6 inch
Maximum Output Power	+3.3V: 12A +5V: 15A +12V: 41A -12V: 0.2A +5Vsb: 3A
Rated Input Voltage/Current	100-240Vac / 6.6A max
Rated Input Frequency	50-60HZ
Inrush current	Less than 30A

2.2.9 Thermal

Active cooling with up to 6 fans is integrated in the chassis.

The hardware acceleration cards described in Section 2.2.2.2 use passive cooling and a custom heatsink and is equipped with temperature sensors. It is designed to operate in temperatures ranging from -5°C to +55°C.

2.2.10 Environmental and Regulations

The O-DU₇₋₂ hardware system is RoHS Compliant. The power supply unit is EMC FCC/CISPR Class B compliant.

Table 2-10 lists the environmental related features and parameters.

Table 2-10: Environmental Features

Item Name	Description
Operating Temperature	-5°C to +55°C
Non-operating Temperature	-40°C to 70°C
Operating Relative Humidity	8% to 90% (non-condensing)
Non-operating Relative Humidity	5% to 95% (non-condensing)

The hardware accelerator described in Section 2.2.2.2 is designed to operate in indoor environments and in temperatures ranging from -5°C to +55°C.

2.3 O-RU₇₋₂ Hardware Reference Design

In this section we describe, the HW reference design of all O-RU₇₋₂ components including block diagram, HW components, synchronization and timing, mechanical design, power supply, thermal, and environment requirements.

2.3.1 O-RU₇₋₂ High-Level Functional Block Diagram

Figure provides a high-level functional block diagram depicting the major HW/SW components. It also highlights the internal/external interfaces that are required. This document shows how to implement the system defined by the IPC-HAR [6] document.

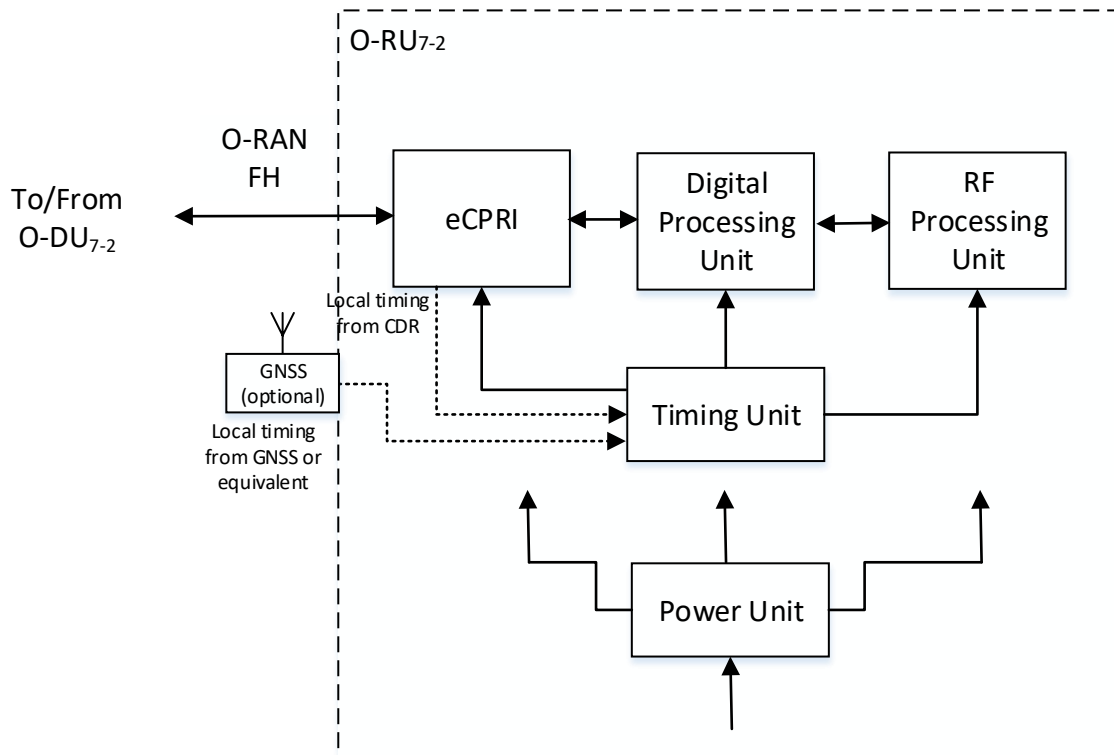


Figure 2-12: High-Level Functional Block Diagram

2.3.2 O-RU₇₋₂ Hardware Components for FPGA based Design

General block diagrams for 2T2R and 4T4R O-RU₇₋₂ examples are shown subsequently.

Figure 2-13 shows a 2T2R implementation and Figure 2-14 shows a 4T4R implementation. In each of these diagrams, the Digital Processing Unit is further detailed in 2.3.2.1. Items under the umbrella of RF Processing Unit, including the Transceiver, RFFE and other RF items are reviewed in additional detail in 2.3.2.2. Clock and Synchronization are reviewed in 2.3.4. The Power Unit is detailed in 2.3.7.

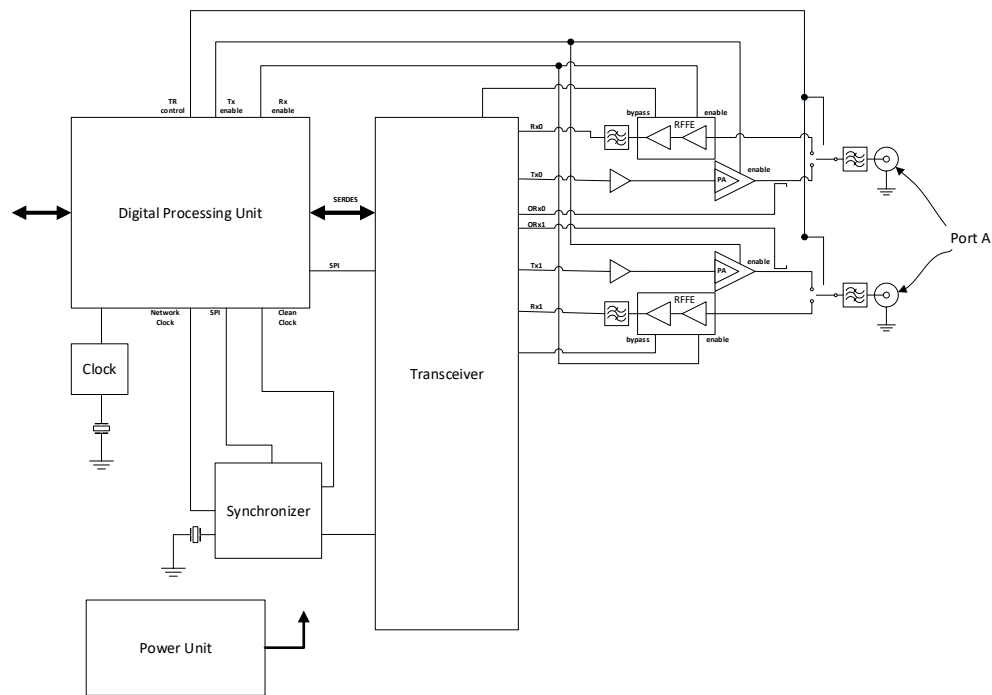


Figure 2-13: 2T2R General Block Diagram with TR switch

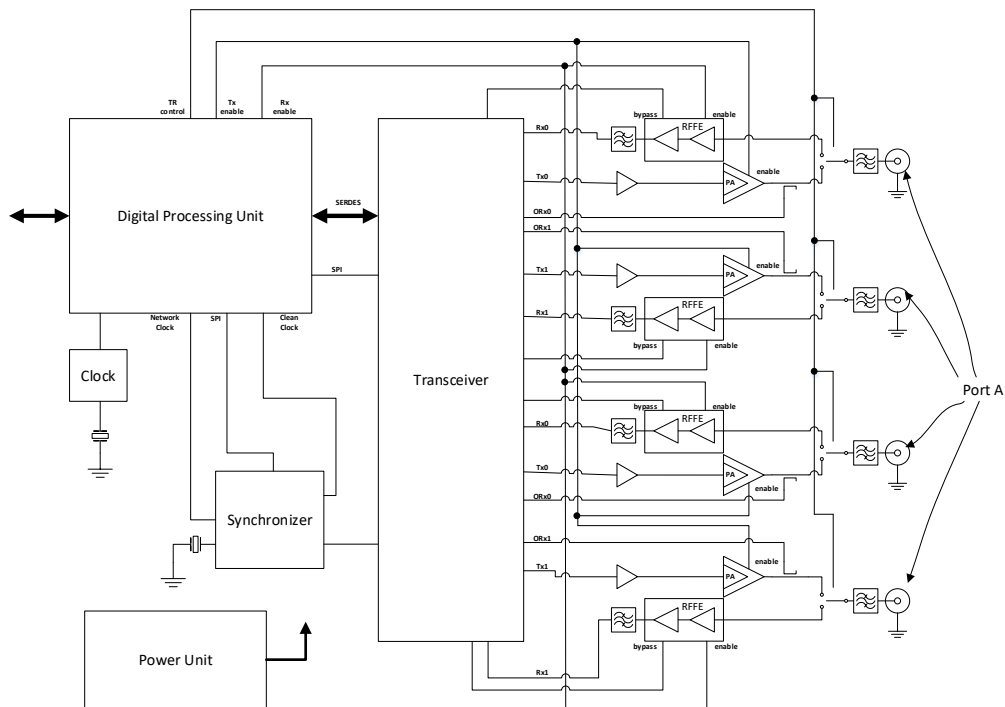


Figure 2-14: 4T4R General Block Diagram with TR switch

In following section, we describe their functionality, interfaces and performance of every block in the RF Processing Unit. Since device integration is an ongoing activity, chip boundaries may be fluid and some functionalities may move from one block to another or entire functionalities may be absorbed into other blocks. The sections below describe the functional blocks independent of which physical device they may reside in.

2.3.2.1 Digital Processing Unit

The digital processing unit of O-RU₇₋₂ is mainly for performing tasks related to FH interface, lower L1, RF interface, and OAM.

i. FPGA Solution Design 1

The digital processing unit of O-RU₇₋₂ is mainly for performing FH interface, lower L1, RF interface, and OAM functions.

a. FPGA Requirement

The following items are the main requirement for the O-RU₇₋₂:

- Interface requirement: One lane of bi-direction Serdes targeting eCPRI will be 10Gbps for FH split option 7-2. Four lanes of bi-direction JESD204B Serdes will be used for 2T2R. Eight lanes of bi-direction JESD204B Serdes will be used for 4T4R
- Resource requirement: FPGA resource requirements for 2T2R and 4T4R are shown in Table 2-11 and Table 2-12.

Table 2-11: Resource requirement for 2T2R O-RU₇₋₂

Module	FF	LUT	BRAM18	DSP
Model A	343800	171900	1000	900
IFFT & CP+	7763	5323	27	23
IFFT & CP-	7763	5323	27	23
PRACH frontend	4000	3000	32	50
DUC	4820	4346	0	108
DDC	4820	4346	0	108
CFR	11470	6136	22	36
DPD	34269	13250	188	87
JESD204B	4314	4285	0	0
Fronthaul(eCPRI)	10000	8000	1	0
Other	8000	5000	100	12
Total	97219	59009	397	447

Table 2-12: Resource requirement for 4T4R O-RU₇₋₂

Module	FF	LUT	DSP	BRAM18
Resource Mapping	10000	8000	0	102
PreCoding & pilot lns	5000	4000	16	4
IFFT+CP	6466	3952	15	75
Resource De-Mapping	10000	8000	0	181

FFT+CP	6466	3952	15	75
PRACH Filtering	8000	6000	36	16
Sub-Total	45932	33904	82	453
ChFIR	2832	2686	84	0
DUC	5148	3756	36	0
CFR	25676	12372	96	48
DPD	45770	17643	105	217
AGC	1000	600	8	4
DDC	1716	1252	12	0
UL_ChFIR	2832	2686	84	0
JESD2048	9900	11127	0	0
eCPRI	15000	14000	0	50
total	155806	100026	507	772

- Processor requirement: For device model A, the processor will be used, one is used for device control and management plane functions, the other one is for Digital Pre-Distortion (DPD) feedback path. For device model C, dual-core ARM cortex-A53 will be used, similarly one is used for device control and management plane functions, the other one is for DPD feedback path.
- Power requirement: Power estimations for 2T2R and 4T4R are shown in Figure 2-15 and Figure 2-16.

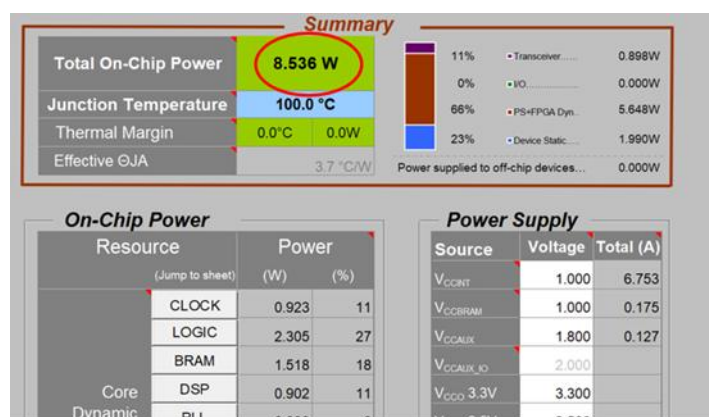


Figure 2-15: Power estimation for 2T2R O-RU₇₋₂

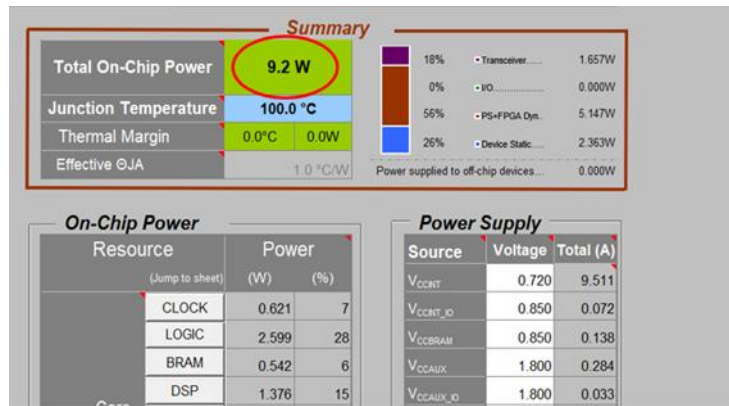


Figure 2-16: Power estimation for 4T4R O-RU₇₋₂

- Speed grade, environmental requirement: -2L, 0°C to 110°C

b. FPGA Design

This solution of the digital processing unit incorporates FPGA and a processor. The FPGA handles high speed digital processing such as low L1, FH, DDC, DUC, CFR and so on. All functions are listed in the previous section. The processor is used for hardware device configuration and the OAM function. The FPGA and the processor core can be integrated into one SOC or implemented into two devices. Here the FPGA and the processor core are integrated into one SOC device.

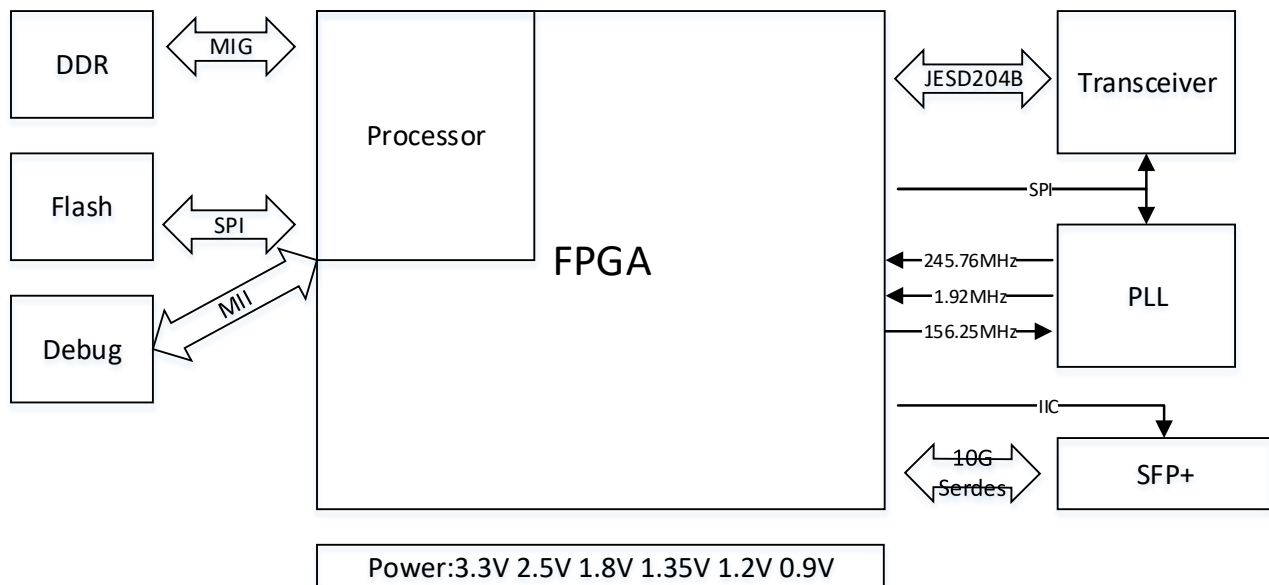


Figure 2-17: FPGA Reference Design Diagram

For the processor portion, the internal RAM resource may not be enough. So the external DDR is needed to let the processor handle more RAM consuming functions such as operation system or stack protocols like Network Configuration Protocol (NetConf) Client. For the O-RU₇₋₂ design, the DDR3 with 256Mb*16bit memory capacity is enough. The interface between the DDR and processor can be memory interface generator (MIG). The external Flash is used to store operation system related files, calibration information of the RF portion, NetConf related files, FPGA firmware and so on. For this O-RU₇₋₂ design, the flash with

2Gb memory capacity is enough. The online debug function is performed by external Ethernet PHY with an RJ45 connector. This allows the administrator to visit the internal function of the O-RU₇₋₂ and control it. The interface between the DDR and processor can be Media-Independent interface (MII). The Ethernet PHY device can be very general 100M Ethernet Transceiver. The FPGA has one Serdes lane connected to optical module to perform the fronthaul link between O-RU₇₋₂ and O-DU₇₋₂ /FHM₇₋₂. Another 4 Serdes lanes of the FPGA are needed to connect one transceiver of the O-RU₇₋₂ to transmit or receive IQ sample by the interface of JESD204B while the FPGA needs synchronized clock signals to work well. The interface between PLL and FPGA should be Low-Voltage Differential Signaling (LVDS).

2.3.2.2 RF Processing Unit

For the RF processing unit of O-RU₇₋₂, it will perform functionalities of ADC, DAC, LO, down converter, up converter etc.

2.3.2.2.1 Transceiver Reference Design

For the O-RU₇₋₂ the sampling function and frequency conversion function can be performed by transceiver. The purpose of using the transceiver is to saving power and size of the PCB. The Transceiver is to convert between high-speed baseband data and a low-level RF for both transmit and receive signal chain. In addition, the transceiver is responsible for orchestration of control signals not limited to the PA enable, LNA enable, LNA bypass as well as other required system level signals.

a. Hardware Requirements

Include the requirements for this component.

- Interface Requirements: The interface requirements for the Transceivers are list in Table 2-13.

Table 2-13: Interface requirements of the Transceiver

Item Name	Description
High Speed Data	High speed data represents the baseband information being transmitted or received. Depending on the configuration of the ORAN device, various bandwidths may be supported leading to a range of payload rates. Options for data include parallel data paths, JESD204B and JESD204C. Up to 8 lanes in each direction may be supported although fewer is preferred. Options such as DPD and numeric precision will impact the payload rate. Several options are shown in the following table. All data represents IQ 16-bit (N=16) precision. Some devices support IQ 12 bit (N'=12) which may reduce the required data rates accordingly. The tables below assume 1 ORx for 2 TRx. From Table 2-14 and Table 2-15, the number of lanes required may be determined by dividing the total bit rates shown by the capacity of a lane, typically 12.5 GBPS for JESD204B and 25 GBPS for JESD204C.
Reference Clock (Device Clock)	The transceiver should receive a reference for internal clock and LO synthesis needs. This reference clock can function as the JESD204 Device Clock where the interface is by SERDES. The specific clock frequency is determined by the operation mode of the transceiver and may range from 1Hz upward.
SYSREF	If the transceiver supports SERDES, then it should accept a SYSREF signal from the clock or data source as appropriate. The number and configuration for the SYSREF is dependent on the operating mode of the transceiver.

SYNCB	If the transceiver supports SERDES, then it should also support a SYNCB for each link as appropriate.
Control	Control of the transceiver is by way of 3 or 4 wire SPI or IIC functioning as a slave. Support for 1.8V control is required and tolerance of 3.3V is preferred. The transceiver may optionally include a separate SPI master for control of peripheral devices as required.
GPIO	The transceiver may optionally include GPIO for controlling peripherals including but not limited to PAs, LNAs and other devices. These GPIOs should at a minimum support 1.8V outputs but the specifics will be determined by the connected devices. The GPIO should also support input from peripheral devices. Input should at a minimum support 1.8V logic with tolerance of 3.3V preferred.
Tx Enable	The transceiver should provide an output to support enabling and disabling the external devices in the transmit chain such as a TxVGA (optional) and PA.
Rx Enable	The transceiver should provide an output to support enabling and disabling the external devices in the transmit chain such as a RF Front End Module or LNA.
LNA Bypass	The transceiver should provide an output to support bypassing the LNA appropriately in the condition of a large blocker if so required.
RF Outputs	RF outputs including the main Tx signal should support 50 ohm or 100 ohms signalling. These outputs can be either single ended or differential.
RF Inputs	RF inputs including the main Rx and the Observation Rx (ORx) (for DPD) should support 50 ohm or 100 ohms signalling. These inputs can be either single ended or differential. The device should support at least 1 ORx.

Table 2-14: Payload with DPD in Digital Device (GBPS Rx/Tx)

Bandwidth (MHz)	Parallel	JESD204B	JESD204C	Parallel	JESD204B	JESD204C
	2T2R			4T4R		
20	2.95/5.9	3.69/7.37	3.04/6.08	5.9/11.8	7.37/14.75	6.08/12.17
50	5.9/11.8	7.37/14.75	6.08/12.17	11.8/23.59	14.75/29.49	12.17/24.33
100	11.8/23.59	14.75/29.49	12.17/24.33	23.59/47.19	29.49/58.98	24.33/48.66

Table 2-15: Payload with DPD in Transceiver (GBPS Rx/Tx)

Bandwidth (MHz)	Parallel	JESD204B	JESD204C	Parallel	JESD204B	JESD204C
	2T2R			4T4R		
20	1.97/1.97	2.46/2.46	2.03/2.03	3.93/3.93	4.92/4.92	4.06/4.06
50	3.93/3.93	4.92/4.92	4.06/4.06	7.86/7.86	9.83/9.83	8.11/8.11
100	7.86/7.86	9.83/9.83	8.11/8.11	15.73/15.73	19.66/19.66	16.22/16.22

- Algorithm Requirements: The transceiver is required to provide appropriate algorithms to sustain RF operation including but not limited to Rx AGC, Tx Power control. DPD and CFR may be included either in the transceiver or in the digital baseband device making the trade-offs in Payload shown in Table 2-11 and Table 2-12.
- Device Configuration: The transceiver should support either 2T2R or 4T4R. In addition, at least one ORx path should be supported. Additional ORx paths are allowed as required for the application.
- Power Dissipation: Total dissipation of the TRx should be less than 6W for 4T4R.
- RF Specifications are given in Table 2-16.

Table 2-16: RF specifications in Transceiver

Parameter	Symbol	Min	Typ	Max	Unit	Test Conditions/Comments
TRANSMITTERS						
Center Frequency		650		6000	MHz	
Transmitter Synthesis Bandwidth				450	MHz	
Transmitter Large Signal Bandwidth				200	MHz	
Transmitter Attenuation Power Control Range		0		32	dB	Signal-to-noise ratio (SNR) maintained for attenuation between 0 dB and 20 dB
Transmitter Attenuation Power Control Resolution			0.05		dB	
Adjacent Channel Leakage Ratio (ACLR)						20 MHz LTE at -12 dBFS
				-66	dB	75 MHz < f ≤ 2800 MHz
In Band Noise Floor						0 dB attenuation; in band noise falls 1 dB for each dB of attenuation for attenuation between 0 dB and 20 dB
Out of Band Noise Floor			-154.5		dBm/Hz	600 MHz < f ≤ 3000 MHz
Maximum Output Power			-153		dBm/Hz	0 dB attenuation; 3 × bandwidth/2 offset
Third Order Output Intermodulation Intercept Point	OIP3		6		dBm	600 MHz < f ≤ 3000 MHz
				27	dBm	0 dB transmitter attenuation
						600 MHz < f ≤ 5700 MHz

Parameter	Symbol	Min	Typ	Max	Unit	Test Conditions/Comments
Error Vector Magnitude (Third Generation Partnership Project (3GPP) Test Signals) 1900 MHz LO 3800 MHz LO	EVM		0.6 0.53		% %	50 kHz RF PLL loop bandwidth 300 kHz RF PLL loop bandwidth
OBSERVATION RECEIVER Center Frequency Gain Range	ORx	450	30	6000	MHz dB	IIP3 improves decibel for decibel for the first 18 dB of gain attenuation; QEC performance optimized for 0 dB to 6 dB of attenuation only
Analog Gain Step			0.5		dB	For attenuator steps from 0 dB to 6 dB
Receiver Bandwidth Maximum Useable Input Level	P _{HIGH}			450	MHz	0 dB attenuation; increases decibel for decibel with attenuation; continuous wave corresponds to −1 dBFS at ADC
Integrated Noise			−11 −58.5 −57.5		dBm dBFS dBFS	75 MHz < f ≤ 3000 MHz 450 MHz integration bandwidth 491.52 MHz integration bandwidth
Third-Order Input Intermodulation Intercept Point Narrow Band	IIP3					Maximum observation receiver gain; test condition: P _{High} - 11 dB/tone
1900 MHz			15		dBm	
2600 MHz			16.5		dBm	
3800 MHz			18		dBm	
Wide Band						
1900 MHz			13		dBm	IM3 products > 130 MHz at baseband; test condition: P _{High} - 11 dB/tone ; 491.52 MSPS
2600 MHz			11		dBm	
3800 MHz			13		dBm	
Third-Order Intermodulation Product	IM3					
			−70		dBc	600 MHz < f ≤ 3000 MHz
Spurious-Free Dynamic Range	SFDR		64		dB	Non IMx related spurs, does not include HDx; (P _{HIGH} − 11) dB input signal
Harmonic Distortion Second Order Harmonic Distortion Product	HD2		−80		dBc	(P _{HIGH} − 11) dB input signal In band HD falls within ±100 MHz

Parameter	Symbol	Min	Typ	Max	Unit	Test Conditions/Comments
Third-Order Harmonic Distortion Product	HD3		-73		dBc	Out of band HD falls within ± 225 MHz
			-70		dBc	In band HD falls within ± 100 MHz
			-65		dBc	Out of band HD falls within ± 225 MHz
RECEIVERS						
Center Frequency				6000	MHz	
Gain Range			30		dB	
Analog Gain Step			0.5		dB	Attenuator steps from 0 dB to 6 dB
			1		dB	Attenuator steps from 6 dB to 30 dB
Receiver Bandwidth				200	MHz	
Maximum Useable Input Level	P_{HIGH}					0 dB attenuation, increases decibel for decibel with attenuation; continuous wave = 1800 MHz; corresponds to -1 dBFS at ADC
			-11		dBm	75 MHz < f $\leq$ 3000 MHz
Noise Figure	NF		12		dB	0 dB attenuation, at receiver port 600 MHz < f $\leq$ 3000 MHz
Input Third-Order Intercept Point	IIP3					
Difference Product, 2600 MHz (Wideband)	IIP3, d		17		dBm	Two ($P_{HIGH} - 9$) dB tones near band edge
2600 MHz (Midband)			21		dBm	
Sum Product, 2600 MHz (Wideband_	IIP3, s		20		dBm	Two ($P_{HIGH} - 9$) dB tones, at bandwidth/6 offset from the LO
HD3	HD3					($P_{HIGH} - 6$) dB continuous wave tone at bandwidth/6 offset from the LO
			-66		dBc	600 MHz < f $\leq$ 4800 MHz

b. Hardware Design

For the O-RU₇₋₂ the sampling function and frequency conversion function can be performed by transceiver. Usually the transceiver is integrated by ADC, DAC, LO, down converter, up converter and so on. The block diagram of transceiver design is shown in Figure 2-18.

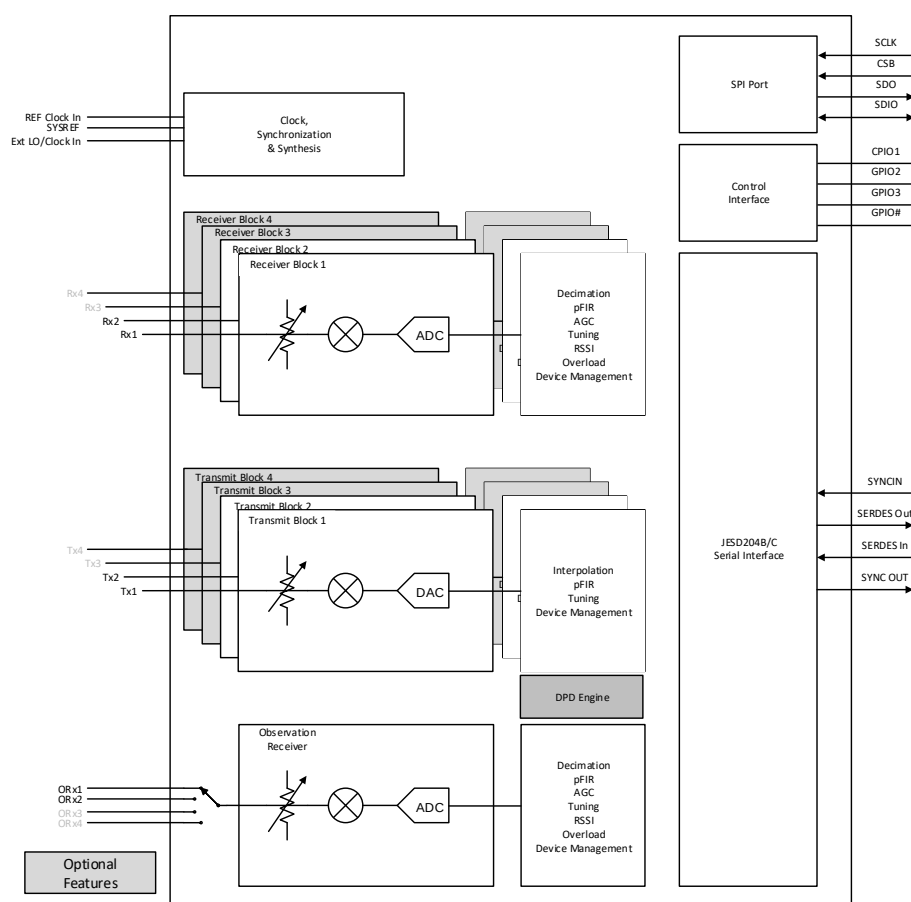


Figure 2-18: Transceiver Reference Design Diagram (Optional elements are highlighted in Grey)

The definition of the input/output lines are as follows:

- REF Clock In is the differential reference input to the on-chip synthesizer. This may also function as the SERDES Device Clock.
- SYSREF is a JESD204B/C differential synchronization signal.
- Ext LO/Clock Input is an optional input that can be used as an alternate LO or clocking signal.
- Rx1 through Rx4 are differential receiver inputs to their respective cores.
- Tx1 through Tx4 are differential transmitter output from their respective cores.
- ORx1 through ORx4 are differential observation receiver inputs.
- SCLK is the serial control device clock input.
- CSB is the active low device select for the control interface.
- SDO is the serial data output for the control interface. This pin may be omitted for 3-pin control implementations.
- SDIO is a bidirectional serial data interface. In 4-pin mode, this pin functions as the serial data input.
- GPIO1 – GPIO# are general purpose IO signals used for interrupts, enables, test mode and resets. These are used as interface and control for peripheral devices including controllers, TxVGAs, PA, LNA and similar devices where SPI control is too slow.
- SYNCIN are differential pins associated with the receiver channels of the JESD204 interface. In not used, they are typically grounded. Up to 4 pair may be supported.
- SERDES Out are differential JESD204B/C data output interfaces. Up to 8 lanes may be active.
- SERDES In are differential JESD204B/C data input interfaces. Up to 8 lanes may be active.

- SYNCOUT are differential pins associated with the transmitter channels of the JESD204 interface. If not used, do not connect. Up to 4 pair may be supported.

2.3.2.2.2 Power Amplifier (PA) Reference Design

The Power Amplifier boosts the RF output to the level required for the base station class.

a. Hardware Requirements

The PA should have large enough gain to reduce the need for an additional driver. This will reduce cost and PCB space. The output power should be at least 27dBm (30dBm for 500mW/port to compensate for the loss of switch and antenna filter). The ACLR should be greater than 47dBc according to the related 3GPP test mode. DPD is needed to reduce the power consumption. The P1 dB requirement is closely related to the DPD algorithm.

Table 2-17: PA Requirements

Frequency band	Band41	Band78	Additional Information
Gain	>33dB	>33dB	
P3dB	>34dBm	>34dBm >37dBm(500mW/port)	ACLR>=47dBc with DPD@100MHz NR 27dBm (30dBm for 500mW/port)
Input return loss	<-15dB	<-15dB	
Output return loss	<-15dB	<-15dB	
Switching Speed	<1us	<1us	
HD2	>28dBc	>28dBc	CW 27dBm
HD3	>30dBc	>30dBc	CW 27dBm

- Interface Require

The interface requirements of the transmit PA are listed in Table 2-18.

Table 2-18: Interface requirements of the transmit PA

Item Name	Description
Enable	The enable input should be compatible with 1.8V logic and tolerate 3.3V as required. A logic high enables the PA. Logic low disables the device and places it in a minimum dissipation mode.
RF Outputs	RF outputs support 50-ohm single ended to properly interface to a directional coupler, isolator, switch or antenna.
RF Inputs	RF inputs should support 50 ohm, single ended match to the transceiver output or preamp.
RF power detector	Power detector (optional)

- Power Dissipation: Dissipation while enabled should not exceed ~3W.

b. Hardware Design

RF_{in} is the input to the PA, RF_{out} is the output of the PA. V_{cc} and V_{bias} are the power inputs of the PA. PAEN is the control pin to disable or enable the PA. The input and output should match to 50 ohm as much as possible to reduce the reflection. V_{cc} and V_{bias} need capacitors to reduce the DC power ripples and give a short route to reduce the RF energy leakage. Table 2-19 shows the details.

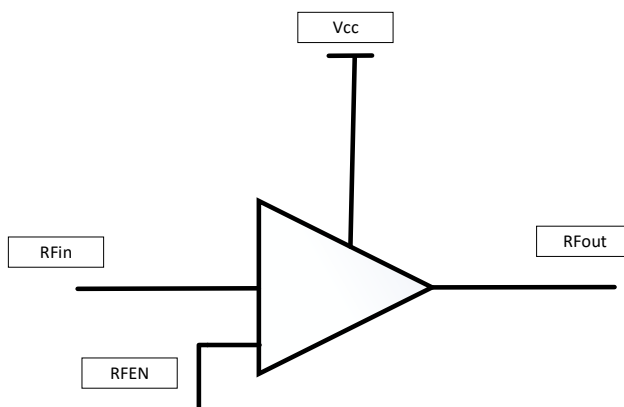


Figure 2-19: PA Reference Design Diagram

2.3.2.2.3 Low Noise Amplifier (LNA) Reference Design

The purpose of the LNA is to boost the Rx signal to a level that can nominally interface directly to the transceiver. This block will typically be a 2-stage amplifier with a 2nd stage bypass. The frontend will also include a TR switch to shunt any Tx signal to a termination away from the amplifier.

a. Hardware Requirements

The requirements of the LNA are listed here.

- Interface Requirements: The interface requirements of the transmit LNA are listed in Table 2-19.

Table 2-19: One stage LNA Requirements

Parameter	Band 41	Band 78	Additional Information
NF	<1dB	<1dB	
Gain	>15dB	>15dB	
P1dB	>18dBm	>18dBm	
Input return loss	<-12dB	<-12dB	
Output return loss	<-15dB	<-15dB	
Switching Speed	<1us	<1us	
IP3	>30dBm	>30dBm	2 Tone CW -15dBm
K	>1	>1	up to f = 12.5 GHz
Max input power	>15dBm	>15dBm	

- Device Configuration: Single or dual device.
- Power Dissipation: Less than 500 mW for a dual when device is fully enabled in receive mode.
- RF Specifications: For LNA unit, it should have larger enough gain to reduce extra driver amplifier for cost and PCB space. The NF figure of LNA should small enough to overcome the loss of switch and filter.

Table 2-20: Two stage LNA Requirements

Frequency band	Band41	Band78	Additional Information
NF	<1 dB	<1.5 dB	
Gain	High gain ≥ 32 dB Low gain ≥ 16 dB	High gain ≥ 32 dB Low gain ≥ 16 dB	
P1dB	>18dBm	>18dBm	
Input return loss	<-12dB	<-12dB	
Output return loss	<-15dB	<-15dB	

Switching Speed	<1us	<1us	
IP3	>30dBm	>30dBm	2 Tone CW -15dBm
K	>1	>1	up to f = 12.5 GHz
Max input power	>15dBm	>15dBm	

b. Hardware Design

The reference designs for one stage LNA and two-stage LNA are given.

- One stage LNA

RFin is the input port of the LNA. RFout is the output of the LNA. Vcc is the power input of the LNA. LNA EN is the control pin to disable or enable the LNA. The input match to 50 ohm as much as possible to reduce the reflection. But for output of the LNA, it is hard to get best Noise figure and the output return loss. Compromise is needed between NF and return loss. Usually the return loss should be around -10dB or -12dB. Also should be very careful to assure the stability of the LNA in a large bandwidth. Vcc needs capacitors to reduce the DC power ripples and give a short route to reduce the RF energy leakage. Figure 2-20 shows the details.

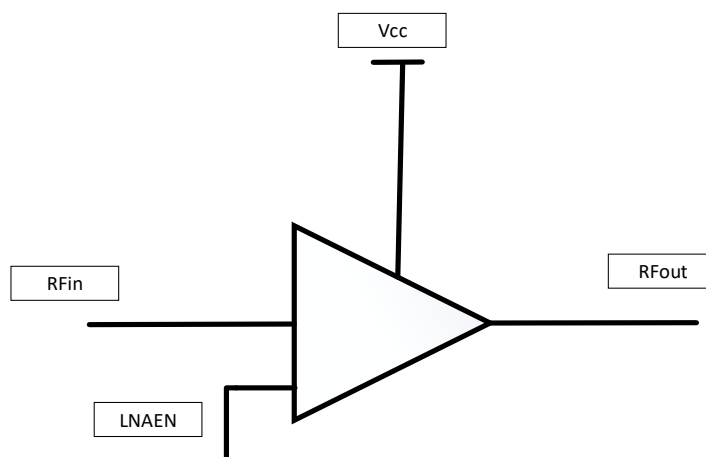


Figure 2-20: One Stage LNA Diagram

- Two stage LNA

RFin is the input port of the LNA. RFout is the output of the LNA. Vcc is the power input of the LNA. LNA EN is the control pin to disable or enable the LNA. The input matches to 50 ohm as much as possible to reduce the reflection. But for output of the LNA, it is hard to get best Noise figure and the output return loss. Compromise is needed between NF and return loss. Usually the return loss should be around -10dB or -12dB. Also should be very careful to assure the stability of the LNA in a large bandwidth. Vcc needs capacitors to reduce the DC power ripples and give a short route to reduce the RF energy leakage. Figure 2-21 shows the details.

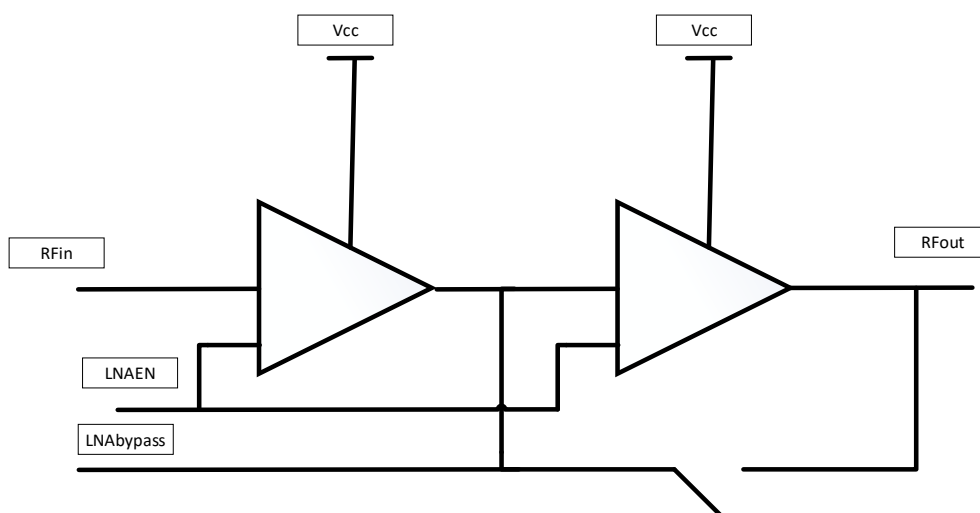


Figure 2-21: Two-Stage LNA Diagram

2.3.2.2.4 RF Switch Reference Design

For TDD use, the TX and RX links work spiritedly by time duplex. The switch is used to change the RF link according to the TDD duplex. In the TDD TX mode, the switch is switched to connect PA and antenna. In TDD RX mode, the switch is switched to LNA and antenna.

a. Hardware Requirements

The requirements of the RF switch are listed here.

- Interface Requirements

The interface requirements of the switch are listed in Table 2-21.

Table 2-21: Interface requirements of the RF switch

Item Name	Description
Control	Control the switch working mode
RF COM	The port connected to the antenna RF inputs should support 50 ohm, single ended to match the antenna.
RF Outputs	The port connected to the LNA. RF inputs should support 50 ohm, single ended to match the LNA.
RF Inputs	The port connected to the PA. RF inputs should support 50 ohm, single ended to match the PA output.

- RF Specifications

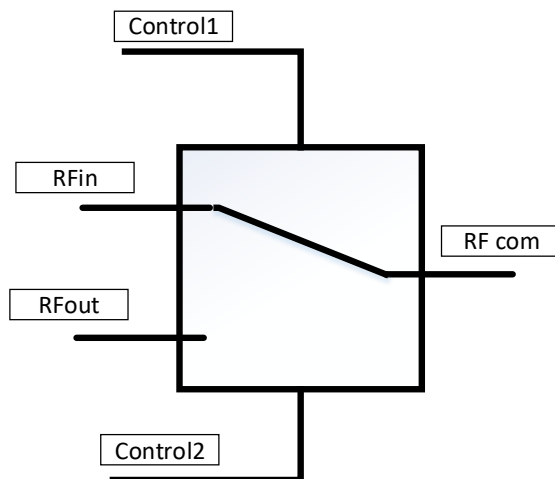
For switch, it should have larger P1dB to not degrade the ACLR or damage the switch itself. The output power should be at least 26dBm (29dBm for 500mW/port) count on the loss of antenna filter so it must have low loss. Also the isolation needs to be high to protect the LNA. Higher HD2 and HD3 are needed to reduce the out of band emissions.

Table 2-22: RF Switch Requirements

Frequency band	Band 41	Band 78	Additional Information
Loss	<0.8dB	<0.8dB	
P1dB	>36dBm	>36dBm >39dBm(500mW/port)	
Input return loss	<-15dB	<-15dB	
Output return loss	<-15dB	<-15dB	
Switching Speed	<1us	<1us	
Isolation	>30dB	>30dB	
HD2	>70dBc	>70dBc	CW 27dBm
HD3	>70dBc	>70dBc	CW 27dBm

b. Hardware Design

RF Com is the input port of the switch, also can be the output of the switch. RF in is the input of the switch. RF out is the output of the switch. Control 1 and control 2 are the control pin to switch the RFin to RF Com or from RF Com to RF out. Figure 2-22 shows the details.


Figure 2-22: RF Switch Reference Design Diagram
2.3.2.2.5 Antenna Reference Design

The antenna is used to radiate the TX power to the air and receive the RX power from the air. For the indoor Picocell scenario, the isotropic antenna is the first choice.

a. Hardware Requirements

The following table shows the antenna requirement for the O-RU₇₋₂.

Table 2-23: Antenna Requirements

Frequency band	Band 41	Band 78
VSWR	≤ 1.8	≤ 1.8
Power capacity	≥ 1 W	≥ 2 W
Gain	≥ 2 dBi	≥ 2.5 dBi
Beam width on vertical direction	$\geq 35^\circ$	$\geq 35^\circ$
pattern roundness on horizontal direction	± 3 dB	± 3 dB

b. Hardware Design

One possible choice of the isotropic antenna is the whip antenna. Following figure shows a design of one kind of whip antenna.


Figure 2-23: Whip Antenna

2.3.3 O-RU₇₋₂ Hardware Components for SDR based Design

General block diagrams for 2T2R and 4T4R O-RU₇₋₂ examples for FPGA based design are shown in 2.3.2. The block diagrams for SDR-based design are very similar to those and are shown below.

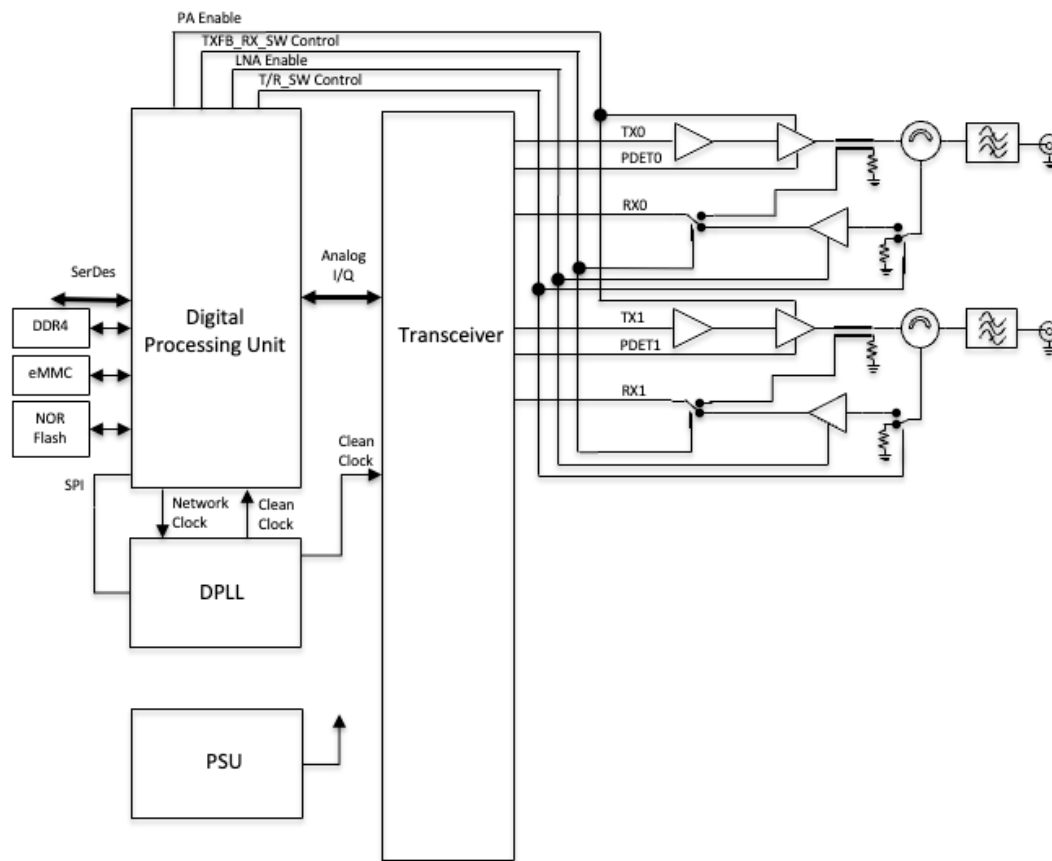


Figure 2-24: 2T2R General Block Diagram for SDR based design

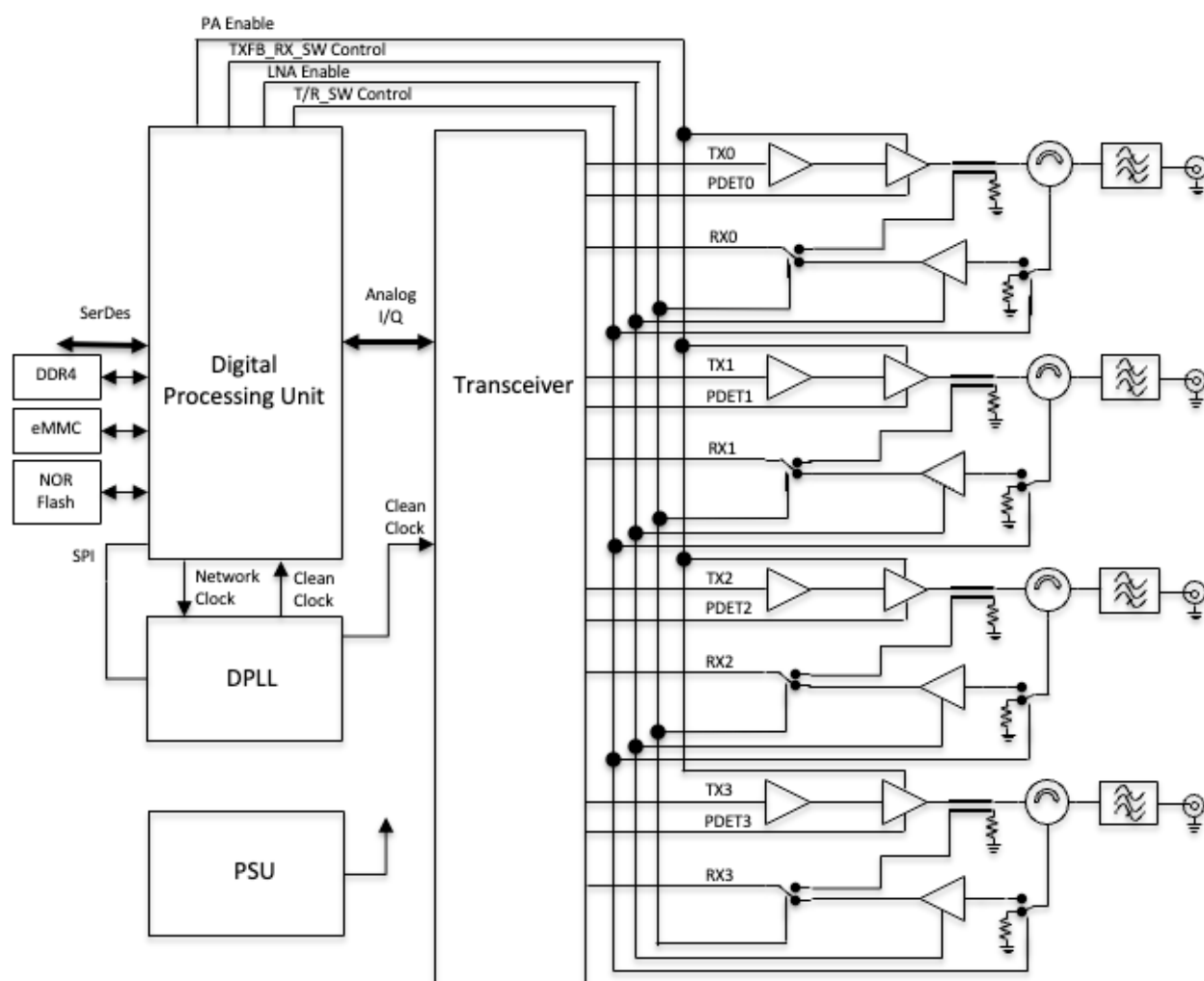


Figure 2-25: 4T4R General Block Diagram for SDR based design

2.3.3.1 Digital Processing Unit

The digital processing unit of O-RU₇₋₂ is mainly for performing tasks related to FH interface, lower L1, RF interface, and OAM.

i. SDR Solution Design

The Software Defined Radio SOC unit of O-RU₇₋₂ is mainly responsible for FH interface, Low PHY, RF I/Q interface, and management functions.

a. SDR SoC requirement:

The following items are required for the O-RU₇₋₂:

- Interface requirement: One lane of bi-directional 10 Gbps SerDes for FH Split 7.2, 4-Port ADC/DAC IQ pairs for 4T4R and 2-port ADC/DAC IQ pairs for 2T2R.

- Resource Requirement: Built-in DSP, RTOS and processors to implement 2T2R and 4T4R ORU 7.2 Low PHY features.

b. SDR SoC Design:

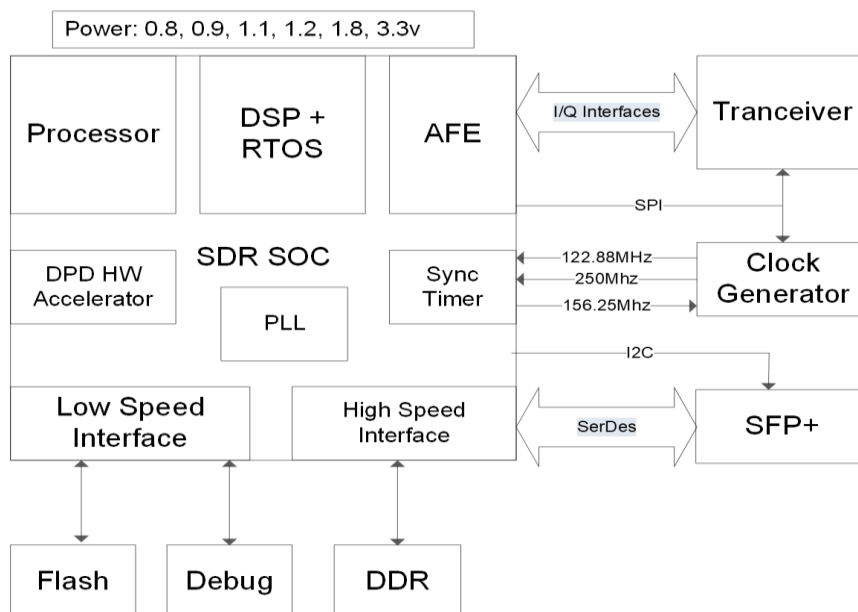


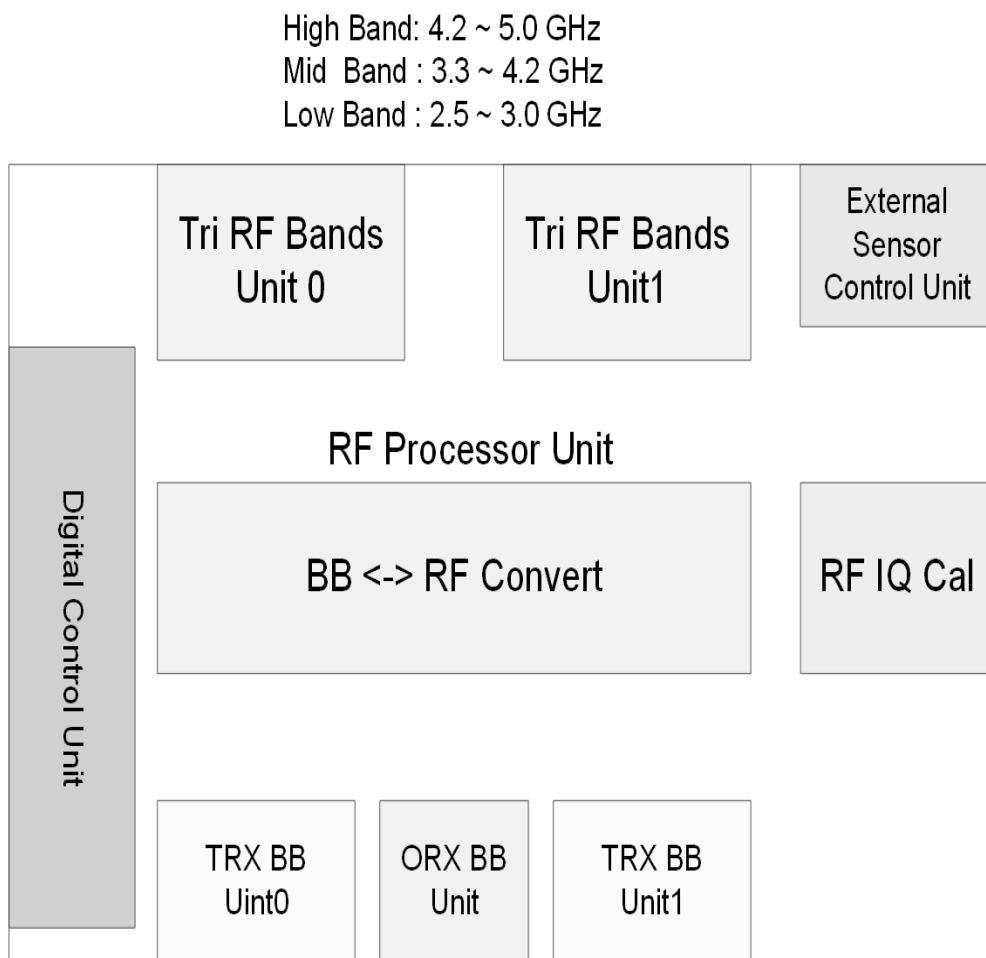
Figure 2-26: SDR SoC Design

1

2.3.3.2 RF Processing Unit

The RF processing unit of O-RU₇₋₂ performs functionalities of local oscillator, down converter, up converter etc.

2.3.3.2.1 Transceiver RFIC Reference Design



5

Figure 2-27: Transceiver RFIC Reference Design

The key features of the transceiver RFIC are as follows:

- Tri-Band 2x2 5G NR sub-6 GHz RF transceiver
- Flexible analog baseband filter - 50 MHz - 400 MHz modulation in TX mode
- Flexible analog baseband filter - 20 MHz - 200 MHz modulation in RX mode
- Observation path for DPD support
- Analog differential I/Q interface to baseband
- Programmable RFIC control, configuration and calibration

13

The key receiver features are as follows:

- Direct conversion, low noise and high dynamic range receive chains
- High speed AGC interface (Rx gain control) to baseband

The key transmitter feature is as follows:

- Direct conversion with adjustable gain on the transmission chains

It includes an auxiliary ADC that supports:

- Calibration of internal circuits
- External voltage and temperature measurements
- External PA Power Envelope Measurements

The RFIC includes:

- Interfaces for configuration and bi-directional data with baseband IC

2.3.3.2.2 Power Amplifier (PA) Reference Design

The Power Amplifier boosts the RF output to the level required for the base station class.

a. Hardware Requirements

The PA is the final stage that amplifies the downlink RF signal. Considering the requirement of the conducted output power of 24 dBm at the antenna connector and the maximum 3 dB insertion loss between the PA and the antenna, including the directional coupler, the circulator, and the band-pass filter, the PA output power should be at least 27 dBm (same output power as in 2.3.2.2.2). The ACLR should be less than -35 dBc (before applying the DPD) to ease the hardware processing requirements of the DPD algorithm and still be able to meet the 3GPP TS38.104 ACLR requirements (clause 6.6.3). To support TDD mode, the PA should be capable of swiftly turning ON and OFF. The switching time should be less than 1 usec.

Table 2-24: PA Requirements

Parameter	Specification	Additional Information
Gain	>32dB	
Linear Output Power (P1 dB)	>27dBm	ACLR<-35dBc
Input VSWR	<2	
HD2	>50dBc	CW 27dBm
HD3	>50dBc	CW 27dBm

- Interface Requirements

The interface requirements of the transmit PA are listed in Table 2-25.

Table 2-25: Interface requirements of the transmit PA

Item Name	Description
Enable	The enable input should be compatible with 1.8V logic and tolerate 3.3V as required. A logic high enables the PA. A logic low disables the PA and places it in a minimum dissipation mode.
RF Outputs	RF outputs support 50-ohm single ended to properly interface to a directional coupler, isolator, switch or antenna.
RF Inputs	RF inputs should support 50-ohm, single ended match to the transceiver output or preamp.
RF power detector	Power detector outputs a DC level according to the RF power and then the DC level is converted to a binary value by an ADC. The RF power can be normalized to compensate for the power deviation caused by temperature deviation.

- Power Dissipation: Dissipation while enabled should not exceed ~3.6W.

b. Hardware Design

Input is the RF input and output is the RF output of the PA. Vcc is the power supply of PA and the internal bias circuits. Enable is the control pin to enable or disable the PA. Pdet is the power detector output connected to ADC input. Ensure that the RF input and output ports see 50 ohms terminations for best PA performance. Bulk capacitors should be used to reduce the ripples of Vcc and decoupling capacitors to decouple the RF energy to ground plane. Ensure the grounding is good enough. Otherwise, the PA is subject to oscillations. Figure 2-28 shows the details.

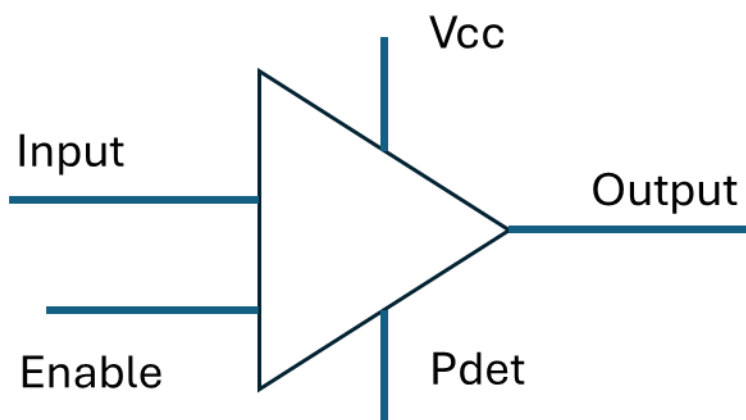


Figure 2-28: PA Reference Design

2.3.3.2.3 Directional Coupler

a. Hardware Requirements

O-RU needs to comply with 3GPP TS38.104 ACLR requirements (clause 6.6.3). Directional Coupler is a necessary component to implement the DPD feedback path. With the directional coupler, the TX RF signal is coupled back to the RX port for DPD calculation. Its input pin is connected to the output pin of PA, and the

coupled pin is connected to the RX with a 50 ohms attenuator. The isolated pin is terminated with a 50 Ω resistor. Minimum 18 dB directivity is required to reject the self-interference due to limited antenna isolation.

Table 2-26: Directional Coupler Requirements

Parameter	Specification	Unit
Coupling	20 +/-1.5	dB
Maximum Operating Band Insertion Loss	0.1	dB
Minimum Operating Band Return Loss	18	dB
Minimum Directivity	18	dB
Minimum Average Input Power	10	W
Operating Temperature	-55 ~ 140	°C

b. Hardware Design

The Input of the directional coupler is connected to the output of the PA. The Direct port is connected to port-1 of the circulator. The Coupled port is connected to the common port of the RF switch. The Isolated port is terminated by a 50 ohms resistor. Make sure all the RF ports see 50 ohms terminations for best directivity performance.

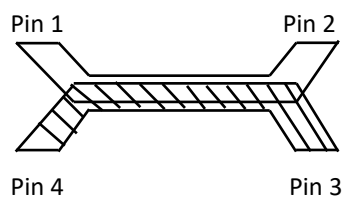


Figure 2-29: Directional Coupler

Table 2-27: Directional Coupler Pin Connectivity

Pin 1	Pin 2	Pin 3	Pin 4
Input	Direct	Isolated	Coupled

1

2.3.3.2.4 Circulator

a. Hardware Requirements

The circulator is used to increase the isolation among antennas. Using the circulator, the DPD feedback path's signal-to-noise ratio can be improved by up to 18 dB, which eases the burden on the DPD algorithm. Two types: clockwise and counter-clockwise, are available depending on the trace routing. Minimum 18 dB isolation is required to reject the self-interference due to limited antenna isolation.

8

Table 2-28: Circulator Requirements

Parameter	Value	Additional Information
Insertion Loss	<0.2dB	
Isolation	>20dB	
Return loss	<-20dB	
Average Power	>1Watt	
Peak Power	>10Watts	
IMD3	<-65dBc	2x10W, 1MHz Spacing
2 nd Harmonic	<-50dBc	
3 rd Harmonic	<-60dBc	

9

b. Hardware Design

Port-1 of the circulator is connected to the output of the directional coupler. Port-2 is connected to the BPF. Port-3 is connected to the common port of the RF switch. Make sure all the RF ports see 50 ohms terminations for best isolation performance. The RF switch should be switched to the port of 50 ohms termination during TX period.

10
11
12
13
14

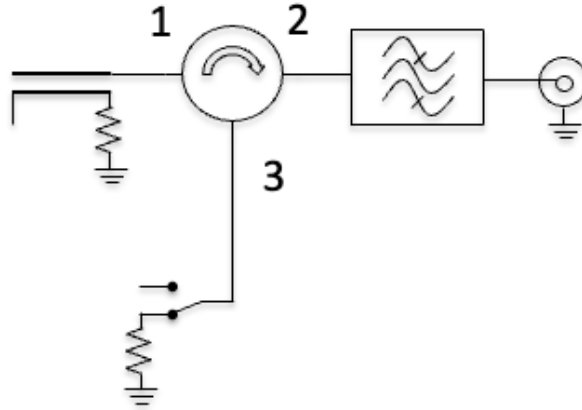


Figure 2-30: Circulator

2.3.3.2.5 Band Pass Filter

a. Hardware Requirements

The BPF placed between the circulator and the antenna rejects the spurs and the harmonics in the TX path and suppresses the unwanted blockers in the RX path. Low insertion loss is required to meet TX output power and RX sensitivity requirements. High rejection at the band-edge is required to suppress the Out-of-Band blocking signal specified in 3GPP TS 38.104 clause 7.5.

The BPF placed between the LNA output and RFIC input is optional and can be added if further suppression at the band-edge is required.

High peak power handling is required to withstand the high peak-to-average ratio of OFDM signal.

Table 2-29: Band Pass Filter Requirements

Parameter		@ 25°C	@ -40°C ~ 105°C	Unit
Maximum Passband Insertion Loss in 20 MHz BW		1.5	2.0	dB
Maximum Passband Ripple		1.0	1.2	dB
Minimum Passband Input/Output Return Loss		12		dB
Minimum Attenuation	1 MHz to (f _{UL_low} - Δf _{OOB}) MHz	18		dB
	(f _{UL_high} + Δf _{OOB}) MHz to (2*f _{UL_low}) MHz	18		
	(2*f _{UL_low}) MHz to (2*f _{UL_high}) MHz	30		
	(3*f _{UL_low}) MHz to (3*f _{UL_high}) MHz	20		
Average Input Power		8		W
Peak Input Power		80		W
Operating Temperature		-40 ~ 105		°C

Note 1: Δf_{OOB} is the maximum offset of the out-of-band boundary from the uplink operating band edge

Note 2: $\Delta f_{\text{OOB}} = 20$ MHz if $(f_{\text{UL_high}} - f_{\text{UL_low}})$ MHz is no larger than 200 MHz; $\Delta f_{\text{OOB}} = 60$ MHz if $(f_{\text{UL_high}} - f_{\text{UL_low}})$ MHz is larger than 200 MHz

b. Hardware Design

Input and Output are the RF ports of the BPF. Make sure the RF input and output ports see 50 ohms terminations for best BPF performance.

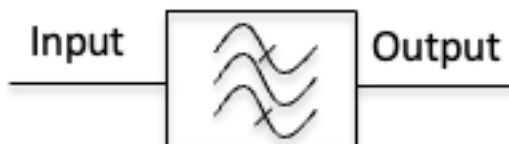


Figure 2-31: Band Pass Filter

2.3.3.2.6 Low Noise Amplifier (LNA) Reference Design

The LNA is required at the RX path prior to the RF transceiver. It not only boosts the RX signal but also reduces the cascaded noise figure for best RX sensitivity. Since the BPF may not reject the interference over the air, the LNA must have enough OIP3 to tolerate the interference to mitigate the de-sensitization. Another benefit of higher OIP3 is higher maximum input power of the wanted signal.

a. Hardware Requirements

The requirements of the LNA are listed here.

- Interface Requirements: The interface requirements of the transmit LNA are listed in Table 2-30.

Table 2-30: LNA Requirements

Parameter	Value	Additional Information
NF	<1dB	
Gain	>20dB	
OP1dB	>17dBm	
Input return loss	<-10dB	
Output return loss	<-10dB	
Switching Speed	<1us	
OIP3	>27dBm	2 Tone CW -20dBm
K	>1	up to $f = 12.5$ GHz

- Power Dissipation: Less than 330 mW when device is fully enabled in receive mode.

b. Hardware Design

Input is the RF input of the LNA and output is the RF output of the LNA. Vcc is the power supply of the LNA and the internal bias circuits. Enable is the control pin to enable or disable the LNA. Make sure the RF input and output ports see 50 ohms terminations for best LNA performance. Bulk capacitors should be used to reduce the ripples of Vcc and decoupling capacitors to decouple the RF energy to ground plane. Minimize the insertion loss from the antenna port to the RF input of the LNA as much as possible to reduce the cascaded noise figure. Ensure the grounding is good enough. Otherwise, the LNA is subject to oscillations. Figure 2-32 shows the details.

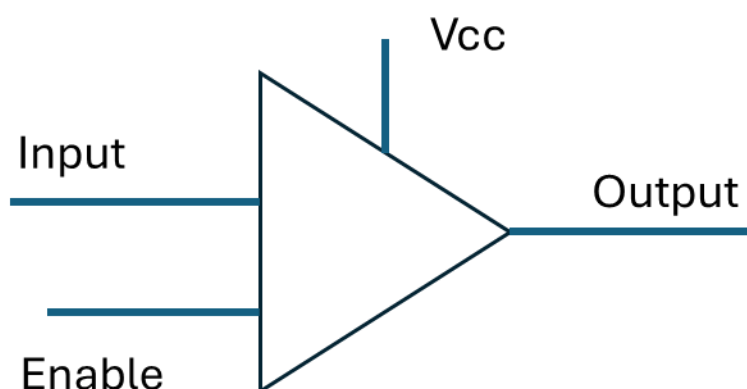


Figure 2-32: LNA

2.3.3.2.7 RF Switch Reference Design

The RF switch is used to change the RF path from a common port. The insertion loss at the through path should be as small as possible, while the insertion loss at the blocked path should be as large as possible. The first use case is to switch TX and RX path from the antenna port. The second use case is to switch the TX feedback and RX from the RX input of RF transceiver. In the first use case, the common port of the RF switch is connected to port-3 as shown in Figure 2-30. During TX duration, the RF switch is switched to the port of 50 ohms termination.

a. Hardware Requirements

The requirements of the RF switch are listed here.

- Interface Requirements:

The interface requirements of the switch are listed in Table 2-31 and Table 2-32.

Table 2-31: Interface requirements of the RF switch for use case 1

Item Name	Description
Control	Control the switch working mode
RF COM	The port connected to port-3 of the circulator. RF inputs should support 50 ohm, single ended to match the circulator.
RF 1	The port connected to the LNA. RF inputs should support 50 ohm, single ended to match the LNA.

RF 2	The port connected to the 50 ohms termination. RF inputs should support 50 ohm, single ended to match the 50 ohms termination.
------	---

Table 2-32: Interface requirements of the RF switch for use case 2

Item Name	Description
Control	Control the switch working mode
RF COM	The port connected to RX input of RF transceiver. RF inputs should support 50 ohm, single ended to match the RF transceiver.
RF port 1	The port connected to the coupling port of the directional coupler. RF inputs should support 50 ohm, single ended to match the directional coupler.
RF port 2	The port connected to the LNA output. RF input should support 50 ohm, single ended to match the LNA output.

- RF Specifications

The switch should have larger P1dB to not degrade the ACLR or damage the switch itself. The output power should be at least 27dBm (30dBm for 500mW/port) considering the loss of antenna filter, and it must have low loss. Also, the isolation needs to be high to protect the LNA. Higher HD2 and HD3 are needed to reduce the out-of-band emissions.

Table 2-33: RF Switch Requirements

Parameter	Value	Additional Information
Loss	<0.5dB	
P0.1dB	>30dBm	
Input return loss	<-15dB	
Output return loss	<-15dB	
Switching Speed	<0.5us	
Isolation	>30dB	
HD2	>70dBc	CW 27dBm
HD3	>70dBc	CW 27dBm

b. Hardware Design

RFcom is the common port of the RF switch which is switched to RF1 or RF2 by CTRL pin. Vcc is the power supply of the transistor and control logic. DC blocks are required at each RF port if non-zero DC exists. Decoupling capacitors are added at Vcc to decouple the RF energy to ground plane. Figure 2-33 shows the details.

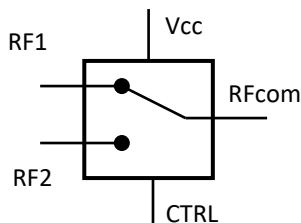


Figure 2-33: SPDT RF Switch Reference Design

2.3.3.2.8 Antenna Reference Design

The Antenna reference design is similar to the one described in 2.3.2.2.5.

2.3.4 Synchronization and Timing

The purpose of the Clocking device is to accept the network reference clock, typically 1 PPS, and generate a jitter free reference clock(s) for other devices in the system including the RF transceiver and digital block. The clock is typically part of an IEEE 1588 implementation either controlled by an external stack or implemented in the clock device itself which could be part of the baseband implementation.

a. Hardware Requirements

Hardware requirements are:

- Interface Requirements: The interface requirements of the transmit Clocking are listed below.

Table 2-34: Interface requirements of the clocking

Item Name	Description
Reference Clock Input	The clock device should receive a network reference clock from the FPGA/ASIC. This typically could be a 1pps, 10 MHz or other standard reference as determined by the specific implementation. More than one input is allowed that may be selected between when the reliability of one reference is compromised. Standard differential clocking should be used to preserve signal integrity.
Control	Control of the transceiver is by way of 3 or 4 wire SPI or IIC functioning as a slave. Support for 1.8V control is required and tolerance of 3.3V is preferred. Typically, the clock devices will be part of an IEEE1588v2 protocol loop controlled by way of this control interface or other GPIO lines as required by the hardware implementation.
Clock Outputs	One or more clock outputs are required to drive the digital device and transceiver clock inputs. Each output should be independently programmable in frequency to suit the application. The outputs should nominally be differential to preserve clocking edge and to maximize tolerance of high common mode signals.
RF Inputs	RF inputs should support 50 ohm or 100 ohms, single ended or differential signalling to match the transceiver output.

- Power Dissipation: Total dissipation should be about less than 2W.
- RF Specifications: Clocking RF requirements are given in the following table.

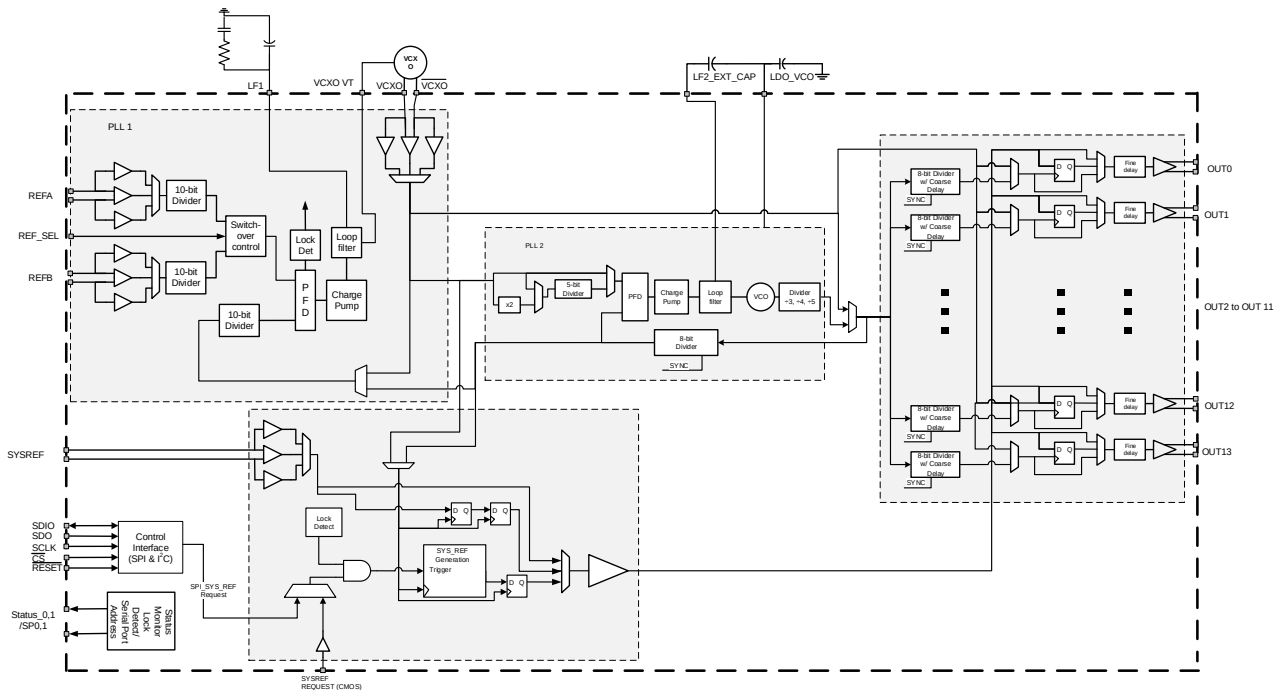
Table 2-35: Clocking RF requirements

Absolution Phase Noise, Dual Loop Mode - LVDS OUTPUT	Typical	Additional Information
$f_{OUT} = 122.88 \text{ MHz}$		Example is using an external 122.88 MHz VCXO; reference = 122.88 MHz; channel divider = 10 or 1; PLL2 loop bandwidth (LBW) = 450 kHz.
10 Hz Offset	−86 dBc/Hz	
100 Hz Offset	−106 dBc/Hz	
1 kHz Offset	−126 dBc/Hz	
10 kHz Offset	−135 dBc/Hz	
100 kHz Offset	−139 dBc/Hz	
800 kHz Offset	−147 dBc/Hz	
1 MHz Offset	−148 dBc/Hz	
10 MHz Offset	−157 dBc/Hz	
40 MHz Offset	−158 dBc/Hz	
Absolute Phase Noise, Single Loop Mode – LVDS OUTPUT	Typical	
LVDS OUTPUT		
$f_{OUT} = 122.88 \text{ MHz}$		Example is using an external 122.88 MHz reference; reference = 122.88 MHz; channel divider = 10; PLL2 LBW = 450 kHz.
10 Hz Offset	−111 dBc/Hz	
100 Hz Offset	−113 dBc/Hz	
1 kHz Offset	−123 dBc/Hz	
10 kHz Offset	−135 dBc/Hz	
100 kHz Offset	−140 dBc/Hz	
800 kHz Offset	−147 dBc/Hz	
1 MHz Offset	−148 dBc/Hz	
10 MHz Offset	−157 dBc/Hz	
40 MHz Offset	−157 dBc/Hz	

CLOCK OUTPUT ADDITIVE TIME JITTER (BUFFER MODE)	Typical	Example is using an external 122.88 MHz source driving VCXO inputs (distribution section only, does not include PLL and VCO)
$f_{OUT} = 122.88 \text{ MHz}$	79 fS	Integrated BW = 200 kHz to 5 MHz
	101 fS	Integrated BW = 200 kHz to 10 MHz
	140 fS	Integrated BW = 12 kHz to 20 MHz
	187 fS	Integrated BW = 10 kHz to 40 MHz
	189 fS	Integrated BW = 1 kHz to 40 MHz
	176 fS	Integrated BW = 1 MHz to 40 MHz

b. Hardware Design

For the clocking function, it is usually performed by a synchronization IC which may include one or more PLLs. And it also can supply numbers of output port to support different frequency clocking.



bidirectional serial data pin for both SPI and I2C. SDO is the serial data out pin for 4-wire mode. OUT0 through OUT13 are differential programmable output clock signals. These are the primary outputs of this device and provide high performance clock signals to the transceiver, baseband device and other key elements. This device is also responsible for providing SYSREF to various devices in the system if required. It may also use an external SYSREF as a source that may be retimed for local timing if necessary. STATUS0/SP0 is a lock detect and other status signal. STATUS1/SP1 is a lock detect and other status signal. SYSREF_IN is an external SYSREF input clock is a differential input representing the JESD204 synchronization from the external source.

2.3.5 External Interface Ports

a. Hardware Requirements

The following table shows the external ports or slots provided by O-RU₇₋₂.

Table 2-36: External Port List

Port Name	Feature Description
Fronthaul interface	One 10Gbps SFP+ transceiver interface
	Or one RJ45 Ethernet interface
Debug interface	RJ45 for debug usage
Power interface	Two-pin male plug for power cable

b. Hardware Design

- SFP+ case and connector: The SFP+ case and connector are standardized component on the market; following figure describe the form factor of one SFP case which is integrated with connector.

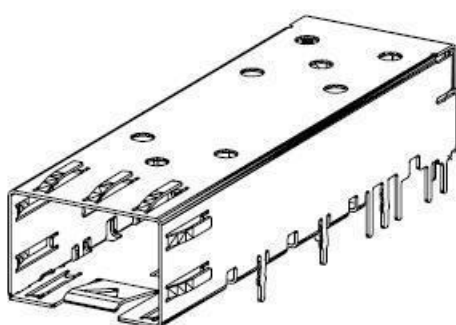


Figure 2-35: SFP+ case and connector

- RJ45 Ethernet interface: The RJ45 Ethernet interface is a standardized component on the market; following figure describes the form factor of one RJ45 interface.

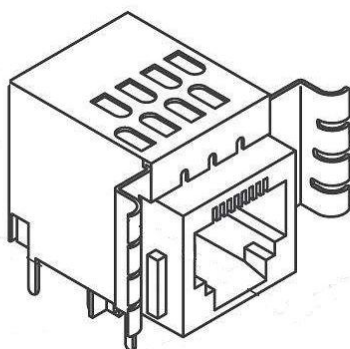


Figure 2-36: RJ45 interface

2.3.6 Mechanical

The shell of O-RU₇₋₂ is shown in the following figure. The O-RU₇₋₂ should be quiet, so it depends on the natural heat dissipation method. Usually the bottom of the shell is built by metal. All hot components should make their surfaces contact to metal shell through thermal pad.

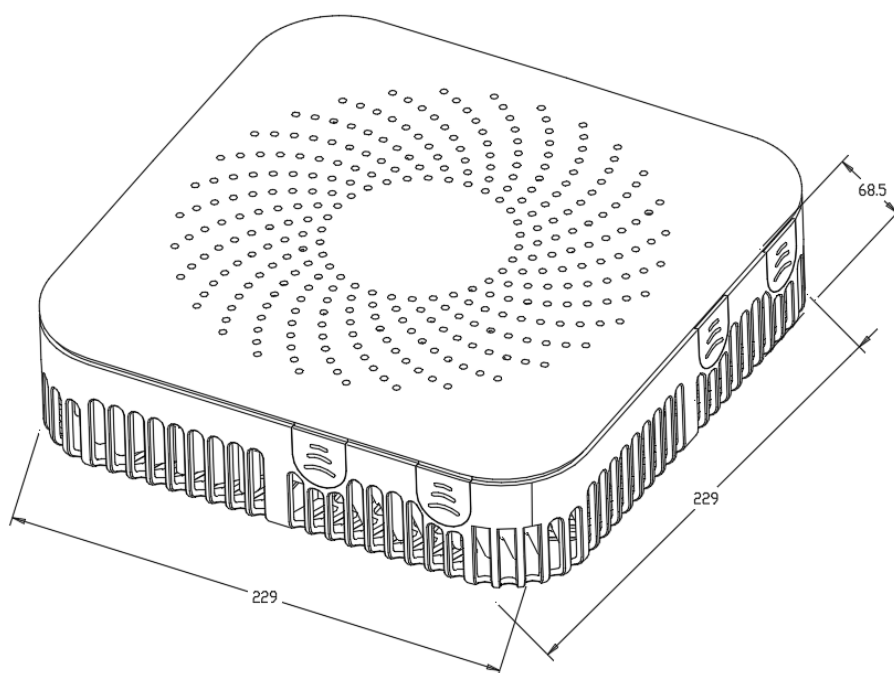


Figure 2-37: O-RU₇₋₂ Shell Mechanical Diagram

2.3.7 Power Unit

For Picocell, for the power solution of the O-RU₇₋₂ can be over Ethernet cable (cat5E, cat6A) by POE (POE+, POE++) or directly power cable with fibre. And for the board power solution, LDO and DCDC can be used.

a. Hardware Requirements

For PA unit, it should have larger enough gain to reduce extra driver amplifier for cost and PCB space. The output power should be at least 27dBm count on the loss of switch and antenna filter. The ACLR should larger than 47dBc according to the related 3GPP test mode. To reduce the power consumption, DPD is needed. And the P1 dB requirement is closely related to the DPD algorithm.

Table 2-37: power unite requirement for 2T2R and 4T4R O-RU₇₋₂

Module	Power consumption			
	2T2R	4T4R	Unit	
FPGA	9	11	W	
Transceiver	5.74	11.48	W	
PLL	1.78	1.78	W	
PA	5	10	W	0.75 TDD ratio
LAN	0.15	0.3	W	0.25 TDD ratio
Ethernet PHY	1	1	W	
DDR	0.9	0.9	W	
Flash	0.1	0.1	W	
others	1.65	1.65	W	
component total power consumption	25.3	38	W	
O-RU ₇₋₂ power consumption	29.7	45	W	on board power efficiency 0.85

Table 2-38: Power unit function

Function	Priority	
Enable/Disable	Mandatory	
Power good	Optional	
Input voltage	Mandatory	48V
Output voltage	Mandatory	0.85V 1.2V 1.3V 1.5V 1.8V 2.5V 3.3V 5V 12V etc.

b. Hardware Design

The block diagram of POE reference design is shown in Figure 2-28. Data Pair and Spare Pair are the POE connections on the data transformers used to source the power over Ethernet. V_{AUX} is a local backup power source if desired. Isolated Output is the isolated raw output from the PoE sub-circuit.

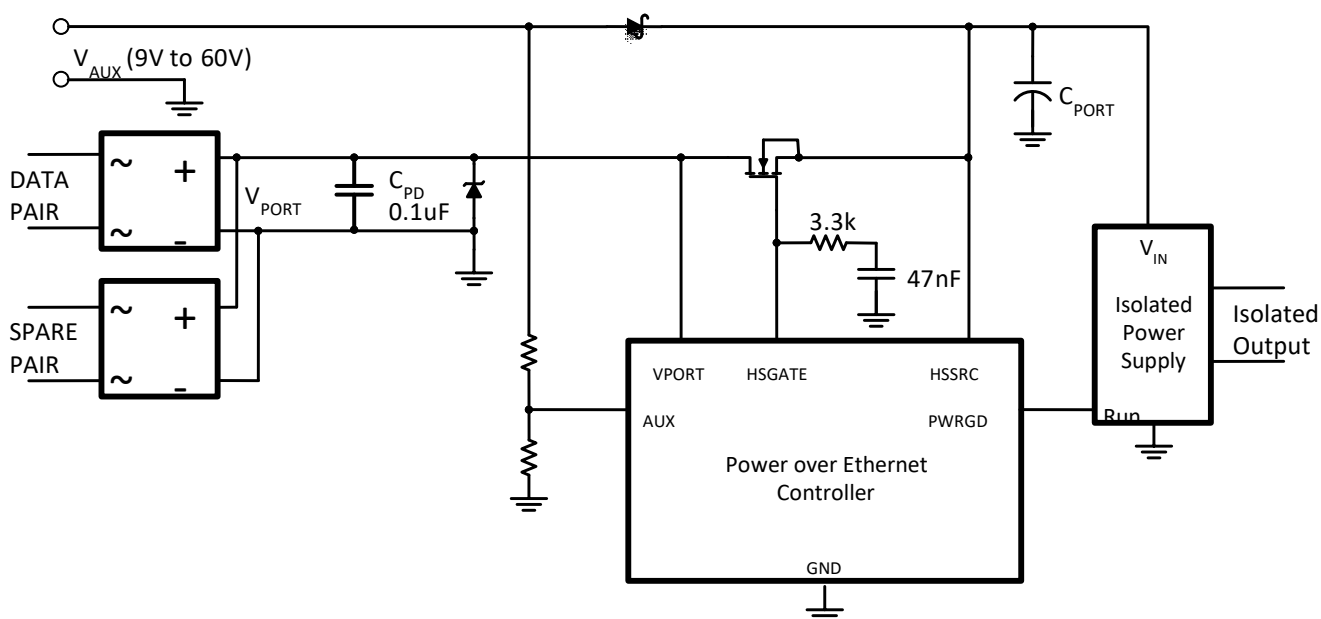


Figure 2-38: PoE Reference Design Diagram

2.3.8 Thermal

O-RU₇₋₂ will use passive cooling.

2.3.9 Environmental and Regulations

The O-RU₇₋₂ hardware system is RoHS Compliant. The power supply unit is EMC FCC/CISPR Class B compliant. Table 2-39 lists the environmental related features and parameters.

Table 2-39: Environmental Features

Item Name	Description
Operating Temperature	-5°C to +55°C
Non-operating Temperature	-40°C to 70°C
Operating Relative Humidity	8% to 90% (non-condensing)
Non-operating Relative Humidity	5% to 95% (non-condensing)

2.4 FHM₇₋₂ Hardware Reference Design

The FHM₇₋₂ is used to bridge the connection between O-DU₇₋₂ and O-RU₇₋₂. It needs to power the O-RU₇₋₂ and reduces the front haul bandwidth.

2.4.1 FHM₇₋₂ High-Level Functional Block Diagram

Figure 2-39 shows the FHM₇₋₂ block diagram for this reference design. Following sections below describe the functionality, interface and performance for each respective block of the Digital Processing Unit. As device integration is an ongoing activity, chip boundaries may be fluid and some functionality may move from one block to another or entire functionalities may be absorbed into other blocks. The sections below describe the functional blocks independent of which physical device they may reside in.

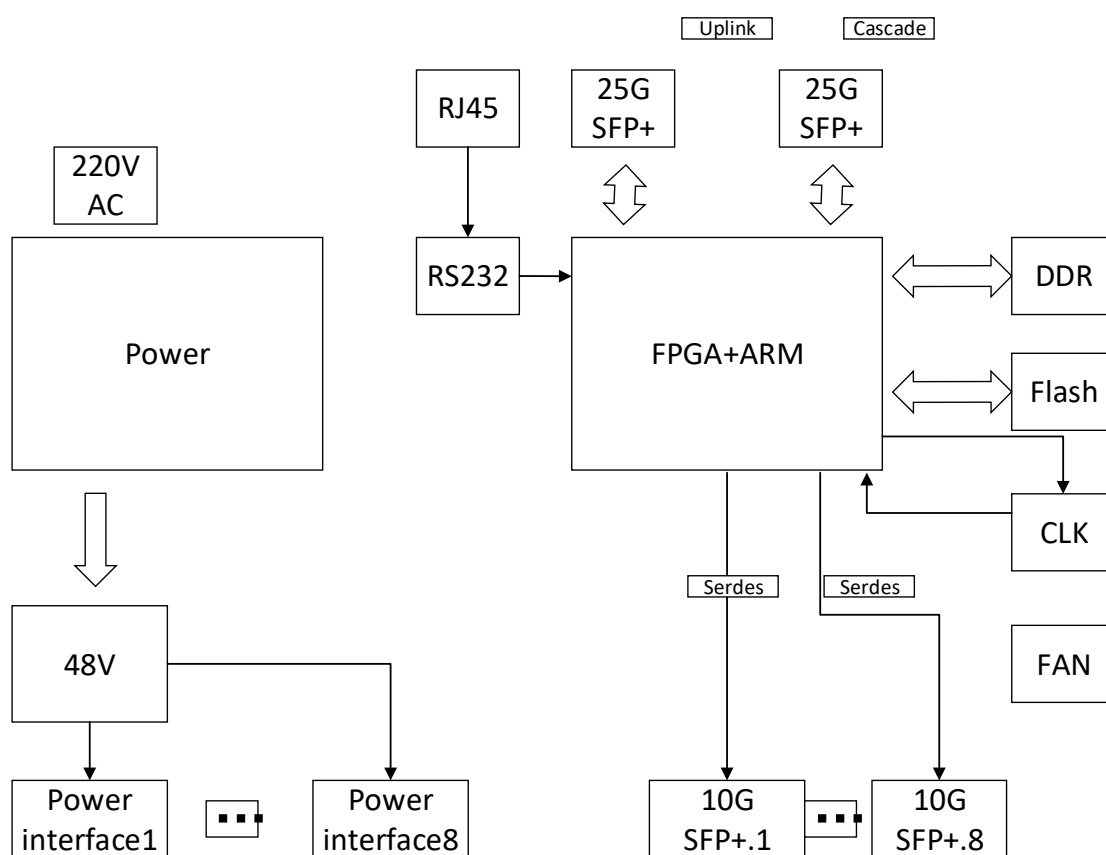


Figure 2-39: FHM₇₋₂ General Block Diagram

2.4.2 FHM₇₋₂ Hardware Components

2.4.2.1 Digital Processing Unit

The FHM₇₋₂ need a lot of high-speed interfaces. And for the shared cell application, the uplink needs combining function.

i. FPGA Solution Design 1

In this section, the requirement and reference design based on FPGA are described.

a. FPGA Requirement

The requirements are as follows:

- Interface requirements are shown in Table 2-40.

Table 2-40: Interface requirements of the FPGA

Serdes lane	At least 8 lanes of bi-direction10Gb Serdes
	At least 2 lanes of bi-direction25Gb Serdes

b. FPGA Design

This unit handles multiple high-speed interfaces and digital signal processing. The main chipset is based on FPGA with ARM multi-core processor while FPGA is responsible for high speed data processing and ARM cores are mainly for configuration and management.

The FPGA function is shown below:

For the uplink processing, the eCPRI interface module is used to receive eCPRI link form O-RU₇₋₂ and generate the reference frequency and time slot for time and frequency synchronization. Then the eCPRI de-frame module gets the uplink IQ bit stream of the time domain the carrier from O-RU₇₋₂. If the uplink IQ stream is compressed, then it needs the de-compression module to translate them into original bit width.

After that all the O-RU₇₋₂ uplink IQ streams are combined and the precondition for the combined function is that each stream from different O-RU₇₋₂ is aligned in time domain.

Then the time domain IQ date goes to the Low PHY function module to change to frequency domain IQ samples by FFT. Other functions are not specified in detail here.

If the FHM₇₋₂ can support two separate cells, then the two cell uplink signal should be aggregated or interleaved together. Then capture the IQ streams in the eCPRI frame and transmitted by eCPRI interface to the O-DU₇₋₂.

For the downlink processing, the procedure is much the same with uplink but with an inverse sequence. The difference is that the downlink signal is duplicated with 8 copies and sends to different O-RU₇₋₂.

There is also an eCPRI OAM module; actually this module will work with the processor (Arm) to accommodate the OAM of different O-RU₇₋₂ and the FHM₇₋₂ itself.

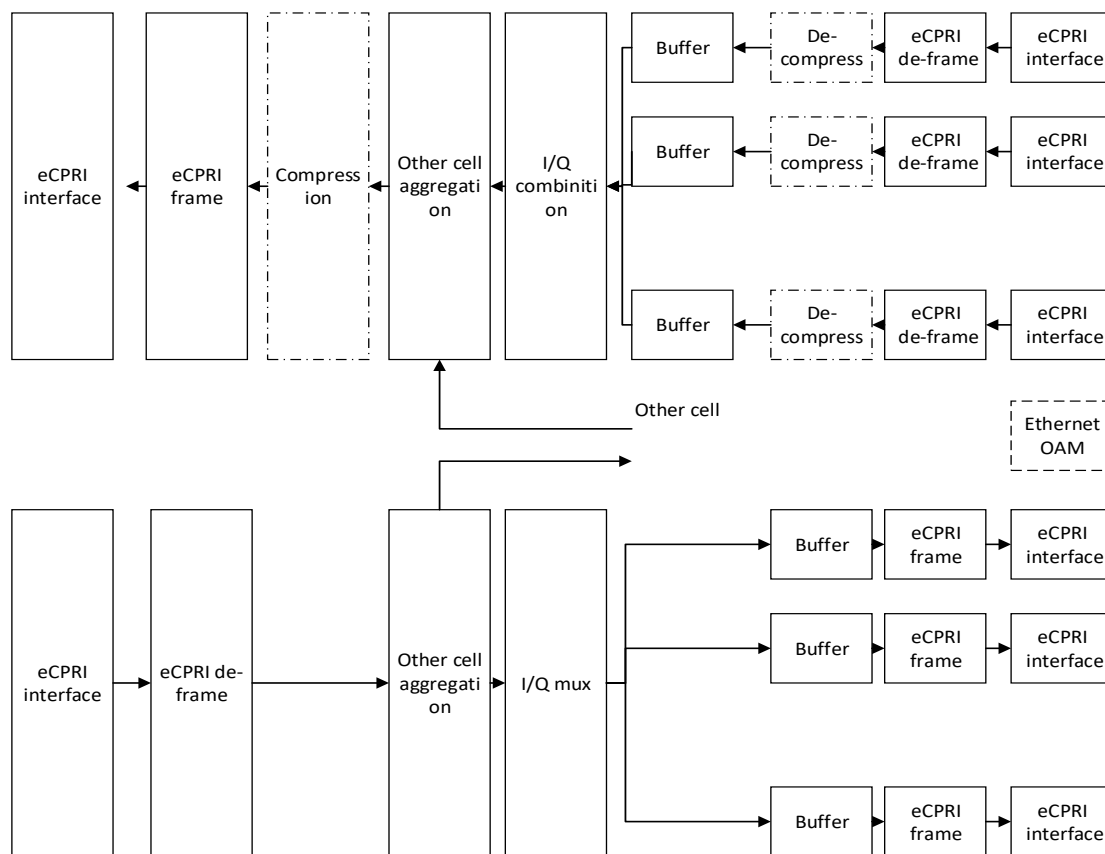


Figure 2-40: FHM₇₋₂ Digital Processing Block Diagram

The resources needed for the FPGA are listed below:

Table 2-41: FPGA Resource usage for FHM₇₋₂

Device	FF	LUT	URAM	BRAM	DSP
IQ MUX	1000	1000	0	0	0
Cascade	2000	2000	0	0	0
IQ combine	2000	2000	0	32	0
eCPRI deframe*9	63000	63000	0	180	0
eCPRI frame*9	63000	63000	0	180	0
Fronthaul (eCPRI)x10	100000	80000	0	10	0
compress*1	3000	3000	0	24	32
de_compress*8	24000	24000	0	192	208
ethernet_cm	3000	3000	0	15	0
Other	20000	15000	0	200	0

2.4.3 Synchronization and Timing

In this section we describe the internal and external timing and synchronization that are managed by this entity.

a. Hardware Requirements

- CLK requirement

The FHM₇₋₂ may support some kinds of synchronization method:

- 1) IEEE 1588/SyncE
- 2) Support GPS/GLONASS/GALILEO/BEIDOU

So the PLL must support 1pps or eCPRI CDR as the reference frequency.

Table 2-42: Requirements of the PLL device

Device	Description
1PPS	Supported
Synchronizer number	at least 1
Output channel	At least 5
VCO integrated	Supported

b. Hardware Design

This unit is to recover the clock from external source and generate the synchronized clock to other devices. Upon scenarios, there will be external sync source via eCPRI.

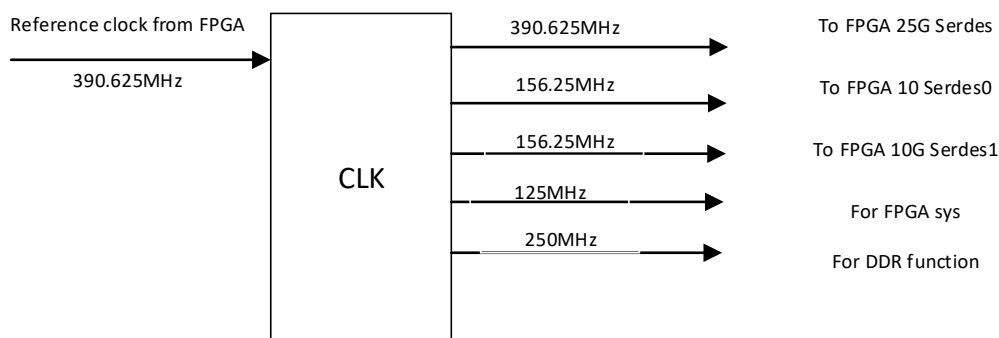


Figure 2-41: CLK reference design for FHM₇₋₂

For the general FPGA device, each bank may have 4 Serdes channels. It is better to have separated CLK to each bank. So two CLK of 156.25 MHz are needed for the FHM₇₋₂, which is used for the 8 10G Serdes interfaces with O-RU₇₋₂. One 390.625MHz is used for 2 25G Serdes interface with O-DU₇₋₂ or cascaded FHM₇₋₂. One 125MHz is used for FPGA system. One 250MHz CLK for DDR device function. One 390.625MHz is used for 25G eCPRI

interface between O-DU₇₋₂ and FHM₇₋₂. The input of the CLK module comes from FPGA CDR function which gets reference CLK from the eCPRI line rate. Usually, one PLL device which is integrated with VCO is needed.

2.4.4 External Interface Ports

List and provide description of all external interfaces.

a. Hardware Requirements

The following table shows the external ports or slots provided by FHM₇₋₂.

Table 2-43: External Port List

Port Name	Feature Description
Fronthaul interface	Eight 10Gbps SFP+ transceiver interfaces
	Two 25Gbps SFP+ transceiver interfaces
Debug interface	RJ45 for debug usage
Power interface	PLUG AC FEMALE for power cable

b. Hardware Design

- SFP+ case and connector

The SFP+ case and connector are standardized components on the market; following figure describes the form factor of one SFP case which is integrated with connector.

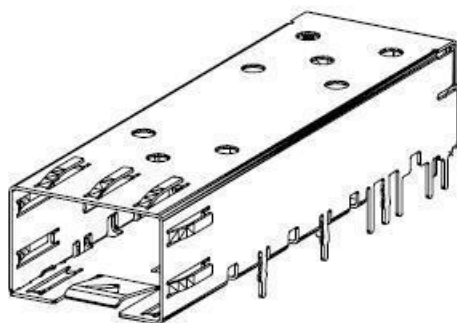


Figure 2-42: SFP+ case and connector

- RJ45 Ethernet interface

The RJ45 Ethernet interface is a standardized component on the market; following figure describes the form factor of one RJ45 interface.

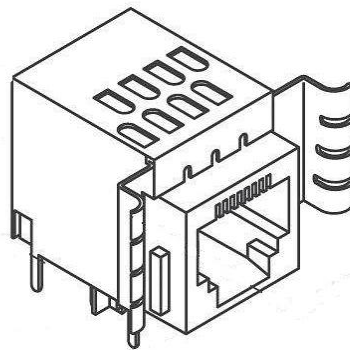


Figure 2-43: RJ45 interface

- Power interface

The 220V AC power connector is a standardized component on the market; following figure describes the form factor of one 220V AC power connector.

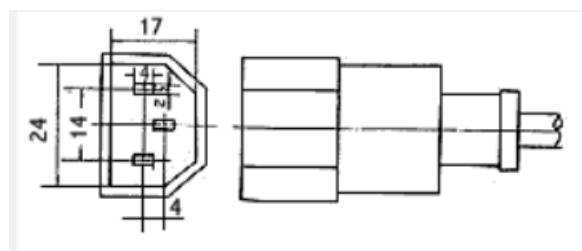


Figure 2-44: AC power interface

2.4.5 Mechanical

The 1U rack mount chassis can contain the layout of the power unit and processing unit. The shell of FHM₇₋₂ is shown in the following figure. The power consumption of FHM₇₋₂ is huge, so it may need a fan to accelerate the heat dissipation.

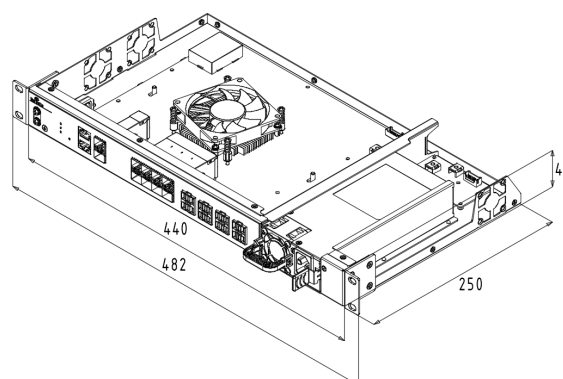


Figure 2-45: Shell Mechanical Diagram

2.4.6 Power

At minimum, fully describe the power consumptions for this white box. Include all AC/DC input or outputs and their ratings. Not mandatory but if you like you can add summary of all component's power requirement and overall white box.

a. Hardware Requirements

- Power requirement: The power solution is divided into two parts.
 1. Input power module: This power module must support AC to DC conversion. Usually it converts 220V AC to 48V DC.
 2. Output power module: It will supply power to the O-RU₇₋₂ connected to the FHM₇₋₂ and also the device on the FHM₇₋₂ itself.

Table 2-44: Requirements of the power unit

Item	Description
Input voltage	AC:100V~264V
Output voltage	DC: 48V
Output power	At least 600W

b. Hardware Design

Describe hardware reference design for this power unit.

This unit has two main functions which are internal power supply and external/remote power supply. The input power is normally AC (100V to 250V) but DC input could be optional. For remote power supply, it will support -48V DC via either standalone cable or cat-6A cable.

Usually a separated AC-DC power supply is used for the 220V to 48V conversion. It is very common power supply on the market.

It should have fan to cool itself and handle for easy to plug in and out.

Then the output power is divided into two portions. One for O-RU₇₋₂ power supply, the other is for the device on the FHM₇₋₂ board.

For the on-board power solution, 48V to 12V converter is needed as a standard 1/8 brick module. Then the power voltage is further changed to lower level such as 5V, 3.3V, 1.1V, 1.0V and so on by DC/DC or LDO devices.

2.4.7 Thermal

Active cooling.

2.4.8 Environmental and Regulations

The FHM₇₋₂ will fulfil the requirement in 4.6.3 of [6].

2.5 FHGW_{7-2→8} Hardware Reference Design

The FHGW_{7-2→8} is used to bridge the connection between O-DU₇₋₂ and O-RU₈. It needs to power the O-RU₈ and reduces the front haul bandwidth.

2.5.1 FHGW_{7-2→8} High-Level Functional Block Diagram

Figure 2-46 depicts the FHGW_{7-2→8} block diagram for this reference design. Following sections will describe the functionality, interface and performance for each respective block within the figure. Since device integration is an ongoing activity, chip boundaries may be fluid and some functionality may move from one block to another or entire functionalities may be absorbed into other blocks. The descriptions below describe the functional blocks independent of which physical device they may reside in.

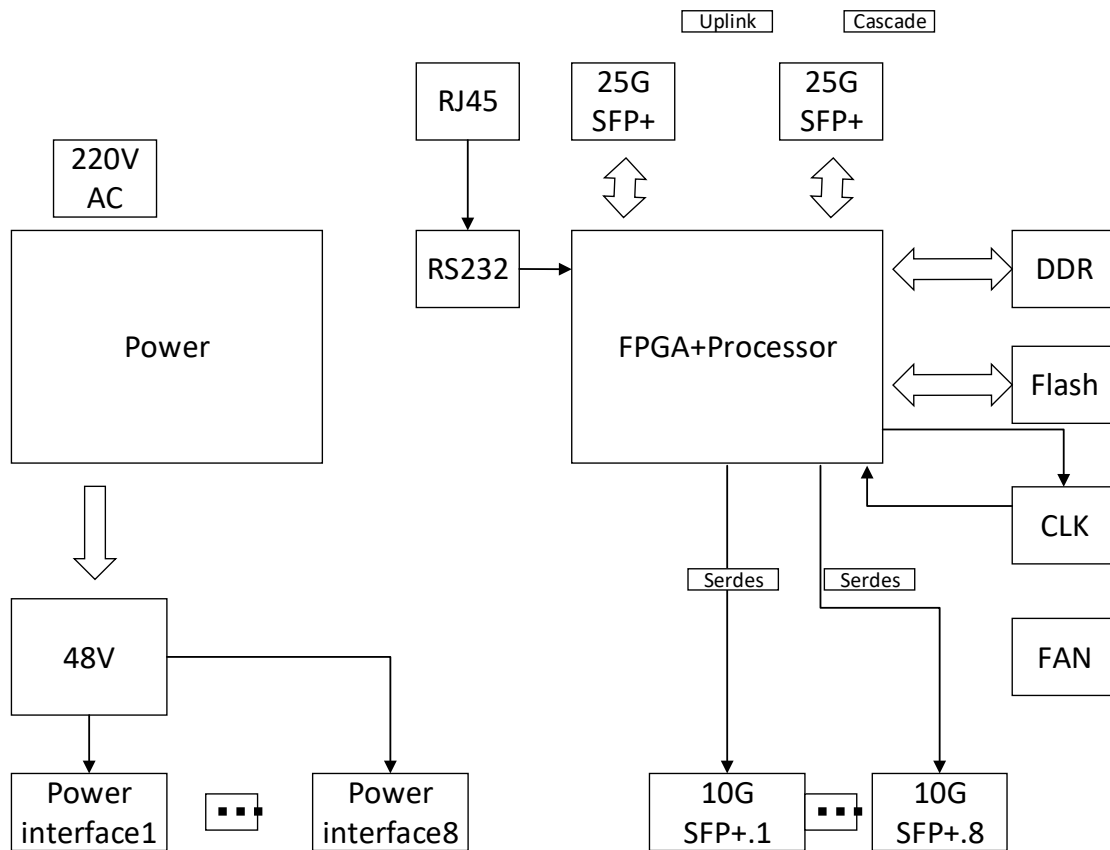


Figure 2-46: FHGW_{7-2→8} General Block Diagram

2.5.2 FHGW_{7-2→8} Hardware Components

2.5.2.1 Digital Processing Unit

The FHGW_{7-2→8} need a lot of high-speed interface. It would need the function of transforming the FH protocol with split Option 8 to one of with split Option 7-2. This requires availability of the Low PHY function as required by option 7-2. For the shared cell application, the uplink will also need a combining function for aggregation and dis aggregation of data.

i. *FPGA Design Solution 1*

In this section, the requirement and reference design based on FPGA are described.

a. *FPGA Requirement*

- Interface requirement: Interface requirements for this FPGA design solution are provided in the following table.

Table 2-45: Interface Requirements of the FPGA

Item Name	Description
Serdes lane	At least 8 lanes of bi-direction10Gb Serdes
	At least 2 lanes of bi-direction25Gb Serdes

b. *FPGA Design*

This unit handles multiple high-speed interfaces and digital signal processing. The main chipset is based on FPGA with multi-core processor while FPGA is responsible for high speed data processing and processor cores are mainly for configuration and management.

The FPGA function is shown below:

For the uplink processing, the CPRI interface module is used to receive CPRI link from O-RU₈ for generation of reference frequency and time slot for time and frequency synchronization. Then the CPRI de-frame module gets the uplink IQ bit stream of the time domain the carrier from O-RU₈. If the uplink IQ stream is compressed, then it needs the de-compression module to translate them into original bit width. All the O-RU₈ uplink IQ streams are combined and the precondition for the combine function is that each stream from different O-RU₈ is aligned in time domain. Then the time domain IQ data goes to the Low PHY function module to change to frequency domain IQ samples by performing FFT function.

If the FHGW_{7-2→8} can support two separate cells, then the two cell uplink signal should be aggregated or interleaved together. Then FHGW_{7-2→8} should capture the IQ streams in the eCPRI frame and transmit it via eCPRI interface to the O-DU₇₋₂. For the downlink processing, the procedure is very much the same as uplink but in reverse order. The difference is that the downlink signal is duplicated with 8 copies and transmits that to different O-RU₈. There is also one eCPRI/CPRI OAM module. Actually this module will work with the processor to accommodate the OAM of different O-RU₈ and the FHGW_{7-2→8} itself.

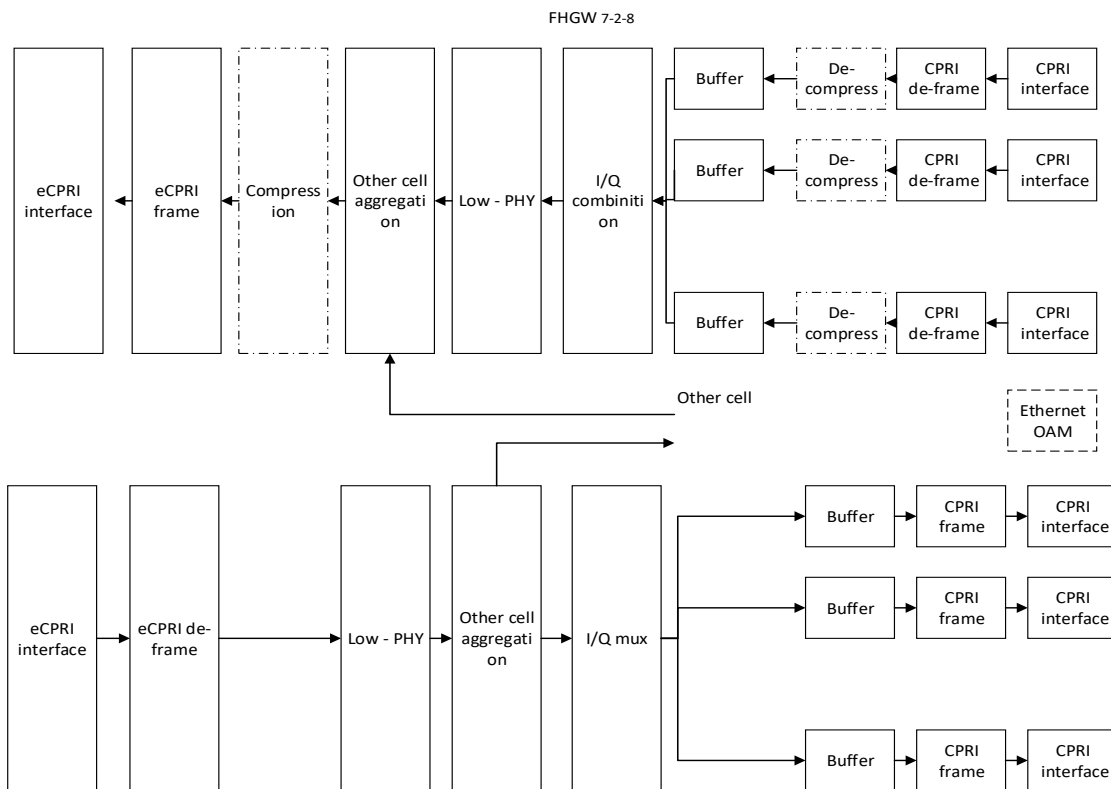


Figure 2-47: FHGW₇₋₂₋₈ Digital Processing Block Diagram

The FPGA resources needed for the FPGA are listed below:

Table 2-46: FPGA Resource Usage for FHGW₇₋₂₋₈

Device	FF	LUT	URAM	BRAM	DSP
IQ MUX	1000	1000	0	0	0
Cascade	2000	2000	0	0	0
IQ combine	2000	2000	0	32	0
eCPRI deframe*2	14000	14000	0	40	0
eCPRI frame*2	14000	14000	0	40	0
CPRI deframe*8	40000	40000	0	0	0
CPRI frame*8	40000	40000	0	0	0
Fronthaul (eCPRI)x2	20000	16000	0	2	0
Fronthaul (CPRI)x8	37890	24804	0	8	0
compress*1	3000	3000	0	24	32
de_compress*8	24000	24000	0	192	208
Ethernet CM	3000	3000	0	15	0
iFFT & CP+	7763	5323	0	27	23
FFT & CP-	7763	5323	0	27	23
PRACH front end	4000	3000	0	32	50
Other	20000	15000	0	200	0

2.5.3 Synchronization and Timing

This section describes the synchronization and timing mechanism that is used in the FHGW_{7-2→8}.

a. Hardware Requirements

- CLK requirement

The FHGW_{7-2→8} may support some kinds of synchronization method:

- 1) IEEE 1588/SyncE
- 2) Support GPS/GLONASS/GALILEO/BEIDOU

So the PLL must support 1pps or eCPRI CDR as the reference frequency.

Table 2-47: Requirements of the PLL device

Item Name	Description
1PPS	Supported
Synchronizer number	at least 1
Output channel	At least 6
VCO integrated	Supported

b. Hardware Design

This unit is to recover clock from external source and generate the synchronized clock to other devices. Upon scenarios, there will be external sync source via eCPRI.

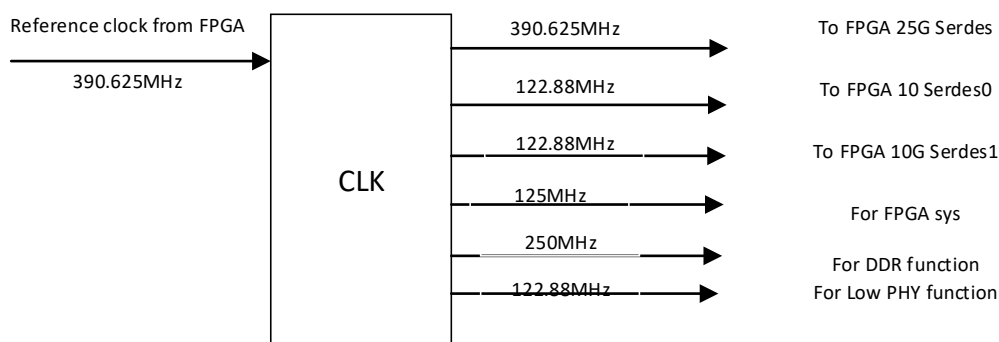


Figure 2-48: CLK reference design for FHGW_{7-2→8}

For general FPGA device, each bank may have four Serdes channels. It is better to have separated CLK signals to each bank. Three CLK signals of 122.88MHz are needed for the FHGW_{7-2→8}. One CLK signal of 125MHz is used for FPGA system, while one CLK signal of 250MHz is used for DDR device function. One CLK signal of 390.625MHz is used for 25G eCPRI interface between O-DU₇₋₂ and FHGW_{7-2→8}, and one CLK signal of 122.8MHz is used for low PHY function. The input of the CLK module comes from FPGA CDR function which can get reference CLK from the sync-plane of eCPRI. Usually, one PLL device integrated with VCO and two separated synthesizers is needed.

2.5.4 External Interface Ports

This section describes the external interface ports that are needed in the FHGW₇₋₂₋₈.

a. Hardware Requirements

The following table shows the external ports or slots provided by FHGW₇₋₂₋₈.

Table 2-48: External Port List

Port Name	Feature Description
Fronthaul interface	Eight 10Gbps SFP+ transceiver interfaces
	Two 25Gbps SFP+ transceiver interfaces
Debug interface	RJ45 for debug usage
Power interface	PLUG AC FEMALE for power cable

b. Hardware Design

- SFP+ case and connector

The SFP+ case and connector are standardized components on the market; following figure describes the form factor of one SFP case which is integrated with connector.

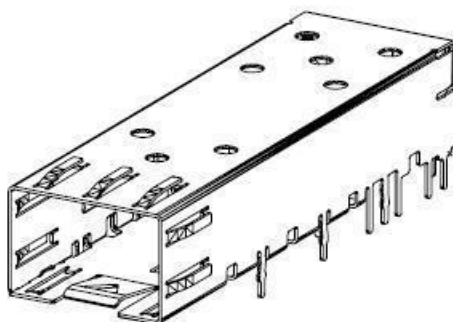


Figure 2-49: SFP+ case and connector

- RJ45 Ethernet interface

The RJ45 Ethernet interface is a standardized component on the market; following figure describes the form factor of one RJ45 interface.

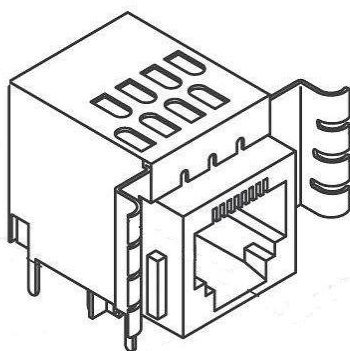


Figure 2-50: RJ45 interface

- Power interface

The 220V AC power connector is a standardized component on the market; following figure describes the form factor of one 220V AC power connector.

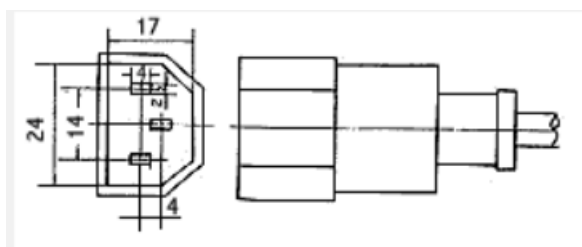


Figure 2-51: AC power interface

2.5.5 Mechanical

The 1U rack mount chassis can contain the layout of the power unit and processing unit. The shell of FHGW₇₋₂₋₈ is showing in the following figure. The power consumption of FHGW₇₋₂₋₈ is huge, so it may need a fan to accelerate the heat dissipation.

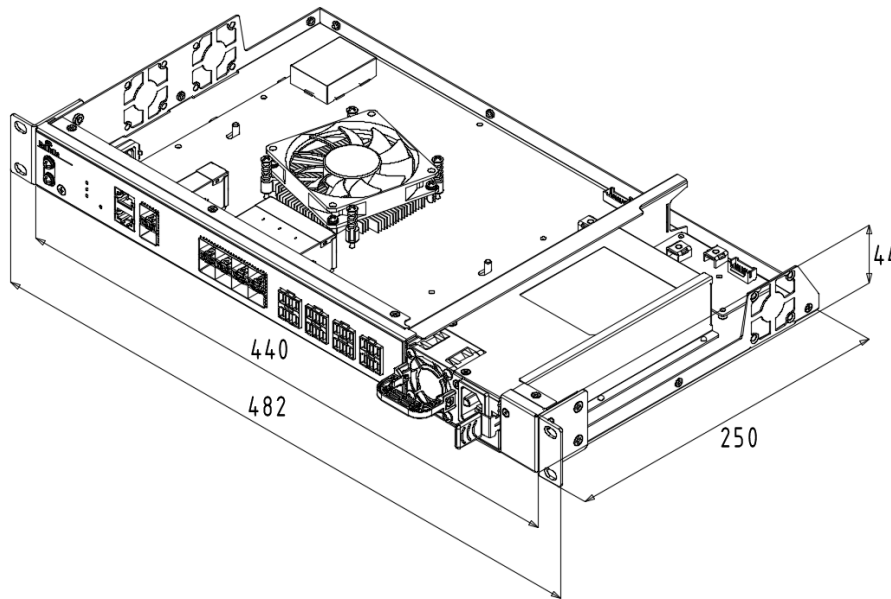


Figure 2-52: Shell Mechanical Diagram

2.5.6 Power

At minimum, fully describe the power consumptions for this white box. Include all AC/DC input or outputs and their ratings. Not mandatory but if you like you can add summary of all component's power requirement and overall white box.

a. Hardware Requirements

- Power requirement: The power solution is divided into two parts.
 1. Input power module
This power module must support AC to DC conversion. Usually it converts 220V AC to 48V DC.
 2. Output power module
It will supply power to the O-RU₈ connected to the FHGW₇₋₂₋₈ and also the device on the FHGW₇₋₂₋₈ itself.

Table 2-49: Power Unit Requirements

Item Name	Description
Input voltage	AC:100V~264V
Output voltage	DC: 48V
Output power	At least 600W

b. Hardware Design

This unit has two main functions which are internal power supply and external/remote power supply. The input power is normally AC (100V to 250V) but DC input could be optional. For remote power supply, it will support -48V DC via either standalone cable or cat-6A cable. Usually a separated AC-DC power supply is used for the 220V to 48V conversion. It is very common power supply on the market. It should have fan to cool itself and handle for easy to plug in and out. Then the output power is divided into two portions. One for O-RU₈ power supply, the other is for the device on the FHGW_{7-2→8} board. For the on-board power solution, 48V to 12V converter is needed as standard 1/8 brick module. Then the power voltage is further changed to lower level such as 5V, 3.3V, 1.1V, 1.0V and so on by DC/DC or LDO devices.

2.5.7 Thermal

Active cooling shall be used for this fronthaul gateway.

2.5.8 Environmental and Regulations

This FHGW_{7-2→8} will fulfil the requirements as described in 4.6.3 of [6].

Annex 1: Parts Reference List

According to WG7 scope and charter, component selection for example implementation of white box hardware is allowed for WG7 but would not be mandatory in any Specification.

Recommended O-DU₇₋₂ parts reference is given in table below:

No.	Part Number	Descriptions
1	Intel® Xeon® Processor D-2183IT	Intel® Xeon® Skylake D D-2100 SoC processor
2	Intel® I350	Intel® 1GbE x4 Ethernet controller
3	Intel® X557-AT2	Intel® 10GbE x2 Ethernet controller
4	Intel® I210	Intel® 1GbE Ethernet controller
5	ASPEED® AST-2500	ASPEED® BMC controller
6	UniIC® SCQ08GU03H1F1C-26V	UniIC® DDR4 8GB
7	Powerleader® MTFDDAK480TDC-1AT1ZABYY	Powerleader® SSD 480GB
8	Accelink® RTX228-551	Accelink® 10Gbps Optical module

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Annex 2 FHGW_{7-2→8} Option 8 CPRI specification reference design

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Please refer to “Annex2 - CPRI specification reference design” in O-RAN.WG7.IPC-HRD-Opt8.0-v03.00, <https://www.o-ran.org/specifications>, reuse all contents except to replace the O-DU₈ with O-DU₇₋₂, and replace the FHM₈ with FHGW_{7-2→8}.

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Annex 3 FHGW_{7-2→8} Option 8 CPRI spec reference design – Management plane

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Please refer to “Annex3 - Option 8 CPRI spec reference design – Management plane” in O-RAN.WG7.IPC-HRD-Opt8.0-v02.00, <https://www.o-ran.org/specifications>, reuse all contents except to replace the O-DU₈ with O-DU₇₋₂, and replace the FHM₈ with FHGW_{7-2→8}.

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Annex 4 FHGW_{7-2→8} management plane reference design

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A4.1 Scope

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This document specifies the management plane reference design for FHGW_{7-2→8}. Except content specify here, other management function is based ORAN WG4 Management Plane Specification and reuse it.

19

A4.2 Reference

20

21

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

22

23

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.

24

- For a specific reference, subsequent revisions do not apply.

25

26

- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document in Release 15.

27

[1] 3GPP TR 21.905: “Vocabulary for 3GPP Specifications”

[2] Interface Specification: “CPRI Specification Common Public Radio Interface (CPRI)”, V7.0, October 2015.

[3] ORAN-WG4.CUS.0-v03.00 Technical Specification, ‘O-RAN Fronthaul Working Group Control, User and Synchronization Plane Specification’. <https://www.o-ran.org/specifications>

[4] ORAN-WG4.MP.0-v03.00 Technical Specification, ‘O-RAN Fronthaul Working Group Control, Management Plane Specification’. <https://www.o-ran.org/specifications>

A4.3 Definitions and Abbreviations

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

Abbreviations	Full name
C&M	Control and Management
CPRI	Common Public Radio Interface
LOF	Lost of Frame
LOS	Lost of Signal
LSB	Least Significant Bit
M-Plane	Management Plane
MSB	Most Significant Bit
O-DU ₇₋₂	O-RAN Distributed Unit with split option 7-2
FHGW _{7-2→8}	FHGW between O-DU ₇₋₂ and O-RU ₈
O-RU ₈	O-RAN Radio Unit with split option 8
RAI	Remote Alarm Indication
S-Plane	Synchronization Plane
SAP	Service Access Point
SDI	SAP Defect Indication
U-Plane	User Plane

Note: If without clear clarification in this Annex, O-DU always refer to O-DU₇₋₂, FHGW always refers to FHGW_{7-2→8}, O-RU always refers to O-RU₈.

A4.4 High Level Description

This specification refers to following architecture model.

From the perspective of the lower layer connection protocol, as shown in the following figure, the architecture model includes:

- Ethernet/VLAN protocol connection is applied between O-DU₇₋₂ and FHGW_{7-2→8}
- Ethernet/VLAN protocol connection is applied between FHGW_{7-2→8} and FHGW_{7-2→8}
- CPRI protocol connection is applied between FHGW_{7-2→8} and O-RU₈

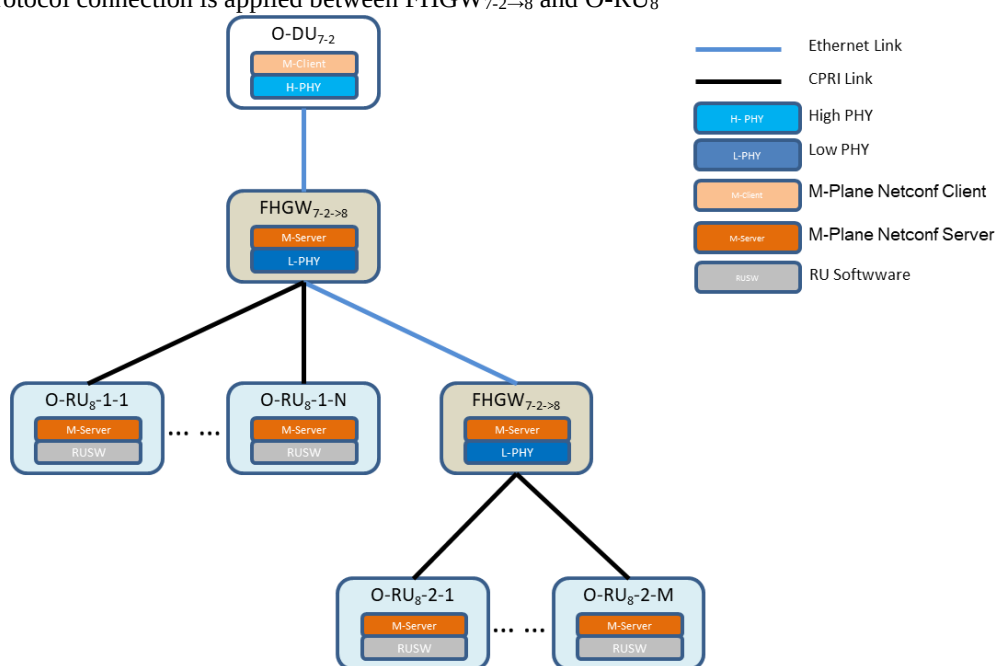


Figure 1 Architecture-lower layer connection

From M-Plane perspective, as shown in the figure below, this architecture model includes:

- NETCONF client is deployed in O-DU₇₋₂
- NETCONF server is deployed in FHGW_{7-2→8} and O-RU₈
- O-DU and FHGW_{7-2→8} are connected with NETCONF management protocol
- O-DU and O-RU₈ are connected with NETCONF management protocol

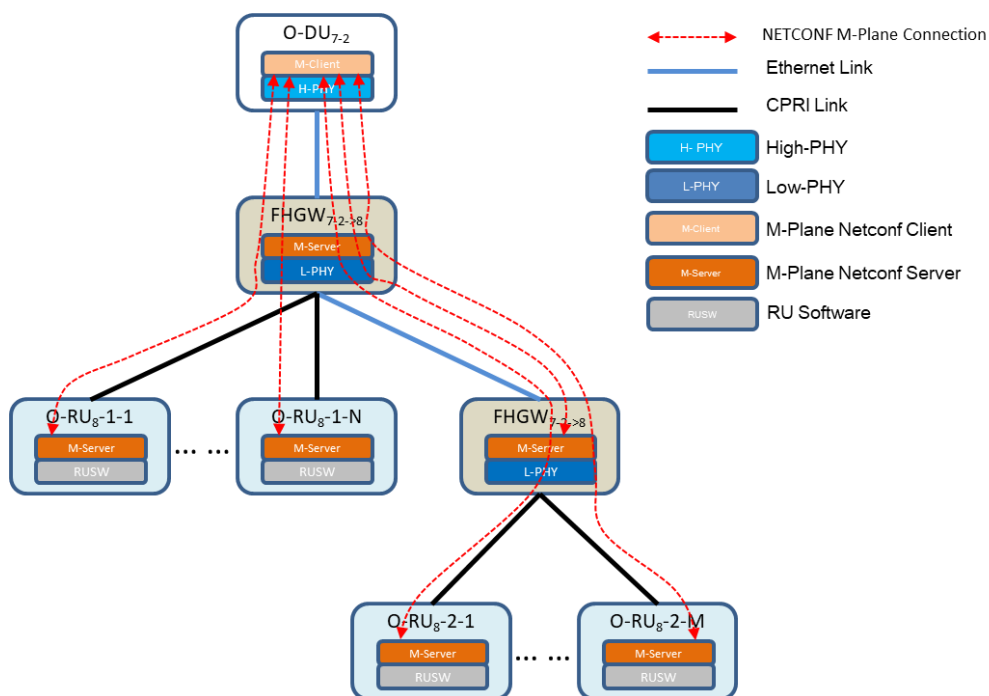


Figure 2 Architecture M-Plane connection

From C-Plane & U-Plane perspective, as shown in the figure below, this architecture model includes:

- High-PHY is deployed in O-DU₇₋₂
- Low-PHY is deployed in O-RU₈
- The O-DU and FHGW_{7-2->8} connect with e-CPRI protocol and split with Option7-2.
- The FHGW_{7-2->8} and FHGW_{7-2->8} connect with e-CPRI protocol and split with Option7-2.
- The FHGW_{7-2->8} and O-RU₈ connect with CPRI protocol and split with Option8.

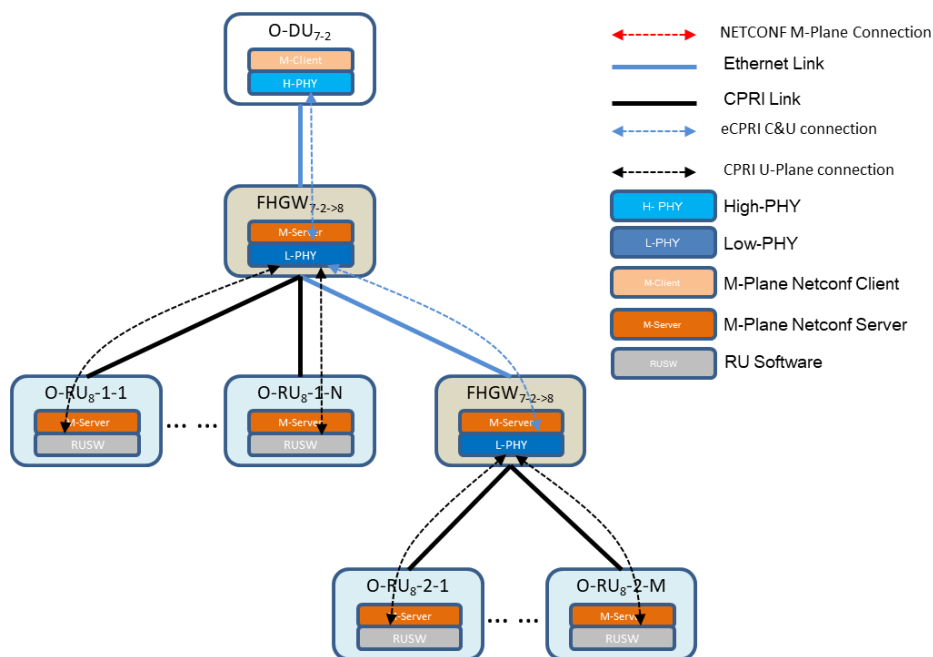


Figure 3 Architecture C&U-Plane connection

The content of this chapter includes the management of FHGW_{7-2→8} in accordance with the description of the previous architecture model chapter

A4.5 Interface Management

A4.5.1 CPRI Interface Management

The basic properties management of the CPRI interface are in the following section, while the details of U-Plane data flow and transceiver parameter acquisition are in the specification of section 4.3.1.

A4.5.1.1 CPRI Interface YANG model

Yang model: o-ran-interfaces-cpri.yang

The basic properties management of CPRI Interface includes:

- The configuration of CPRI protocol version
- The configuration of CPRI interface model
- The configuration of CPRI line speed rate
- The configuration of CPRI link coding parameter
- CPRI start pointer configuration of C-Plane and M-Plane
- read CPRI status

To make it simple, there are some working assumption about the basic property's management:

- 1) CPRI interface mode. It is agreed that the northward interface of CPRI link is the master port mode, and the southward interface is the slave port mode. For example, in the case of direct connection, O-DU or FHGW_{7-2→8} southward CPRI interface is master port mode, O-RU₈ port is slave port mode; while in the case of cascading, the northward port at the level above the O-RU₈ cascade is configured with slave port mode, and the southward port at the level below the O-RU₈ concatenation is configured with master port mode.
- 2) CPRI physical layer is in state A when it is ready. To ensure that CPRI link can reach state B and is in physical layer synchronization state, CPRI port shall be configured as follows:
 - a) CPRI line rate configuration is as follows:

Table 4 The configuration of CPRI line rate

CPRI line bit rate [Mbit/s]	Z	X	W	Y	B
10137.6	0, 1, ..., 149	0, 1, ..., 255	0, 1, ..., 15	0, 1, 2, ..., 19	0, 1, ..., 159
12165.12				0, 1, 2, ..., 23	0, 1, ..., 191
24330.24				0, 1, 2, ..., 47	0, 1, ..., 383

- b) IQ sampling bits width

Table 5 IQ sampling bits width

Direction of link	Symbol for sample width	Range[bits]
Downlink	M	16

Downlink	M	8(with compression algorithm)
Uplink	M'	16
Uplink	M'	8(with compression algorithm)

c) CPRI link coding

1. Link encoding can be configured as:
2. 64B/66B
3. 64B/66B RS-FEC (optional)

d) CPRI protocol version

In order to ensure that the CPRI master and slave ports enter the state C, both ends of the CPRI agreement shall use the CPRI protocol version No. 1. "Protocol Version" is defined in the chapter "L1 Inband Protocol" of the CPRI specification.

Note: The difference between Protocol Version 1 and 2 is that when Protocol Version = 2, all data (except subchannel $N_s = 0$ control word BYTES #Z.X.Y $Y \leq 1$ and subchannel $N_s = 2$) need to add perturbation before linear encoding. With Protocol Version = 1, perturbation is not required. For simplicity, use the Protocol Version = 1 configuration.

3) To ensure that the master and slave ports of CPRI enter the state D, the control and management channel conventions are configured as follows:

- a) the control management channel, namely Ethernet channel rate configuration is as follows:

Table6 Ethernet channel rate configuration

CPRI line bit rate[Mbit/s]	Length of control word[bit]	Control word consisting of BYTES#	Minimum ethernet bit rate[Mbit/s](# Z.194.0=rr111 111)	Maximum ethernet bit rate[Mbit/s](# Z.194.0=rr010 100)
10137.6	128	#Z.X.0,#Z.X.1,#Z.X.2,#Z.X.3,#Z.X.4,#Z.X.5,#Z.X.6,#Z.X.7,#Z.X.8,#Z.X.9,#Z.X.10,#Z.X.11,#Z.X.12,#Z.X.13,#Z.X.14,#Z.X.15	7.68	337.92
12165.12	128	#Z.X.0,#Z.X.1,#Z.X.2,#Z.X.3,#Z.X.4,#Z.X.5,#Z.X.6,#Z.X.7,#Z.X.8,#Z.X.9,#Z.X.10,#Z.X.11,#Z.X.12,#Z.X.13,#Z.X.14,#Z.X.15	7.68	337.92
24330.24	128	#Z.X.0,#Z.X.1,#Z.X.2,#Z.X.3,#Z.X.4,#Z.X.5,#Z.X.6,#Z.X.7,#Z.X.8,#Z.X.9,#Z.X.10,#Z.X.11,#Z.X.12,#Z.X.13,#Z.X.14,#Z.X.15	7.68	337.92

- b) CPRI C-Plane and M-Plane pointer, pointing to the starting location for fast control and management of Ethernet channels. The Ethernet Pointer is configured as 20

- 4) To ensure that the master and slave ports of CPRI enter state E and State F, the O-RU₈ slave port can use the established Ethernet control and management channel to execute the NETCONF Call Home procedure in Chapter 4.
- 5) CPRI state conventions

- 1) CPRI starting state
 - A . Standby
 - B . L1 Synchronization and Rate Negotiation
 - C . Protocol Setup
 - D . C&M Plane (L2+) Setup
 - E . Interface and Vendor specific Negotiation
 - F . Operation
4. Based on the above CPRI startup states, the M-Plane defines quarriable CPRI startup states, including ENABLE and DISABLE. When the equipment at both ends of the CPRI link enters the F state, the startup state of CPRI is ENABLE state, and the startup state of CPRI is DISABLE state if it is in the A-E state.
- 2) CPRI link status

CPRI specification defines the following alarms:

 - A . LOS
 - B . LOF
 - C . SDI
 - D . RAI
5. Based on the above CPRI link alarms, the quarriable CPRI link states, including NORMAL and ABNORMAL, are defined at the time of the alarms. If have alarm for the ABNORMAL, otherwise NORMAL.

A4.5.1.2 Initialization CPRI Interface

Pre-condition:

The M-Plane connection between O-DU and FHGW_{7-2→8} is established, and the FHGW_{7-2→8} southward CPRI interface initialization begins.

Post-condition:

At the end of this procedure, the initialization configuration from the FHGW_{7-2→8} southward to the CPRI interface is completed.

Procedure:

- 1) In the startup phase, according to the definition in the o-ran-interfaces-cpri.yang model, the NETCONF client completed the initialization configuration of basic parameters such as line-bit-rate, sample width, Protocol Version, Pointer P for each FHGW_{7-2→8} southward CPRI interface.
- 2) During the link establishment procedure of CPRI, Protocol Version, Pointer P and other information are included in the control word of CPRI and sent to O-RU₈.

A4.5.1.3 Retrieve Transceiver infomation

Pre-condition:

The M-Plane connection between O-DU and FHGW_{7-2→8} is established.

Post-condition:

At the end of this procedure, O-DU obtains the information of FHGW_{7-2→8} transceiver module.

Procedure:

- 1) During startup or hot plug, The FHGW_{7-2→8} detects transceiver modules and stores data information from transceiver modules in files;
- 2) The NETCONF client obtains the file according to the netconf-yang file upload procedure.
- 3) Exception:
 - If transceiver module is inserted during the file upload procedure, the contents of files before the procedure or failure will be provided.
 - If FHGW_{7-2→8} can not read data from transceiver module, or file does not exist, then FHGW_{7-2→8} does not create file or delete the file created before (note: The file upload will fail because the file does not exist during the file upload procedure.).

A4.5.1.4 Configuration CPRI Interface for U-Plane

Pre-condition:

The M-Plane connection between O-DU and FHGW_{7-2→8} is established.

Post-condition:

At the end of this procedure, the U-Plane from the FHGW_{7-2→8} side is managed and configured. Configuration of the U-Plane data flow path between O-DU and FHGW_{7-2→8} is completed.

Procedure:

- 1) The NETCONF client will configure and manage the U-Plane configuration of the FHGW_{7-2→8} southward CPRI interface according to the newly added transport-flow type in the o-ran-processing-element.yang model. and set the transport-flow-type to "CPRI INTERFACE ", and carry out corresponding CPRI interface configuration of U-Plane.
- 2) eexc-id identifies the end-to-end data flow from the processing end of the O-DU to the end of the O-RU₈ antenna. Configure the data flow path between O-DU and FHGW_{7-2→8} according to eexc information.
- 3) The FHGW_{7-2→8} side completes the mapping from eexc to the FHGW_{7-2→8} southward CPRI port. The specific CPRI interface IQ configuration information includes whether to compress IQ data, starting bit position of IQ data, starting bit position of compression information, AGC bit position etc. After completing the IQ configuration procedure of the CPRI interface, the CPRI mapping relationship on the southward CPRI interface is determined. In the case of different cell bandwidth and antenna configuration, different IQ configuration is required on the CPRI interface.

A4.6 FHGW_{7-2→8} U-Plane Routing and Carrier Configuration

The target of this chapter is similar to "Chapter 12 Details of O-RU₈ Operations" of [4].

U-Plane configuration in this chapter specifically refers to the U-Plane configuration of Option 7-2. Based on the architectural model in Chapter 2. Indoor Pico Cell have a Low PHY function (FFT/iFFT), as well as copy and combine functions, which U-Plane configuration described in this chapter.

This chapter presents the configuration of U-Plane message routing between O-DU and FHGW_{7-2→8}. After completing the U-Plane configuration, routing is established between the O-DU and the FHGW_{7-2→8}'s U-Plane application endpoints, and IQ data can be exchanged using the U-Plane application protocol defined by [3].

The U-Plane message routing configuration includes:

- 1) eAxC-ID address format definition
- 2) U-Plane endpoint addressing
- 3) U-Plane message routing configuration procedure

A4.6.1 General Description

A4.6.1.1 YANG Model Files

o-ran-uplane-conf-hub.yang: defining the U-Plane configuration capability of the FHGW_{7-2→8}.

A4.6.1.2 Specific Conventions

Table 8

	O-RAN M-plane Specification	This Specification
eAxC_ID subfield bit width	Not fixed	Fixed
Non-delay management service	supported	Not supported

A4.6.1.3 YANG Model Detail

U-Plane configuration model is a user plane configuration of the Low PHY deployed in FHGW_{7-2→8}. The YANG model is defined in o-ran-uplane-conf-hub.yang, which mainly includes:

- ✓ List of multiple static TX endpoints: static-low-level-tx-endpoints
- ✓ List of multiple static RX endpoints: static-low-level-rx-endpoints
- ✓ List of multiple configurable TX endpoints: low-level-tx-endpoints. Each configurable TX endpoint low-level-tx-endpoint map to a static TX endpoint static-low-level-tx-endpoint.
- ✓ List of multiple configurable RX endpoints: low-level-rx-endpoints. Each configurable RX endpoint low-level-rx-endpoint map to a static RX endpoint static-low-level-rx-endpoint.
- ✓ List of multiple TX links: low-level-tx-links. Each low-level-tx-link, One end link e-CPRI data flow model eth-flow defined in the o-ran-processing-element.yang, the other end link to configurable TX endpoint low-level-tx-endpoint.
- ✓ List of multiple RX links: low-level-rx-links. Each RX link low-level-rx-link, one end link to e-CPRI data flow model eth-flow defined in the o-ran-processing-element.yang, the other end link to configurable TX endpoint low-level-rx-endpoint.

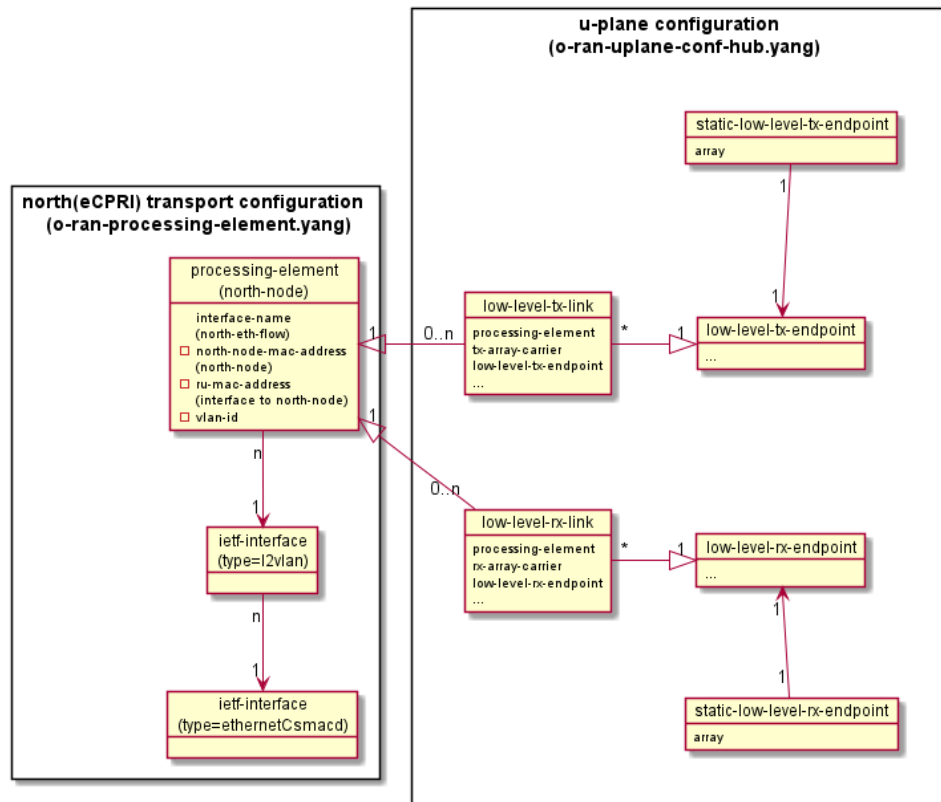


Figure 1 FHGW₇₋₂₋₈ U-Plane Configuration

A4.6.2 Format of eAxC_ID

As defined in [3], eAxC_ID consists of four parameters: DU-PORT, RU-PORT, CC-ID and BAND-SECTOR-ID. The sequence of parameters in eAxC_ID must obey the CUS interface specification definition.

In this specification, eAxC_ID is 16 bit, Each subfield is configured with a fixed length: [DU-PORT]:[BAND-SECTOR]:[CCID]:[RU-PORT] = 2:6:4:4.

<du-port-bitmask> 1100000000000000 **</du-port-bitmask>**

<band-sector-bitmask> 0011111100000000 **</band-sector-bitmask>**

<ccid-bitmask> 0000000011110000 **</ccid-bitmask>**

<ru-port-bitmask> 0000000000001111 **</ru-port-bitmask>**

A4.6.3 U-Plane Endpoint Addressing

For FHGW₇₋₂₋₈ based on e-CPRI Option7-2 to CPRI Option8, eaxc-id is used for resource mapping and U-Plane endpoint addressing.

Different eaxc-id correspond to different logical functions, such as cell, carrier, air separation data flow, etc. It identifies the end-to-end data flow from the processing end of the O-DU to the end of the O-RU₈ antenna.

During the configuration procedure:

O-DU should configure the eaxc-id for FHGW_{7-2→8} U-Plane endpoints so as to realize the mapping between logical functions and FHGW_{7-2→8} physical resources. Refer to the YANG model **low-level-tx-endpoint** and **low-level-rx-endpoint** in the o-ran-uplane-conf-hub.yang.

O-DU should configure the relation between the U-Plane endpoint and the northward e-CPRI data flow for the FHGW_{7-2→8}, to realize the mapping between the northward e-CPRI data flow and the O-RU₈ physical resources.

O-DU should configure the relation between the U-Plane endpoint and the southward e-CPRI data flow for the FHGW_{7-2→8}, to realize the mapping between the data flow to e-CPRI and the physical resources of O-RU₈. See 15.2.4 shared cell for details.

O-DU should configure the relation between the U-Plane endpoint and the southward CPRI data flow for the FHGW_{7-2→8}, to realize the mapping between the southward CPRI data flow and the O-RU₈ physical resource. See 15.2.4 shared cell for details.

In the procedure of data transmission, combined with the 15.2.1.1 CPRI U-Plane data flow model:

In the downlink direction, the FHGW_{7-2→8} obtains the e-CPRI data flow from the e-CPRI interface, finds the corresponding U-Plane Tx endpoint (**low-level-tx-endpoint**) according to eaxc-id, and sends the processed data to the CPRI data flow on the corresponding CPRI interface.

In the uplink direction, after the FHGW_{7-2→8} receives the CPRI data flow from the CPRI interface, it finds the corresponding U-Plane Rx endpoint (**low-level-rx-endpoint**), according to eaxc-id, and sends the processed data to the e-CPRI data flow on the corresponding e-CPRI interface.

The rules for configuring eaxc-id by O-DU are as follows:

- For the same uplink U-Plane data flow,
 - ✓ The low level TX endpoint (**low-level-tx-endpoint**) in FHGW_{7-2→8} should be configured with the same eaxc-id with TX endpoint (tx-endpoint) in O-RU₈;
- For the same downlink U-Plane data flow,
 - ✓ The low level RX endpoint (**low-level-rx-endpoint**) in FHGW_{7-2→8} should be configured with the same eaxc-id with the RX endpoint (rx-endpoint) in O-RU₈;

The FHGW_{7-2→8} should reject configuration requests that violate the above rules.

The diagram below shows an example of eaxc-id assignment in the architecture converting from eCPRI Option7 to CPRI Option8.

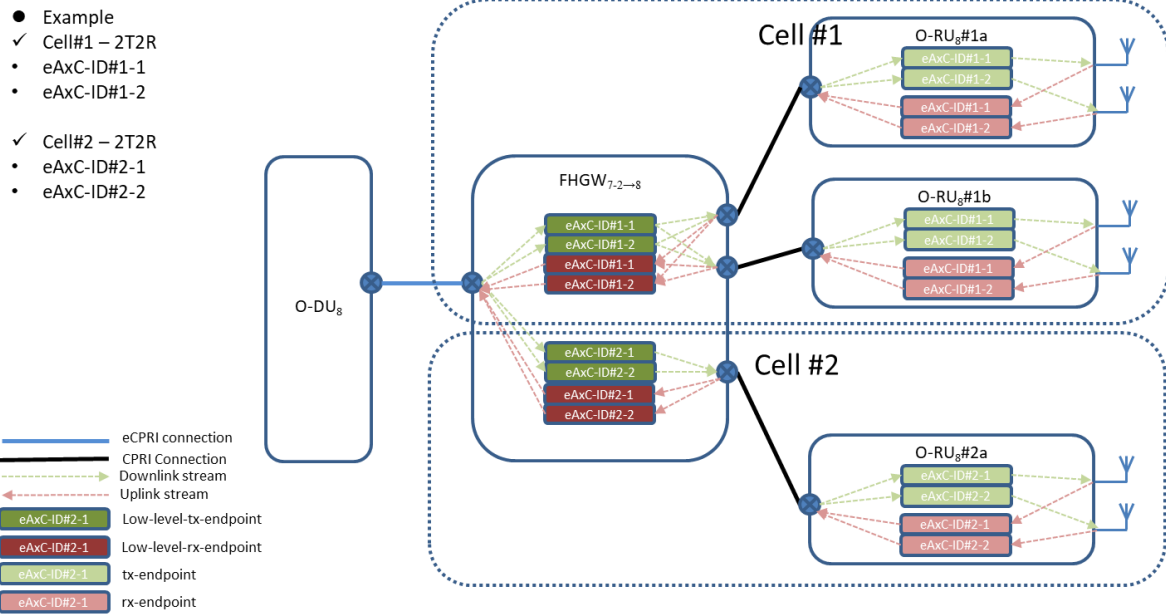


Figure 2 Examples of eaxc-id assignment

A4.6.4 U-Plane Routing and Carrier Configuration

The list YANG model of **low-level-tx-endpoints** and **low-level-rx-endpoints** in the o-ran-uplane-conf-hub.yang, represent the collection of FHGW₇₋₂₋₈ U-Plane endpoints. Each element of the list represents a TX endpoint or RX endpoint. The endpoint parameter (eexc-id) represents the local address of the endpoint, It's an unsigned 16-bit integer, and follows the eexc-id address format defined above.

The NETCONF client is responsible for correlation all to the same interface, and a unique eexc-id address assigned by the **low-level-rx-endpoint** element and the **low-level-tx-endpoint** element referenced by the **level-rx-link/the level-tx-link**.

Note:

- ✓ The **low-level-rx-endpoint** and the **low-level-tx-endpoint** can be assigned to the same eexc-id.
- ✓ The NETCONF server should reject configuration requests that violate the above rules.
- ✓ The default eexc-id parameter has a value of 0.

Pre-condition:

- The M-Plane connection between the NETCONF client and the NETCONF server is established.

Post-condition:

- Assign eexc-id to the U-Plane transceiver endpoint.
- Carrier creation can be further performed.

Procedure:

- 1) The NETCONF client determines the following elements whether exist provided by the NETCONF server:

- **low-level-tx-endpoint** element: by reading o-ran-uplane-conf-hub.yang of the **static-low-level-tx-endpoints** list;
- **low-level-rx-endpoint** element: by reading o-ran-uplane-conf-hub.yang of the **static-low-level-rx-endpoints** list;
- **interface** elements: by reading o-ran-interfaces.yang **interface** list.

2) The NETCONF client determines the ability and parameters of O-RU₈ exposure through the following nodes:

- the **static-low-level-tx-endpoints** and the **static-low-level-rx-endpoints** exposed ability.
- the **endpoint-types** and the **endpoint-capacity-sharing-groups** exposed ability,

Note 1 :

- ✓ The information above shall be considered when the NETCONF client concatenates **low-level-t/rx-endpoints** (where the argument **name** points to **static-level-t/rx-endpoints**).

3) For the elements identified in Step 1), the NETCONF client checks the following relationships:

- **static-low-level-tx-endpoint** element and **interface** element;
- **static-low-level-rx-endpoint** element and **interface** element.

4) For **static-low-level-rx-endpoint**, the NETCONF client determines its ability to support time-management-free and/or time-managed:

- The information of the delay service type supported by the endpoint is exposed through the parameter **delayed-data-transfer-supported** under **static-low-level-rx-endpoint**. and Supporting time management is a mandatory feature.
- Configure the non-time-managed-delay-enabled parameter of low-level-rx-endpoint. the default value of this parameter is FLASE, namely the endpoint support time management service by default.

5) The NETCONF client can optionally check the transmission connection of the NETCONF server, as described in Chapter 4.6 in [3].

Note 2:

- ✓ After this step completes smoothly, the NETCONF client knows which NETCONF servers provide the **low-level-tx-endpoint** and **low-level-rx-endpoint elements** that the Tx/Rx endpoint of the NETCONF client service has access to.

6) The NETCONF client determines accessible the **low-level-rx-endpoint** elements and the **low-level-tx-endpoint** elements, suitable for the target cell configuration (that is, connected to a specific antenna array and able to support the target business type).

7) The NETCONF client assigns an unique eaxc-id to the endpoint identified in Step 1).

Note 3:

- ✓ For all endpoints connected to the same **interface** element, the eaxc-id is enforced to be unique. and is associated with either the **lower-level-rx-endpoint element** or the **lower-level-tx-endpoint element**.

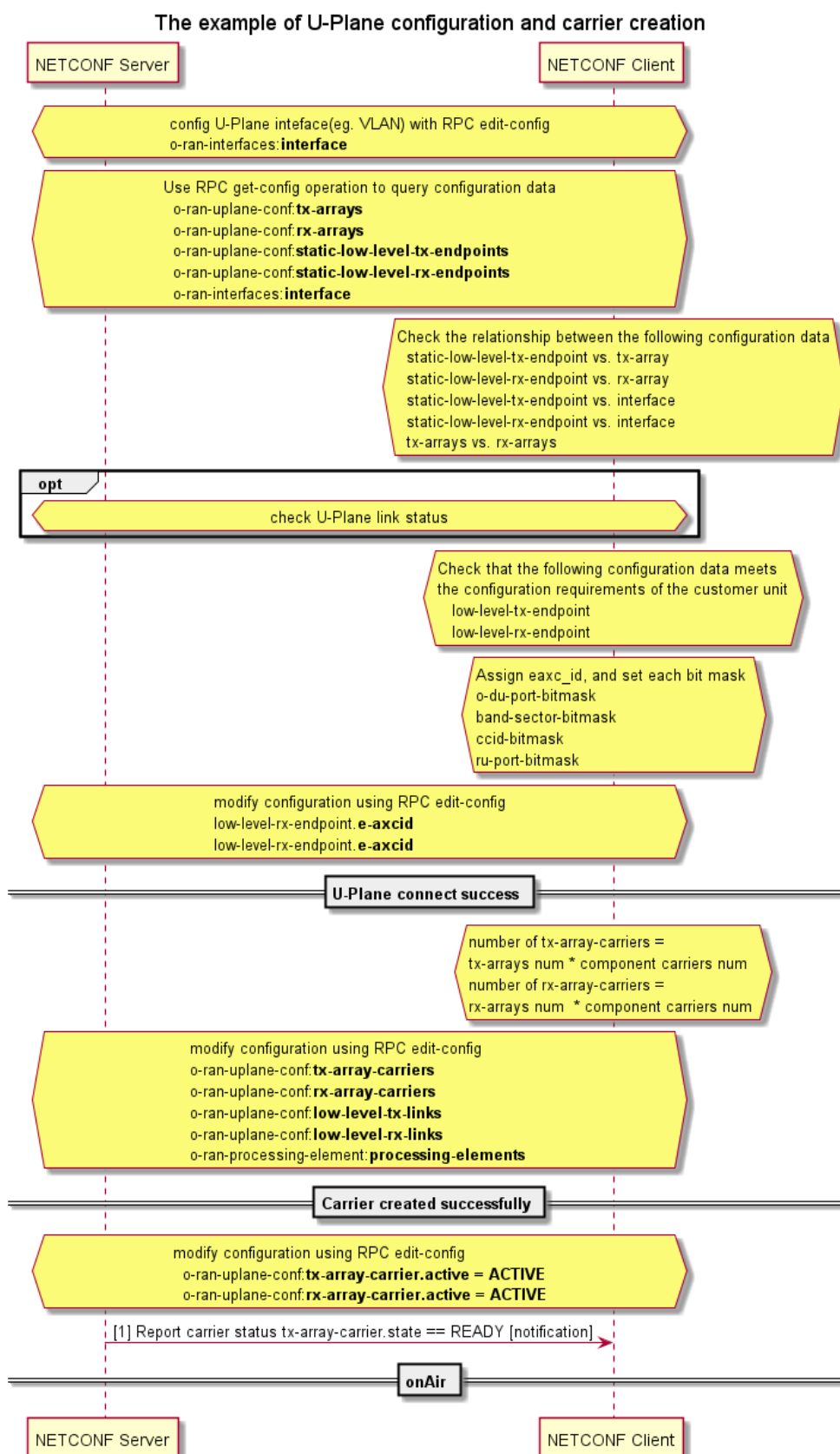


Figure 3 FHGW₇₋₂₋₈ U-Plane Configuration Procedure

A4.7 FHGW_{7-2→8} Delay Management

The fronthaul interface between O-DU and FHGW_{7-2→8} adopts e-CPRI based on Option 7-2x function split defined by ORAN-WG4.CUS.0-v03.00. Its delay management complies with the technical requirements in Section 11.3, "Delay Management of shared Cell".

A4.7.1 Convention

The current delay measurement and management model is implemented according to the FHGW_{7-2→8} functional model given in Section 15.2.4. The functional model supports the following combine types:

Combine type1: Uplink data from southward CPRI interfaces are combined in CPRI domain, after processed by the Low PHY and output in the format of Option7-2x at the FHGW_{7-2→8} northward interface, namely uplink data of CPRI domain combined within the FHGW_{7-2→8}.

Combine type 2: Southward Option7-2x port (e-CPRI) Northward data can be combined in the 7-2x data domain. This mode is the FHM mode of O-RAN Option7-2x shared cell. This method supports the combine of 1~n sub-FHGW_{7-2→8} uplinked data at the FHGW_{7-2→8}.

Corresponding to **Combine type 1**, the combine delay of Option8 field is reflected as a part of TBDelayUL. Corresponding to **Combine type 2**, the combine delay of option7-2x field is reflected as T_Comb.

Unsupported combines:

Data combine between Option8 southward port and Option7-2X southward port is not currently supported.

A4.7.2 Delay Parameters

Based on e-CPRI's definition of reference points, take the 2-level FHGW_{7-2→8} cascade as an example, and the delay reference points are shown in the figure below.

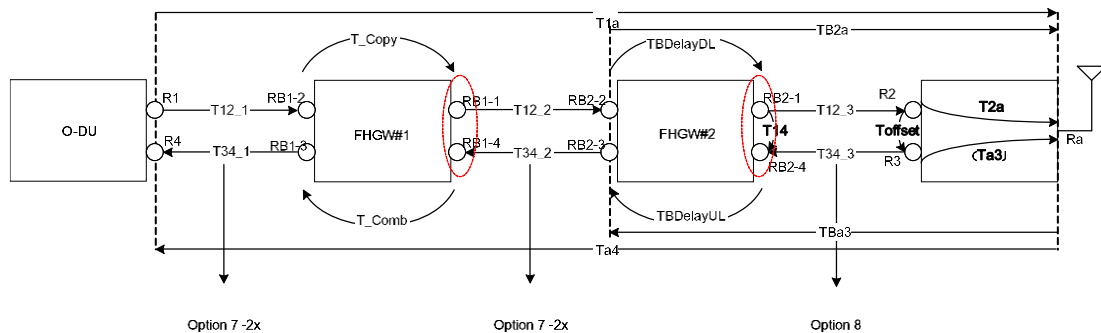


Figure 4 Definition I of Time delay reference point and delay parameter

N FHGW _{7-2→8} cascades	symbols of figure		Time delay parameter model		Specification
			DL	UL	
e-CPRI network transmission delay (For the NTH fronthaul connection, n=1...N)	T12_n T34_n		T12_n_min T12_n_max	T34_n_min T34_n_max	Configure parameters. O-DU parameter acquisition method: Method 1: Predefined Method 2: Delay measurement Notify O-RU ₈ via M-Plane
O-DU TX/RX window	T1a Ta4	C Plane	T1a_min_cp_dl T1a_max_cp_dl	T1a_min_cp_ul T1a_max_cp_ul	Intermediate values. According to the receive/send window of FHGW _{7-2→8} and network transmission delay calculation
		U Plane	T1a_min_up T1a_max_up	Ta4_min Ta4_max	
O-DU delay profile	NA		T1a_max_up _{O-DU} TXmax _{O-DU}	Ta4_max _{O-DU} RXmax _{O-DU}	The capability parameter of O-DU. O-DU can be told to O-RU ₈ via M-Plane
FHGW _{7-2→8} #1 Copy/combine processing delay on southward cascade port	T_Copy T_Comb		T_Copy	T_Comb	Capability parameters (not configurable). The FHGW _{7-2→8} is reported to the O-DU via the M-Plane
	NA		NA	Ta3_prime_max T_Waiting	Configure parameters. O-DU is configured to the FHGW _{7-2→8} via the M-Plane
FHGW _{7-2→8} #2 Processing delay on southward directly connected port FHGW _{7-2→8} delay profile	TBDelayDL TBDelayUL	C Plane	TBDelay_min_cp_dl TBDelay_max_cp_dl	TBDelay_min_cp_ul TBDelay_max_cp_ul	Capability parameters (not configurable). Set values for bandwidth, subcarrier intervals, and ports It may include: The IQ data received from the northward interface in the cache is extracted to the Low PHY processing unit; Low PHY treatment Make a copy of the data to send to the southward interface
		U Plane	TBDelay_min_up_dl TBDelay_max_up_dl	TBDelay_min_up_ul TBDelay_max_up_ul	
	NA		Tcp_adv_dl		Capability parameters (not configurable) The descending C-Plane is relative to the advance of the -U-Plane Make a copy of the data to send to the southward interface
FHGW _{7-2→8} #2+O-RU ₈ equivalent delay	TB2a TBa3	C Plane	TB2a_min_cp_dl TB2a_max_cp_dl	TB2a_min_cp_ul TB2a_max_cp_ul	Intermediate values.
		U Plane	TB2a_min_up TB2a_max_up	TBa3_min TBa3_max	

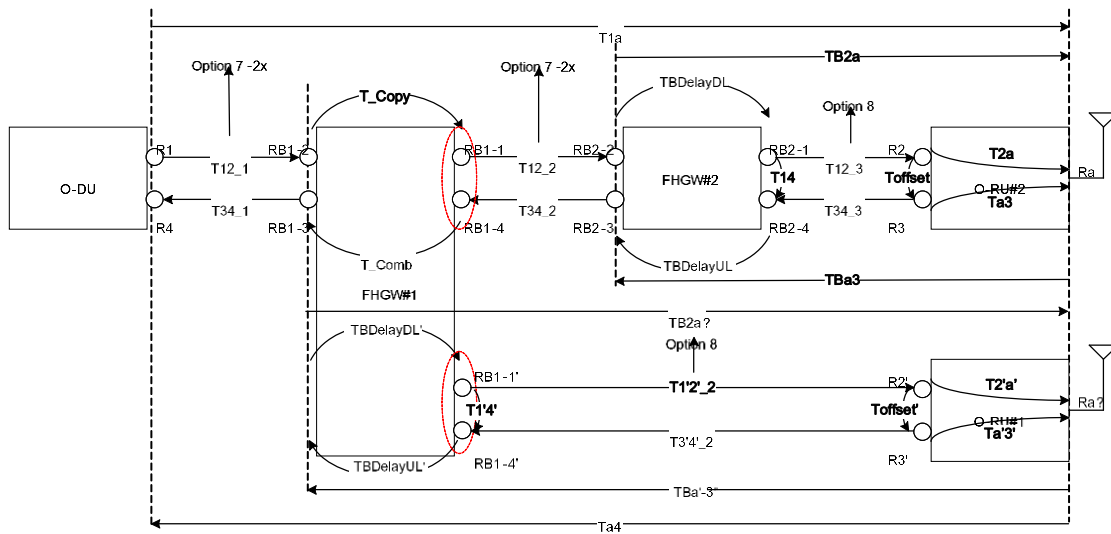


Figure 5 Definition II of delay reference point and delay parameter

The time of delay in Option 7-2 fronthaul changes within a certain range, because of transmission changes in the packet switching network, includes:

Table 1 Delay Parameters

A4.7.3 Relation of Delay Parameters

Table 2 Relation of delay parameters

	DL	UL
CPRI Optical fiber delay calculation	$T_{12_3} = T_{34_3} = (T_{14} - T_{offset})/2$	
e-CPRI MSG5 delay measurement	$t_D = (t_2 - t_{CV2}) - (t_1 + t_{CV1})$	
O-DU TX/RX window calculation	$T_{1a_max_up} \leq T_{B2a_max_up} + T_{12_min}$ $T_{1a_min_up} \geq T_{B2a_min_up} + T_{12_max}$ $T_{1a_max_cp_dl} \leq T_{B2a_max_cp_dl} + T_{12_min}$ $T_{1a_min_cp_dl} \geq T_{B2a_min_cp_dl} + T_{12_max}$	$T_{a4min} \leq T_{Ba3min} + T_{34min}$ $T_{a4max} \geq T_{Ba3max} + T_{34max}$ $T_{1a_min_cp_ul} \geq T_{B2a_min_cp_ul} + T_{12_max}$ $T_{1a_max_cp_ul} \leq T_{B2a_max_cp_ul} + T_{12_min}$
O-DU delay profile constraints	$T_{1a_max_up} - T_{1a_min_up} \geq TX_{maxO-DU}$ $T_{1a_max_upO-DU} \geq T_{1a_max_up}$	$RX_{maxO-DU} \geq T_{a4_max} - T_{a4_min}$ $T_{a4_maxO-DU} \geq T_{a4_max}$
Shared cell hypothesis	$T_{Copy} < T_{Comb}$	
Shared cell configuration parameter relationships		See section 3.14.4.4

A4.7.4 Timing Relations in Uplink and Downlink

The following examples is figure of timing relationships of the U-Plane data transmission.

A4.7.4.1 Timing Relations in Uplink

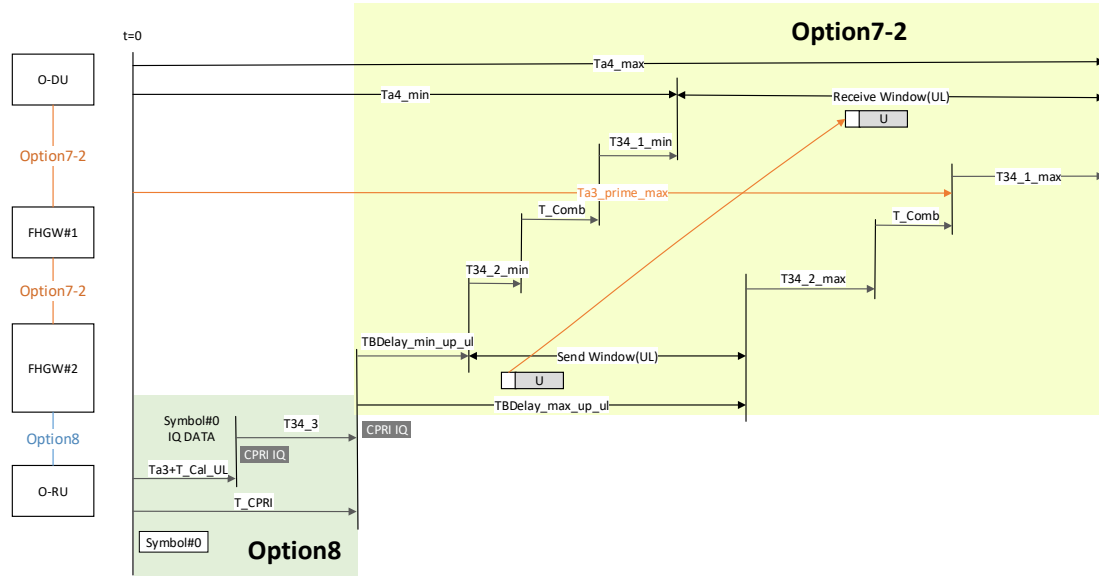


Figure 6 An example of uplink U-Plane (IQ) data timing

UL delay model parameters are shown in the figure above, and the conditions for the time series relationships to be established are as follows:

Condition 1: The introduction of a cascade of FHGW_{7-2→8}s does not change the total delay between O-DU and O-RU₈

Because the processing delay of uplink U-Plane message combining is added to the total prequel delay between O-DU and O-RU₈, the distance between O-DU and O-RU₈ must be reduced in order to maintain the same delay between O-DU and O-RU₈ compared with the cascade node, so as to ensure that O-DU receives UL U-Plane messages in the O-DU receiving window. In other words, the prequel configuration should satisfy:

$$T34_max \geq T34_1_max + T_Comb(\text{for FHGW}_{7-2 \rightarrow 8\#1}) + T34_2_max$$

Generalize to the case that there are N FHGW_{7-2→8} cascading in the fronthaul network,

$$\sum_{i=1}^N T34_i_max \leq T34_max - \sum_{i=1}^{N-1} T_Comb(\text{for O - Hub}\#i)$$

Among which:

- T34_n_{max} is the maximum transmission delay between FHGW_{7-2→8}#1 and O-DU (n=1), and the maximum transmission delay between FHGW_{7-2→8}#n and FHGW_{7-2→8}#n-1 (n=2..., N);

- T_comb is the processing delay of FHGW_{7-2→8} up-combine procedure, which depends on the capability of FHGW_{7-2→8} and is reported through M-Plane.
- T34_max is the maximum uplink transmission delay between O-DU and the final FHGW_{7-2→8}.

Condition 2: UL U-Plane data sent by The FHGW_{7-2→8} to the northward node for combining does not exceed Ta3_prime_max at the latest.

In order to ensure that O-DU receives UL U-Plane messages in the O-DU receiving window, the latest time that FHGW_{7-2→8#1} can send to UL U-Plane messages combined with O-DU is "Ta4_max-T34_1_max". Because T34_1_max depends on the location of FHGW_{7-2→8#1}, you need to tell FHGW_{7-2→8#1} the latest time to send. This time is defined as Ta3_prime_max, which needs to be configured to FHGW_{7-2→8#1} through M-Plane, and satisfies:

$$Ta3_prime_max \text{ (for FHGW}_{7-2 \rightarrow 8\#1}) \leq Ta4_max - T34_1_max$$

Generalize to the case that there are N FHGW_{7-2→8} cascaded in the fronthaul network,

$$Ta3_prime_max \text{ (Cascading FHGW}_{7-2 \rightarrow 8\#n})$$

$$\leq Ta4_max - \sum_{i=1}^n T34_i_max - \begin{cases} \sum_{i=1}^{n-1} T_Comb \text{ (for cascading FHGW}_{7-2 \rightarrow 8\#i}), & n \geq 2 \\ 0, & n = 1 \end{cases}$$

Condition 3: The FHGW_{7-2→8}'s wait time for combine processing should be greater than or equal to the time it receives all UL data

Consider T_Comb, receiving UL U-Plane messages before "Ta3_prime_max-T_comb" is limited to UL message combination. In other words, even if FHGW_{7-2→8#1} did not receive all UL U-Plane messages before "Ta3_prime_max-t_comb", FHGW_{7-2→8#1} should combine any UL messages received and send them to O-DU before Ta3_prime_max. This time "Ta3_prime_max (configuration) -T_comb (ability)" is defined as T_Waiting and is counted in FHGW_{7-2→8#1}. In order to combine all UL U-Plane messages, the configured Ta3_prime_max should meet the following criteria:

$$T_Waiting = Ta3_prime_max \text{ (for FHGW}_{7-2 \rightarrow 8\#1}) - t_comb \text{ (for FHGW}_{7-2 \rightarrow 8\#1}) \geq TBa3_max + T_34_2_max$$

Generalize to the case that there are N FHGW_{7-2→8} cascading in the fronthaul network,

$$TBa3_max + \sum_{i=n+1}^N T34_i_max + \sum_{i=n}^{N-1} T_Comb \text{ (for FHGW}_{7-2 \rightarrow 8\#i}) \leq Ta3_prime_max \text{ (for FHGW}_{7-2 \rightarrow 8\#n})$$

A4.7.4.2 Timing Relations in Downlink

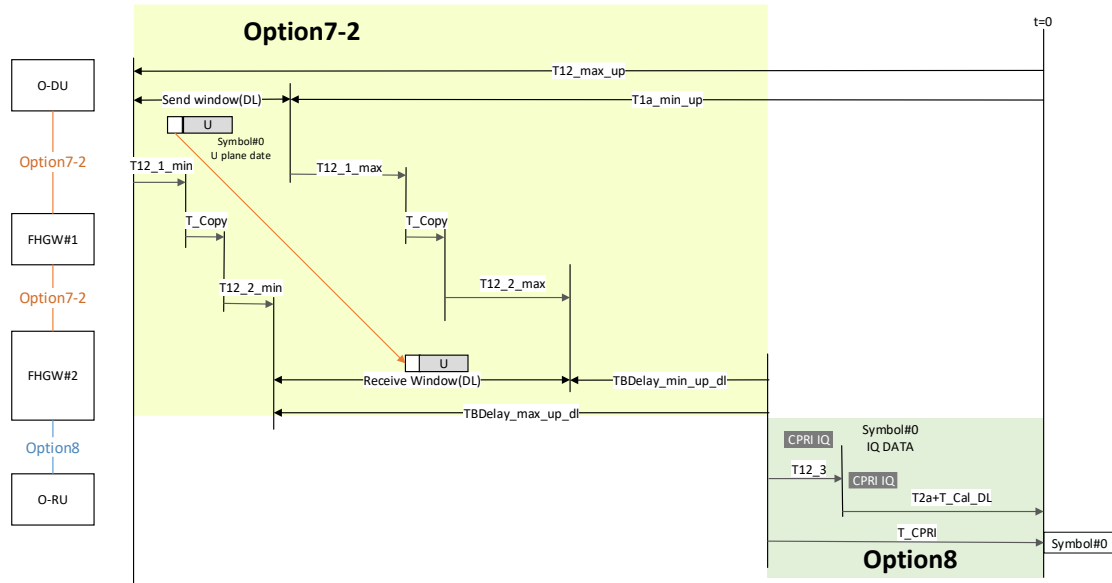


Figure 7 Example of timing sequence of downlink U-Plane (IQ) data

7. The DL temporal relationship as shown in the figure above is valid under the following conditions:

8. Condition 1: The maximum transmission delay of uplink and downlink is equal, namely,

$$9. T_{12_max} = T_{34_max}$$

$$10. T_{12_i_max} = T_{34_i_max} \ (i=1 \sim N)$$

11. Condition 2: $T_{Comb} > T_{Copy}$, the uplink combine delay is greater than the downlink copy delay

Let's say that the UL combine is larger than the DL copy delay. Therefore, as long as the existing network and the group of the FHGW_{7-2→8}#1 configuration meets the UL delay management requirements, the DL delay management of the shared cell does not require additional requirements, which is that the downlink data can be received in the receiving window of the FHGW_{7-2→8}#2. Although DL does not need additional requirements, the O-DU knows that T_{Copy} is beneficial to the optimization of the O-DU fan window. Therefore, T_{Copy} is defined as the ability parameter of the FHGW_{7-2→8}.

A4.7.5 Delay Management Procedure

The procedure of delay management is shown in the figure below.



2

3

4

6

7

- 14

- 2) Increase the delay margin corresponding to x KM fiber based on the measurement and calculation of the first O-RU₈ found in topology;
- 3) Select the maximum delay range after finding all O-RU₈;
- 4) The delay is adjusted when O-RU₈ is added and not adjusted when O-RU₈ is withdrawn,

Uplink direction:

The principle is similar to the downlink case, except that the T_Cal_UL setting only involves O-RU₈.

A4.7.7 YANG Model File

o-ran-delay-management-hub.yang

o-ran-shared-cell.yang

A4.8 FHGW_{7-2→8} Shared Cell Configuration

A4.8.1 General Description

A4.8.2 YANG Model File

The FHGW_{7-2→8} in the architecture of e-CPRI Option7-2 to CPRI Option8 needs to support two type of shared cell configurations at the same time:

- FHGW_{7-2→8} directly connected to O-RU₈ shared cell configuration
- FHGW_{7-2→8} cascade FHGW_{7-2→8} shared cell configuration

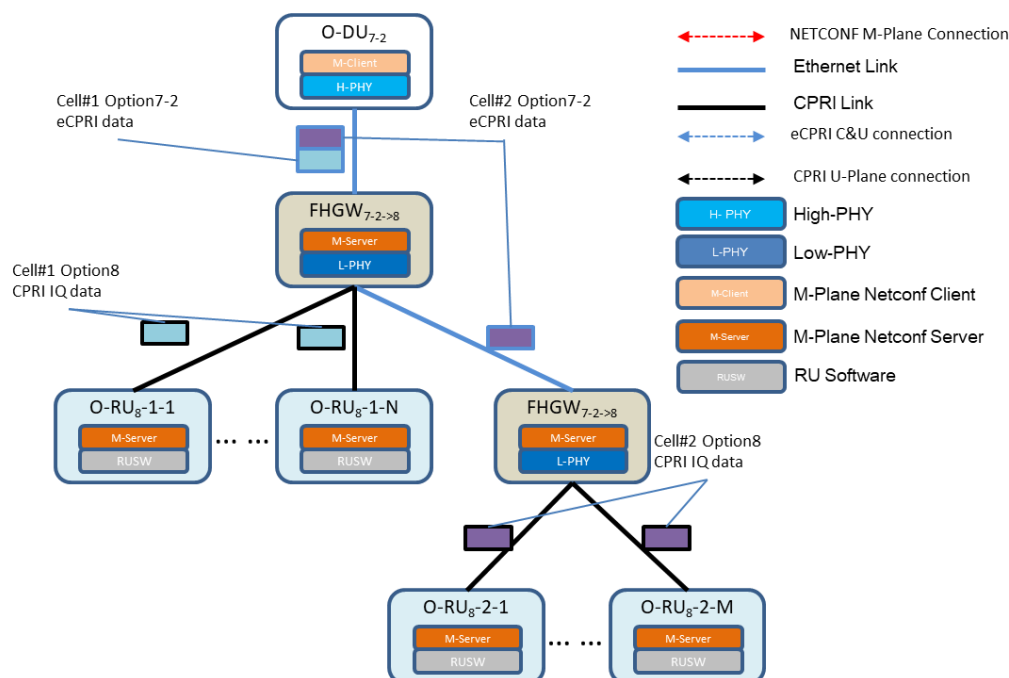


Figure 9 Example of shared cell data flow

A4.8.2.1 Overview of FHGW_{7-2→8} Connect to O-RU₈

In this specification, O-RU₈ under the same FHGW_{7-2→8} can be configured for cell combination. In this configuration, the FHGW_{7-2→8} shared cell information model consists of the following parts:

- ✓ northward e-CPRI U-Plane interface
- ✓ U-Plane configuration, configure the Low PHY
- ✓ copy and combine FHGW_{7-2→8} direct O-RU₈, is used to configure radio unit via CPRI interface connection, and the radio unit combination constitutes the first share of data copy and combine the data model
- ✓ southward CPRI U-Plane interface

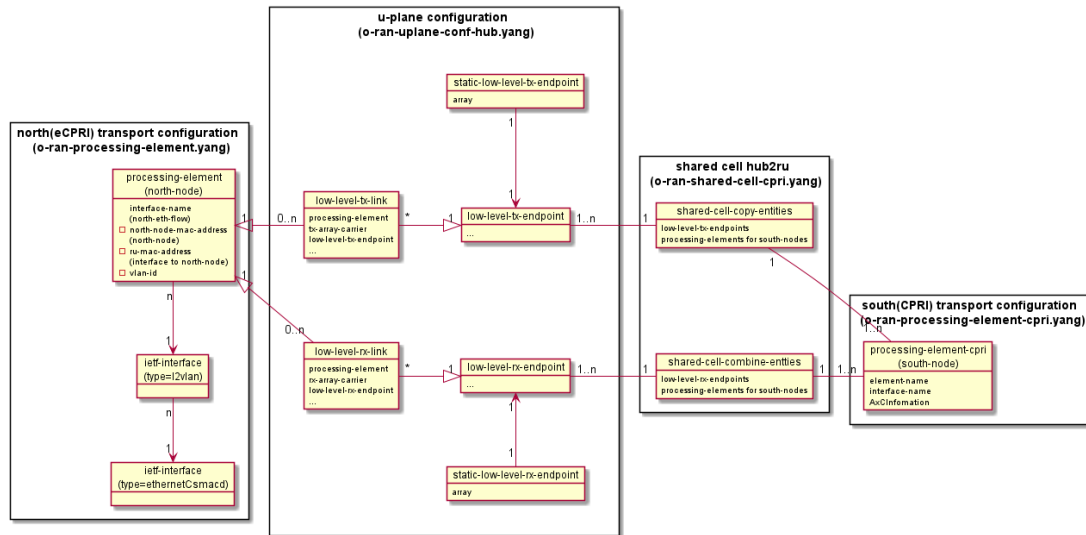


Figure 10 FHGW_{7-2→8} directly connecting O-RU₈ Information

A4.8.2.2 Overview of FHGW_{7-2→8} Cascade to FHGW_{7-2→8}

In this specification, FHGW_{7-2→8} cascading is supported. O-RU₈ under cascading FHGW_{7-2→8}s and directly connected O-RU₈ at the same level are not required to be configured for cell combining. In this configuration, the shared cell information model includes the following parts:

- ✓ northward e-CPRI U-Plane interface
- ✓ copy and combine of FHGW_{7-2→8} cascade to FHGW_{7-2→8}
- ✓ southward e-CPRI U-Plane interface

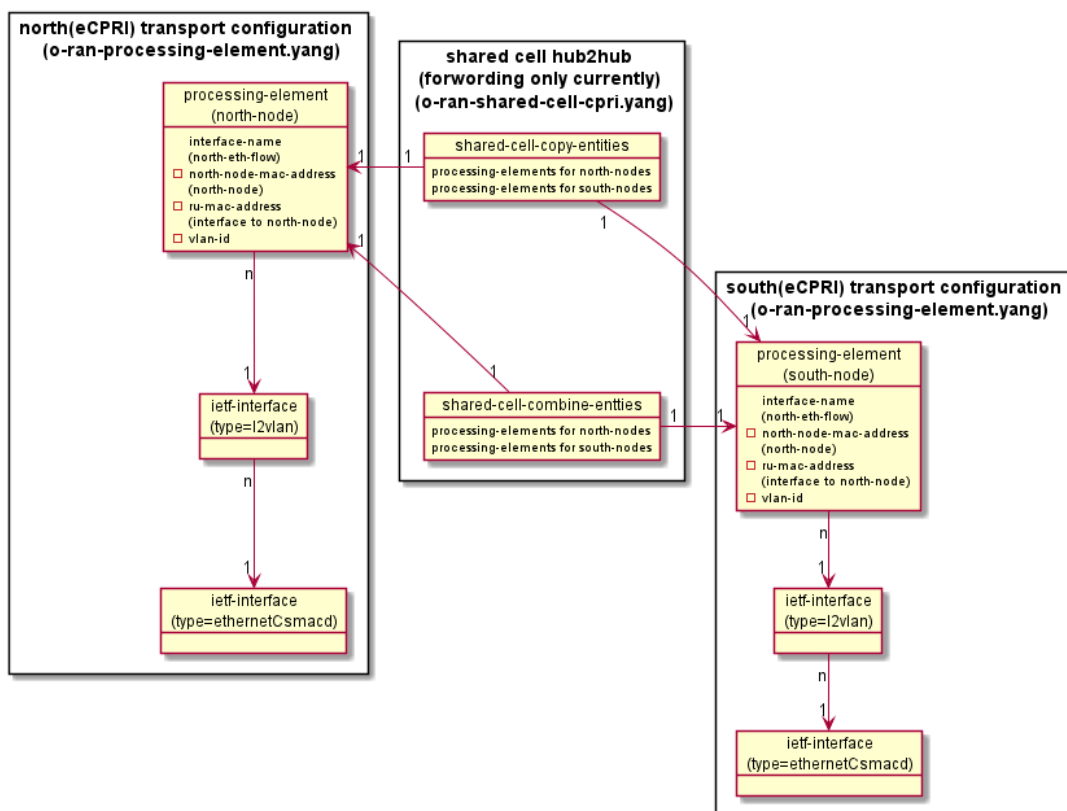


Figure 11 FHGW_{7-2→8} cascade to FHGW_{7-2→8} Information

A4.8.2.3 YANG Model of Northward e-CPRI U-Plane Interface

The FHGW_{7-2→8} northward U-Plane interface YANG model is based on Ethernet and e-CPRI, refer to Chapter 4 of O-RAN.WG4.MP.0-v03.00

A4.8.2.4 YANG Model of Southward e-CPRI U-Plane Interface

The FHGW_{7-2→8} southward cascading FHGW_{7-2→8} U-Plane interface YANG model is based on Ethernet and e-CPRI, refer to Chapter 4 of O-RAN.WG4.MP.0-v03.00

A4.8.2.5 YANG Model of Southward CPRI U-Plane Interface

The FHGW_{7-2→8} southward CPRI U-Plane interface YANG model is based on CPRI, see 3.3.2 for details

A4.8.2.6 YANG Model of U-Plane Configuration

The U-Plane configuration of the FHGW_{7-2→8} contains two parts.

One is Low PHY U-Plane level configuration. In the architecture of this specification, the Low PHY is deployed in the FHGW_{7-2→8} and the High PHY is deployed in the O-DU, so the U-Plane configuration of the Low PHY is required. Please see the details in 15.2.2.3.

Another is shared cell U-Plane configuration. This part similar to the U-Plane configuration in the shared cell section of the O-RAN specification, which is mainly contains the configuration of U-Plane parameters such as eaxc-id related to downlink copy and uplink combine.

To avoid misunderstanding, if not specifically mentioned, U-Plane configuration means Low PHY U-Plane configuration.

A4.8.2.7 YANG Model of Shared Cell Configuration

Two type of shared cell configurations:

- Shared cell configuration of FHGW_{7-2→8} connect to O-RU₈, add a new YANG model file:o-ran-shared-cell-ecpri7-to-cpri8.yang
- Shared cell configuration of FHGW_{7-2→8} cascade FHGW_{7-2→8}, reuse the existing YANG mode file:o-ran-shared-cell.yang

Each type of shared cell contains three parts: the parameters of shared cell capability, the parameters of shared cell downlink copy and the parameters of shared cell uplink combine. The following tables compare the main relations and differences between above two type of shared cell configuration with the option7-2 to Option7-2 FHM shared cell configuration in O-RAN specification.

The parameters of shared cell capability, shared-cell-module-cap:

Table 3 Comparison of parameters of shared cell capability

YANG model of shared cell in ORAN-WG4.MP.0-v03.00	FHGW _{7-2→8} cascade FHGW _{7-2→8} in this Annex	FHGW _{7-2→8} connect to O-RU ₈ in this chapter
t-copy	reuse	Not needed. Reuse the TBDelay-*** in o-ran-delay-management.yang
t-combine	reuse	Not needed. Reuse the TBDelay-*** in o-ran-delay-management.yang
ta3-prime-max-upper-range	reuse	Not needed. The delay of CPRI fronthaul is consistent.
max-number-node-copy-and-combine	reuse	reuse
max-number-eaxcid-copy	reuse	reuse
max-number-eaxcid-combine	reuse	reuse
compression-method-supported	reuse	reuse

Share cell downlink copy parameters, shared-cell-copy-entities:

Table 4 Comparison of parameters of shared cell downlink copy

YANG model of shared cell in ORAN-WG4.MP.0-v03.00	FHGW _{7-2→8} cascade FHGW _{7-2→8} in this Annex	FHGW _{7-2→8} connect to O-RU ₈ in this chapter
---	---	--

	Annex	
north-node-processing-element	reuse	low-level-tx-endpoint
south-node-processing-element	reuse, note 1	south-node-processing-element-cpri
tx-eaxc-id	reuse	Necessary but no need to define, refer to the definition in o-ran-uplane-conf-hub.yang
rx-eaxc-id	reuse	No
downlink-radio-frame-offset	reuse	Necessary but no need to define, refer to the definition in o-ran-uplane-conf-hub.yang
downlink-sfn-offset	reuse	Necessary but no need to define, refer to the definition in o-ran-uplane-conf-hub.yang

Note 1:

In the case that inter-stage RF combine is not supported, in the shared-cell-copy-entity of each copy rule, a northward node processes the element north-node-processing-element, and a southward node processes the element south-node-processing-element.

Shared cell uplink combines parameters, shared-cell-combine-entities:

Table 5 Comparison of parameters of shared cell uplink combine

YANG model of shared cell in ORAN-WG4.MP.0-v03.00	FHGW _{7-2→8} cascade FHGW _{7-2→8} in this Annex	FHGW _{7-2→8} connect to O-RU ₈ in this chapter
north-node-processing-element	reuse	low-level-rx-endpoint
south-node-processing-element	reuse	south-node-processing-element-cpri
rx-eaxc-id	reuse	o-ran-uplane-conf-hub.yang
compression-format	reuse	reuse
ta3-prime-max	reuse	No
downlink-radio-frame-offset	reuse	o-ran-uplane-conf-hub.yang
downlink-sfn-offset	reuse	o-ran-uplane-conf-hub.yang
n-ta-offset	reuse	o-ran-uplane-conf-hub.yang
number-of-prb	reuse	No

A4.8.3 Details of Operation

This section describes the operation details of the shared cell.

A4.8.3.1 Topology Discovery

O-DU can obtain the O-RU₈ topological relation (adjacency relation) of a shared cell. The shared cell topology discovery procedure is filled in to the ethernet forwarding Information Base (FIB) using the LBM/LBR connection check procedure running at the Ethernet layer as defined in Section 3.3.1.5 Connectivity Detection, and FIB is obtained based on O-DU. The difference is that, since the FHGW_{7-2→8} southward is based on CPRI IQ link, only the Ethernet network-based M-Plane has MAC address, therefore, the LBM/LBR procedure in the 3.3.1.5 Connectivity detection

section, the FHGW_{7-2→8} southward is based on M-Plane. Transparent bridge functionality is used for O-RU₈ with shared cell functionality running in FHGW_{7-2→8} and/or cascading mode.

O-DU determines the topology by the following two-step procedure:

- 1) In the first phase, the procedure of using the 3.3.1.5 connectivity detection section is used to execute the Ethernet connection monitoring for all the discovered O-RU₈ MAC addresses. The echo of the Ethernet loopback from the found MAC address should ensure that the transparent bridge in the FHGW_{7-2→8} and the cascading mode is automatically learning to pass the MAC addresses of these devices

Note: In addition to Ethernet connection monitoring, DHCP discovery, Call Home and M-Plane connection establishment can also be used to fill in information in FIB, so maybe O-RU₈ has a northward port to do the above.

- 2) In the second phase, O-DU uses the o-ran-ethernet-forwarding.yang model to discovery which MAC addresses have been learned by the ethernet bridge function. O-DU uses each Ethernet forwarding entry to determine adjacency relationships

The following figure shows an example of the MAC address configuration of the FHGW_{7-2→8} mode and the FHGW_{7-2→8} operation pattern configuration, and the relevant transparent bridge fib table.

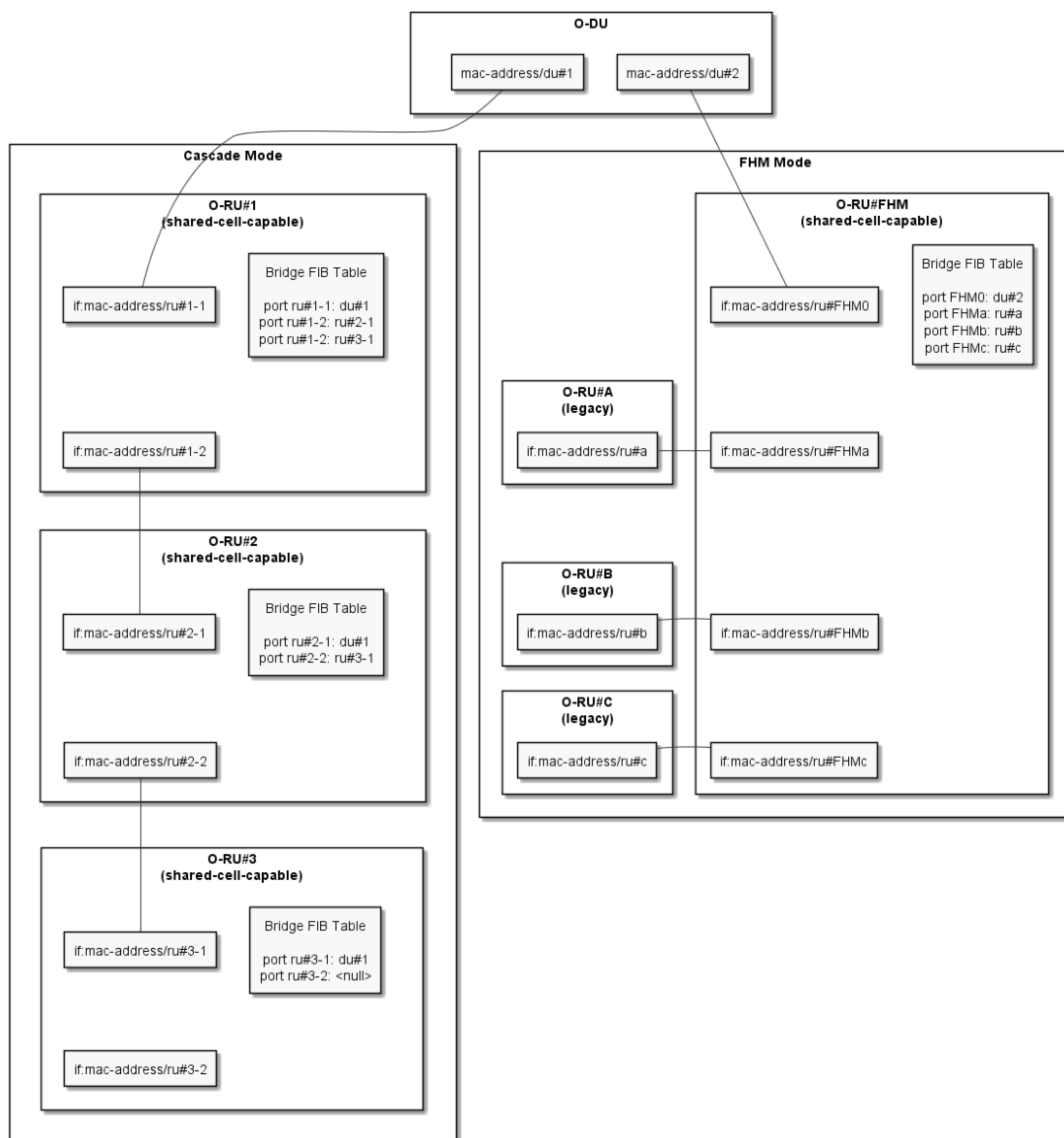


Figure 12 topology discovery

A4.8.3.2 Configuration of Shared Cell

The FHGW_{7-2→8} needs to support two type of shared cell configurations. So does the downlink copy configuration.

- Shared cell downlink configuration of FHGW_{7-2→8} connect to O-RU₈
- Shared cell downlink configuration of FHGW_{7-2→8} cascade to FHGW_{7-2→8}

1 Examples are shown in the figure below

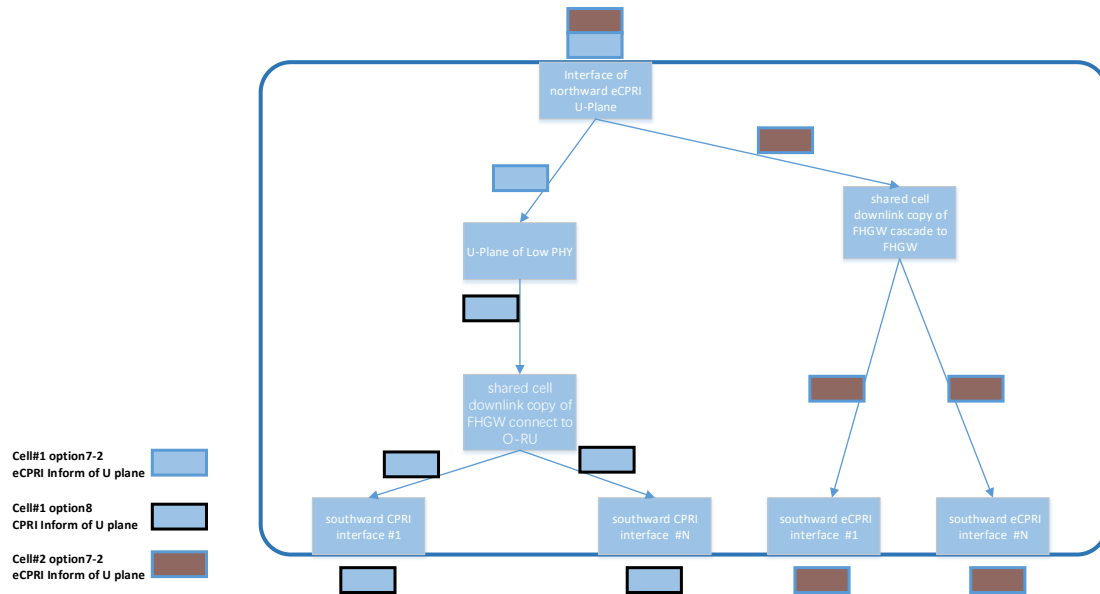


Figure 13 Shared cell downlink copy

5 The FHGW_{7-2→8} needs to support two type of shared cell configurations. So does the uplink combine configuration.

- Shared cell uplink combine configuration of FHGW_{7-2→8} connect to O-RU₈
- Shared cell uplink combine configuration of FHGW_{7-2→8} cascade to FHGW_{7-2→8}

8 Examples are shown in the figure below

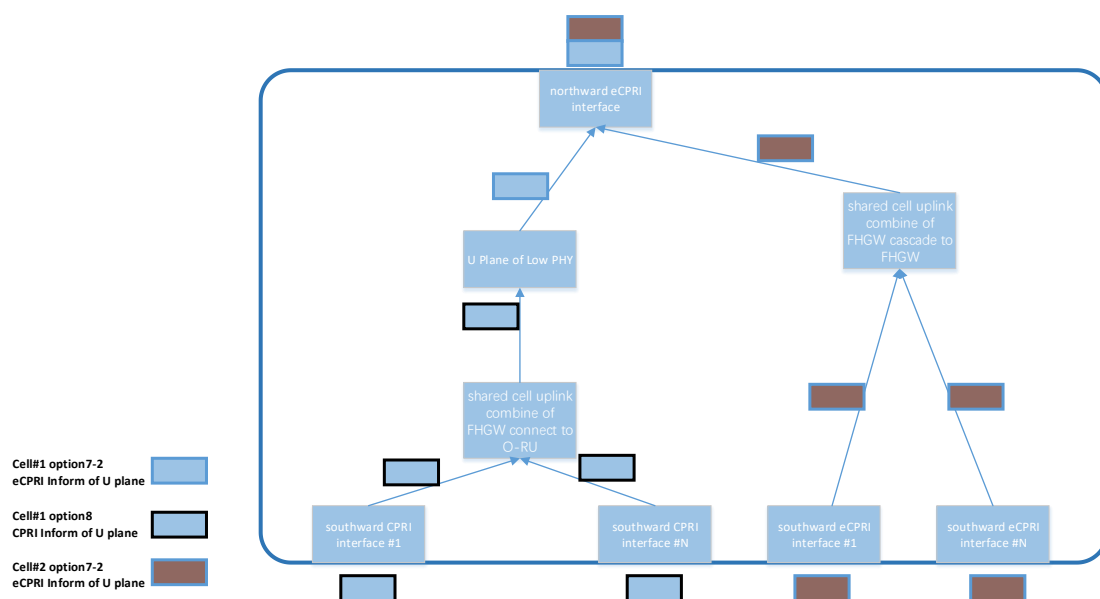


Figure 14 Shared cell uplink combine

A4.8.3.3 Configuration of FHGW_{7-2→8} Connect to O-RU₈

For the configuration of FHGW_{7-2→8} connect to O-RU₈, add a new YANG model file, o-ran-shared-cell-ecpri7-to-cpri8.yang, and eCPRI7-to-CPRI8 mode was added for the shared-cell-copy-combine mode in o-ran-shared-cell.yang, which is used for the configuration of FHGW_{7-2→8} connect to O-RU₈.

- COMMON
- SELECTIVE
- eCPRI7-to-CPRI8

The function of downlink copy of shared-cell-copy-entities-ecpri7-to-cpri8 is configured in two parts.

1) low-level-tx-endpoint

This parameter identifies the data source of the downlink copy of the shared cell, points to the Tx endpoint in the Low PHY U-Plane configuration.

2) south-node-processing-elements-cpri

This parameter identifies the data destination of downlink copy of the shared cell, points to the CPRI flow, which is defined in the CPRI interface YANG model.

The function of uplink combination of shared-cell-combine-entities-option8-cpri is configured in two parts

1) low-level-rx-endpoint

This parameter identifies the data destination of the shared cell uplink combine and points to the Rx endpoint in the Low PHY U-Plane configuration.

2) south-node-processing-elements-cpri

This parameter identifies the data source of the downlink copy data of the shared cell and points to the CPRI-Flow defined in the FHGW_{7-2→8} southward CPRI interface model.

Note: In contrast to the shared cell in O-RAN specification, the downlink copy and uplink combine configuration do not define the shared cell U-Plane configuration part. This is because all the U-Plane configuration information (such as eaxc-id, downlink-radio-frame-offset, downlink-sfn-offset, and N-TA-Offset of uplink only parameter) is fully defined in the low-level-tx-endpoint of Low PHY U-Plane configuration, and the shared cell part does not need to define it again.

A4.8.3.4 Configuration of FHGW_{7-2→8} Cascade to FHGW_{7-2→8}

For the configuration of **FHGW_{7-2→8} cascade to FHGW_{7-2→8}**, continue to reuse the COMMON mode of YANG model of shared-cell-copy-combine in the o-ran-shared-cell.yang. Which consistent with the configuration method of FHM shared cell in the O-RAN specification

A4.9 FHGW_{7-2→8} Startup Example

Table 6 FHGW_{7-2→8} Startup Procedure

FHGW _{7-2→8} Initialization Procedure	Description	Referred Sections
Initialization of M-Plane network layer connection	Initialization of M-Plane network layer connection	Chapter 3.1 of [4]
Clock synchronization	Clock synchronization The M-Plane connection has not been fully established at this stage, and clock synchronization should avoid to depend on configurable parameters, or should use default parameter.	Chapter 10 of [4]
M-Plane Call Home	M-Plane Call Home	Chapter 3.2 of [4]
Establishment of M-plane SSH connection	Establishment of management plane SSH connection	Chapter 3.3 of [4]
Retrieve NETCONF capability set of M-plane	Retrieve NETCONF capability set of M- plane	Chapter 3.5 of [4]
Keeping alive for M-Plane NETCONF connection	Keeping alive for M-Plane NETCONF connection	Chapter 3.6 of [4]
Retrieve the FHGW _{7-2→8} parameters	Retrieve the FHGW _{7-2→8} parameters	Chapter 6 of [4]
Software management	Software management	Chapter 5 of [4]
clock synchronization	clock synchronization	Chapter 10 of [4]
CPRI interface initialization in FHGW _{7-2→8}	CPRI interface initialization in FHGW _{7-2→8}	A2.5
Keeping alive for U-Plane connection	Keeping alive for U-Plane connection in FHGW _{7-2→8}	Chapter 4.10 of [4]
Shared cell and U-Plane configuration	Topology discovery of FHGW _{7-2→8} Read Delay Profile of FHGW _{7-2→8} Northward e-CPRI interface configuration of FHGW _{7-2→8} U-Plane configuration of FHGW _{7-2→8} Shared cell configuration of FHGW _{7-2→8} connect to O-RU ₈ Shared Cell (optional) of FHGW _{7-2→8} cascade to FHGW _{7-2→8} Southward e-CPRI interface configuration of FHGW _{7-2→8} (optional) Southward CPRI interface configuration of FHGW _{7-2→8}	A2.8
Fault management initialization	Fault management initialization	Chapter 8 of [4]
Read status information of the FHGW _{7-2→8}	Read status information of the FHGW _{7-2→8} .	Chapter 6 of [4]
Configure runtime parameters of the FHGW _{7-2→8}	Configure runtime parameters of the FHGW _{7-2→8}	Chapter 6 of [4]

A4.10Appendix - YANG Model Tree Diagram

1 This appendix lists YANG model tree diagrams that are newly defined or modified by this
2 specification; refer to the O-RAN specification for other YANG models.

3 A4.10.1 o-ran-uplane-conf-hub Module

```

4 module: o-ran-uplane-conf-hub
5
6   +--rw eaxc-id-group-configuration {mcap:EAXC-ID-GROUP-SUPPORTED}?
7
8   | +--rw max-num-tx-eaxc-id-groups?    -> /mcap:module-capability/ru-capabilities/eaxcid-grouping-capabilities/max-num-tx-eaxc-id-groups
9
10  | +--rw max-num-tx-eaxc-ids-per-group? -> /mcap:module-capability/ru-capabilities/eaxcid-grouping-capabilities/max-num-tx-eaxc-ids-per-group
11
12  | +--rw max-num-rx-eaxc-id-groups?    -> /mcap:module-capability/ru-capabilities/eaxcid-grouping-capabilities/max-num-rx-eaxc-id-groups
13
14  | +--rw max-num-rx-eaxc-ids-per-group? -> /mcap:module-capability/ru-capabilities/eaxcid-grouping-capabilities/max-num-rx-eaxc-ids-per-group
15
16  | +--rw tx-eaxc-id-group* [representative-tx-eaxc-id]
17
18  | | +--rw representative-tx-eaxc-id  uint16
19
20  | | +--rw member-tx-eaxc-id*       uint16
21
22  | +--rw rx-eaxc-id-group* [representative-rx-eaxc-id]
23
24  | | +--rw representative-rx-eaxc-id  uint16
25
26  | | +--rw member-rx-eaxc-id*       uint16
27
28  +--rw user-plane-configuration
29
30  +--rw low-level-tx-links* [name]
31
32  | +--rw name                string
33
34  | +--rw processing-element   -> /o-ran-pe:processing-elements/ru-elements/name
35
36  | +--rw low-level-tx-endpoint -> /user-plane-configuration/low-level-tx-endpoints/name
37
38  +--rw low-level-rx-links* [name]
39
40  | +--rw name                string
41
42  | +--rw processing-element   -> /o-ran-pe:processing-elements/ru-elements/name
43
44  | +--rw low-level-rx-endpoint -> /user-plane-configuration/low-level-rx-endpoints/name
45
46  | +--rw user-plane-uplink-marking? -> /o-ran-pe:processing-elements/enhanced-uplane-mapping/uplane-mapping/up-marking-name
47
48  +--ro endpoint-types* [id]
49
50  | +--ro id                  uint16
51
52  | +--ro supported-section-types* [section-type]
53
54  | | +--ro section-type      uint8

```

```

1 | | +--ro supported-section-extensions* uint8
2 | | +--ro supported-frame-structures*      uint8
3 | | +--ro managed-delay-support?           enumeration
4 | | +--ro multiple-numerology-supported?   boolean
5 | | +--ro max-numerology-change-duration?  uint16
6 | | +--ro max-control-sections-per-data-section? uint8
7 | | +--ro max-sections-per-symbol?         uint16
8 | | +--ro max-sections-per-slot?           uint16
9 | | +--ro max-remasks-per-section-id?      uint8
10 | | +--ro max-beams-per-symbol?           uint16
11 | | +--ro max-beams-per-slot?             uint16
12 | | +--ro max-prb-per-symbol?             uint16
13 | | +--ro prb-capacity-allocation-granularity* uint16
14 | | +--ro max-numerologies-per-symbol?    uint16
15 | +--ro endpoint-capacity-sharing-groups* [id]
16 | | +--ro id                             uint16
17 | | +--ro max-control-sections-per-data-section? uint8
18 | | +--ro max-sections-per-symbol?         uint16
19 | | +--ro max-sections-per-slot?           uint16
20 | | +--ro max-remasks-per-section-id?      uint8
21 | | +--ro max-beams-per-symbol?           uint16
22 | | +--ro max-beams-per-slot?             uint16
23 | | +--ro max-prb-per-symbol?             uint16
24 | | +--ro max-numerologies-per-symbol?    uint16
25 | | +--ro max-endpoints?                  uint16
26 | | +--ro max-managed-delay-endpoints?    uint16
27 | | +--ro max-non-managed-delay-endpoints? uint16
28 | +--ro endpoint-prach-group* [id]
29 | | +--ro id                             uint16

```

```

1 | +-ro supported-prach-preamble-formats* prach-preamble-format
2
3 | +-ro static-low-level-tx-endpoints* [name]
4
5 | +-ro name string
6
7 | +-ro restricted-interfaces* -> /if:interfaces/interface/name
8
9 | +-ro endpoint-type? -> ../endpoint-types/id
10
11 | +-ro capacity-sharing-groups* -> ../endpoint-capacity-sharing-groups/id
12
13 +-ro static-low-level-rx-endpoints* [name]
14
15 | +-ro name string
16
17 | +-ro restricted-interfaces* -> /if:interfaces/interface/name
18
19 | +-ro endpoint-type? -> ../endpoint-types/id
20
21 | +-ro capacity-sharing-groups* -> ../endpoint-capacity-sharing-groups/id
22
23 | +-ro prach-group? -> ../endpoint-prach-group/id
24
25 +-rw low-level-tx-endpoints* [name]
26
27 | +-rw name -> /user-plane-configuration/static-low-level-tx-endpoints/name
28
29 | +-rw compression!
30
31 | | +-rw iq-bitwidth? uint8
32
33 | | +-rw compression-type enumeration
34
35 | | x--rw bitwidth? uint8
36
37 | | +-rw (compression-format)?
38
39 | | +--:(no-compresison)
40
41 | | +--:(block-floating-point)
42
43 | | | +-rw exponent? uint8
44
45 | | +--:(block-floating-point-selective-re-sending)
46
47 | | | +-rw sres-exponent? uint8
48
49 | | +--:(block-scaling)
50
51 | | | +-rw block-scalar? uint8
52
53 | | +--:(u-law)
54
55 | | | +-rw comp-bit-width? uint8
56
57 | | | +-rw comp-shift? uint8

```

```

1  | |  +--:(beam-space-compression)
2
3  | |  |  +--rw active-beam-space-coefficient-mask*  uint8
4
5  | |  |  +--rw block-scaler?          uint8
6
7  | |  +--:(modulation-compression)
8
9  | |  |  +--rw csf?          uint8
10
11 | |  |  +--rw mod-comp-scaler?      uint16
12
13 | |  +--:(modulation-compression-selective-re-sending)
14
15 | |  +--rw sres-csf?          uint8
16
17 | |  +--rw sres-mod-comp-scaler?    uint16
18
19 |  +--rw frame-structure?          uint8
20
21 |  +--rw cp-type?                  enumeration
22
23 |  +--rw cp-length                 uint16
24
25 |  +--rw cp-length-other           uint16
26
27 |  +--rw offset-to-absolute-frequency-center  int32
28
29 |  +--rw number-of-prb-per-scs* [scs]
30
31 | |  +--rw scs          mcap:scs-config-type
32
33 | |  +--rw number-of-prb  uint16
34
35 |  +--rw e-axcid
36
37 | |  +--rw o-du-port-bitmask  uint16
38
39 | |  +--rw band-sector-bitmask  uint16
40
41 | |  +--rw ccid-bitmask        uint16
42
43 | |  +--rw ru-port-bitmask     uint16
44
45 | |  +--rw eaxc-id            uint16
46
47 |  +--rw downlink-radio-frame-offset  uint32
48
49 |  +--rw downlink-sfn-offset          int16
50
51 +--rw low-level-rx-endpoints* [name]
52
53 |  +--rw name          -> /user-plane-configuration/static-low-level-rx-endpoints/name
54
55 |  +--rw compression
56
57 | |  +--rw iq-bitwidth?          uint8

```

1	+--rw compression-type	enumeration
2	x--rw bitwidth?	uint8
3	+--rw (compression-format)?	
4	+--:(no-compresison)	
5	+--:(block-floating-point)	
6	+--rw exponent?	uint8
7	+--:(block-floating-point-selective-re-sending)	
8	+--rw sres-exponent?	uint8
9	+--:(block-scaling)	
10	+--rw block-scalar?	uint8
11	+--:(u-law)	
12	+--rw comp-bit-width?	uint8
13	+--rw comp-shift?	uint8
14	+--:(beam-space-compression)	
15	+--rw active-beam-space-coeficient-mask*	uint8
16	+--rw block-scaler?	uint8
17	+--:(modulation-compression)	
18	+--rw csf?	uint8
19	+--rw mod-comp-scaler?	uint16
20	+--:(modulation-compression-selective-re-sending)	
21	+--rw sres-csf?	uint8
22	+--rw sres-mod-comp-scaler?	uint16
23	+--rw frame-structure?	uint8
24	+--rw cp-type?	enumeration
25	+--rw cp-length	uint16
26	+--rw cp-length-other	uint16
27	+--rw offset-to-absolute-frequency-center	int32
28	+--rw number-of-prb-per-scs* [scs]	
29	+--rw scs	mcap:scs-config-type

```

| | +--rw number-of-prb    uint16

| | +--rw ul-fft-sampling-offsets* [scs]

| | +--rw scs                mcap:scs-config-type

| | +--rw ul-fft-sampling-offset?  uint16

| +--rw e-axcid

| | +--rw o-du-port-bitmask    uint16

| | +--rw band-sector-bitmask  uint16

| | +--rw ccid-bitmask        uint16

| | +--rw ru-port-bitmask     uint16

| | +--rw eaxc-id             uint16

| +--rw non-time-managed-delay-enabled?  boolean

| +--rw downlink-radio-frame-offset      uint32

| +--rw downlink-sfn-offset              int16

| +--rw n-ta-offset                      uint32

+--rw general-config

+--rw regularization-factor-se-configured?  boolean

+--rw little-endian-byte-order?            boolean

```

A4.10.2 o-ran-shared-cell-ecpri7-to-cpri8 Module

```

module: o-ran-shared-cell-ecpri7-to-cpri8

+--rw shared-cell-cpri7-to-cpri8

+--ro shared-cell-module-cap-ecpri7-to-cpri8

| +--ro max-number-node-copy-and-combine  uint8

| +--ro max-number-eaxcid-copy            uint8

| +--ro max-number-eaxcid-combine         uint8

| +--ro compression-method-supported* [] {FHM-eCPRI7-to-CPRI8}?

+--rw shared-cell-config

+--rw (shared-cell-copy-combine-mode)?

+--:(COMMON)

```

```

1      | +--rw shared-cell-copy-entities* [name]
2
3      | | +--rw name                string
4
5      | | +--rw north-node-processing-element? -> /o-ran-pe:processing-elements/ru-elements/name
6
7      | | +--rw south-node-processing-elements* -> /o-ran-pe:processing-elements/ru-elements/name
8
9      | | +--rw shared-cell-copy-uplane-config {FHM}?
10
11     | |   +--rw tx-eaxc-id* [eaxc-id]
12
13     | |   | +--rw eaxc-id  uint16
14
15     | |   +--rw rx-eaxc-id* [eaxc-id]
16
17     | |   | +--rw eaxc-id  uint16
18
19     | |   +--rw downlink-radio-frame-offset  uint32
20
21     | |   +--rw downlink-sfn-offset          int16
22
23     | +--rw shared-cell-combine-entities* [name]
24
25     |   +--rw name                string
26
27     |   +--rw north-node-processing-element? -> /o-ran-pe:processing-elements/ru-elements/name
28
29     |   +--rw south-node-processing-elements* -> /o-ran-pe:processing-elements/ru-elements/name
30
31     |   +--rw ta3-prime-max?        uint32
32
33     |   +--rw shared-cell-combine-uplane-config {FHM}?
34
35     |   +--rw rx-eaxc-id* [eaxc-id]
36
37     |   | +--rw eaxc-id      uint16
38
39     |   | +--rw compression-method
40
41     |   |   +--rw iq-bitwidth?      uint8
42
43     |   |   +--rw compression-type  enumeration
44
45     |   |   x--rw bitwidth?         uint8
46
47     |   |   +--rw (compression-format)?
48
49     |   |   +--:(no-compression)
50
51     |   |   +--:(block-floating-point)
52
53     |   |   | +--rw exponent?      uint8
54
55     |   |   +--:(block-floating-point-selective-re-sending)
56
57     |   |   | +--rw sres-exponent?  uint8

```

```

1      |      |      +---:(block-scaling)
2      |      |      | +---rw block-scalar?          uint8
3      |      |      +---:(u-law)
4      |      |      | +---rw comp-bit-width?        uint8
5      |      |      | +---rw comp-shift?            uint8
6      |      |      +---:(beam-space-compression)
7      |      |      | +---rw active-beam-space-coefficient-mask*  uint8
8      |      |      | +---rw block-scalar?          uint8
9      |      |      +---:(modulation-compression)
10     |      |      | +---rw csf?                    uint8
11     |      |      | +---rw mod-comp-scaler?        uint16
12     |      |      +---:(modulation-compression-selective-re-sending)
13     |      |      | +---rw sres-csf?                uint8
14     |      |      | +---rw sres-mod-comp-scaler?    uint16
15     |      |      +---rw downlink-radio-frame-offset  uint32
16     |      |      +---rw downlink-sfn-offset        int16
17     |      |      +---rw n-ta-offset                uint32
18     |      |      +---rw number-of-prb              uint16
19     +---:(SELECTIVE)
20
21     augment /sc:shared-cell/sc:shared-cell-config/sc:shared-cell-copy-combine-mode:
22     +---:(eCPRI7-to-CPRI8)
23     +---rw shared-cell-copy-entities-ecpri7-to-cpri8* [name]
24     | +---rw name                string
25     | +---rw low-level-tx-endpoint?      -> /uc-hub:user-plane-configuration/low-level-tx-endpoints/name
26     | +---rw south-node-processing-elements-cpri* -> /o-ran-pe:processing-elements/ru-elements/name
27     +---rw shared-cell-combine-entities-ecpri7-to-cpri8* [name]
28     | +---rw name                string
29     | +---rw low-level-rx-endpoint?      -> /uc-hub:user-plane-configuration/low-level-tx-endpoints/name

```

```
+--rw south-node-processing-elements-cpri* -> /o-ran-pe:processing-elements/ru-elements/name
```

Annex ZZZ: O-RAN Adopter License Agreement

BY DOWNLOADING, USING OR OTHERWISE ACCESSING ANY O-RAN SPECIFICATION, ADOPTER AGREES TO THE TERMS OF THIS AGREEMENT.

This O-RAN Adopter License Agreement (the “Agreement”) is made by and between the O-RAN Alliance and the entity that downloads, uses or otherwise accesses any O-RAN Specification, including its Affiliates (the “Adopter”).

This is a license agreement for entities who wish to adopt any O-RAN Specification.

Section 1: DEFINITIONS

1.1 “Affiliate” means an entity that directly or indirectly controls, is controlled by, or is under common control with another entity, so long as such control exists. For the purpose of this Section, “Control” means beneficial ownership of fifty (50%) percent or more of the voting stock or equity in an entity.

1.2 “Compliant Implementation” means any system, device, method or operation (whether implemented in hardware, software or combinations thereof) that fully conforms to a Final Specification.

1.3 “Adopter(s)” means all entities, who are not Members, Contributors or Academic Contributors, including their Affiliates, who wish to download, use or otherwise access O-RAN Specifications.

1.4 “Minor Update” means an update or revision to an O-RAN Specification published by O-RAN Alliance that does not add any significant new features or functionality and remains interoperable with the prior version of an O-RAN Specification. The term “O-RAN Specifications” includes Minor Updates.

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1.6 “Defensive Suspension” means for the purposes of any license grant pursuant to Section 3, Member, Contributor, Academic Contributor, Adopter, or any of their Affiliates, may have the discretion to include in their license a term allowing the licensor to suspend the license against a licensee who brings a patent infringement suit against the licensing Member, Contributor, Academic Contributor, Adopter, or any of their Affiliates.

Section 2: COPYRIGHT LICENSE

2.1 Subject to the terms and conditions of this Agreement, O-RAN Alliance hereby grants to Adopter a nonexclusive, nontransferable, irrevocable, non-sublicensable, worldwide copyright license to obtain, use and modify O-RAN Specifications, but not to further distribute such O-RAN Specification in any modified or unmodified way, solely in furtherance of implementations of an O-RAN Specification.

2.2 Adopter shall not use O-RAN Specifications except as expressly set forth in this Agreement or in a separate written agreement with O-RAN Alliance.

Section 3: FRAND LICENSE

3.1 Members, Contributors and Academic Contributors and their Affiliates are prepared to grant based on a separate Patent License Agreement to each Adopter under Fair Reasonable And Non- Discriminatory (FRAND) terms and conditions with or without compensation (royalties) a nonexclusive, non-transferable, irrevocable (but subject to Defensive Suspension), non-sublicensable, worldwide patent license under their Necessary Claims to make, have made, use, import, offer to sell, lease, sell and otherwise distribute Compliant Implementations; provided, however, that such license shall not extend: (a) to any part or function of a product in which a Compliant Implementation is incorporated that is not itself part of the Compliant Implementation; or (b) to any Adopter if that Adopter is not making a reciprocal grant to Members, Contributors and Academic Contributors, as set forth in Section 3.3. For the avoidance of doubt, the foregoing licensing commitment includes the distribution by the Adopter’s distributors and the use by the Adopter’s customers of such licensed Compliant Implementations.

3.2 Notwithstanding the above, if any Member, Contributor or Academic Contributor, Adopter or their Affiliates has reserved the right to charge a FRAND royalty or other fee for its license of Necessary Claims to Adopter, then Adopter

is entitled to charge a FRAND royalty or other fee to such Member, Contributor or Academic Contributor, Adopter and its Affiliates for its license of Necessary Claims to its licensees.

3.3 Adopter, on behalf of itself and its Affiliates, shall be prepared to grant based on a separate Patent License Agreement to each Members, Contributors, Academic Contributors, Adopters and their Affiliates under Fair Reasonable And Non-Discriminatory (FRAND) terms and conditions with or without compensation (royalties) a nonexclusive, non-transferable, irrevocable (but subject to Defensive Suspension), non-sublicensable, worldwide patent license under their Necessary Claims to make, have made, use, import, offer to sell, lease, sell and otherwise distribute Compliant Implementations; provided, however, that such license will not extend: (a) to any part or function of a product in which a Compliant Implementation is incorporated that is not itself part of the Compliant Implementation; or (b) to any Members, Contributors, Academic Contributors, Adopters and their Affiliates that is not making a reciprocal grant to Adopter, as set forth in Section 3.1. For the avoidance of doubt, the foregoing licensing commitment includes the distribution by the Members', Contributors', Academic Contributors', Adopters' and their Affiliates' distributors and the use by the Members', Contributors', Academic Contributors', Adopters' and their Affiliates' customers of such licensed Compliant Implementations.

Section 4: TERM AND TERMINATION

4.1 This Agreement shall remain in force, unless early terminated according to this Section 4.

4.2 O-RAN Alliance on behalf of its Members, Contributors and Academic Contributors may terminate this Agreement if Adopter materially breaches this Agreement and does not cure or is not capable of curing such breach within thirty (30) days after being given notice specifying the breach.

4.3 Sections 1, 3, 5 - 11 of this Agreement shall survive any termination of this Agreement. Under surviving Section 3, after termination of this Agreement, Adopter will continue to grant licenses (a) to entities who become Adopters after the date of termination; and (b) for future versions of ORAN Specifications that are backwards compatible with the version that was current as of the date of termination.

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Section 6: INDEMNIFICATION

Adopter shall indemnify, defend, and hold harmless the O-RAN Alliance, its Members, Contributors or Academic Contributors, and their employees, and agents and their respective successors, heirs and assigns (the “Indemnitees”), against any liability, damage, loss, or expense (including reasonable attorneys’ fees and expenses) incurred by or imposed upon any of the Indemnitees in connection with any claims, suits, investigations, actions, demands or judgments arising out of Adopter’s use of the licensed O-RAN Specifications or Adopter’s commercialization of products that comply with O-RAN Specifications.

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Section 8: ASSIGNMENT

Adopter may not assign the Agreement or any of its rights or obligations under this Agreement or make any grants or other sublicenses to this Agreement, except as expressly authorized hereunder, without having first received the prior, written consent of the O-RAN Alliance, which consent may be withheld in O-RAN Alliance’s sole discretion. O-RAN Alliance may freely assign this Agreement.

Section 9: THIRD-PARTY BENEFICIARY RIGHTS

Adopter acknowledges and agrees that Members, Contributors and Academic Contributors (including future Members, Contributors and Academic Contributors) are entitled to rights as a third-party beneficiary under this Agreement, including as licensees under Section 3.

Section 10: BINDING ON AFFILIATES

Execution of this Agreement by Adopter in its capacity as a legal entity or association constitutes that legal entity’s or association’s agreement that its Affiliates are likewise bound to the obligations that are applicable to Adopter hereunder and are also entitled to the benefits of the rights of Adopter hereunder.

Section 11: GENERAL

This Agreement is governed by the laws of Germany without regard to its conflict or choice of law provisions.

This Agreement constitutes the entire agreement between the parties as to its express subject matter and expressly supersedes and replaces any prior or contemporaneous agreements between the parties, whether written or oral, relating to the subject matter of this Agreement.

Adopter, on behalf of itself and its Affiliates, agrees to comply at all times with all applicable laws, rules and regulations with respect to its and its Affiliates' performance under this Agreement, including without limitation, export control and antitrust laws. Without limiting the generality of the foregoing, Adopter acknowledges that this Agreement prohibits any communication that would violate the antitrust laws.

By execution hereof, no form of any partnership, joint venture or other special relationship is created between Adopter, or O-RAN Alliance or its Members, Contributors or Academic Contributors. Except as expressly set forth in this Agreement, no party is authorized to make any commitment on behalf of Adopter, or O-RAN Alliance or its Members, Contributors or Academic Contributors.

In the event that any provision of this Agreement conflicts with governing law or if any provision is held to be null, void or otherwise ineffective or invalid by a court of competent jurisdiction, (i) such provisions will be deemed stricken from the contract, and (ii) the remaining terms, provisions, covenants and restrictions of this Agreement will remain in full force and effect.
